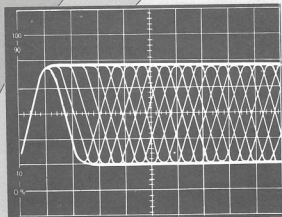


Technical Guide

COMPACT
disc
DIGITAL AUDIO

Vol. 17

Panasonic



AUTO PRODUCTS DIVISION
Matsushita Communication Industrial Co., Ltd.

CONTENTS

CIRCUIT OPERATION

1. TRAVERSE MECHANISM and OPTICAL PICKUP	1
1-1 Traverse Mechanism	1
1-2 3-Beam Type and Single Beam Type	2
1-3 Tracking Error Generation	3
2. BLOCK DIAGRAM and CD SYSTEM OPERATION	5
2-1 Block Diagram	5
2-2 CD System Operation	6
3. LOADING MECHANISM and OPERATION	8
3-1 Disk Loading Mechanism	8
3-2 Loading Operation	9
3-3 Eject Operation	10
3-4 Disk Loading Circuit	11
4. POWER CIRCUIT	13
4-1 Required Power for CD Circuit	14
4-2 Power Supply Control	14
5. MICROCOMPUTER PERIPHERAL CIRCUITS	15
5-1 Reset Circuit	15
5-2 Keyboard Interface Circuit	17
5-3 LCD and LED Display Circuit	18
6. AN8370S CIRCUIT OPERATION	20
6-1 Control Logic Circuit	21
6-2 Laser APC Circuit	23
6-3 RF Amplifier and Dropout Detector Circuits	23
6-4 Auto Focus Circuit and Focus Error Circuit	25
6-5 Tracking Error Circuit and Kick Hold Circuit	27
6-6 Focus Tracking Drive Circuit	31
7. TRAVERSE SERVO CIRCUIT OPERATION	32
7-1 Traverse Detector Circuit	33
7-2 Mode Select Circuit	33
7-3 Traverse Drive Circuit	33
8. AN8371S CIRCUIT OPERATION	34
8-1 EFM Signal	34
8-2 PCK Signal	35
8-3 Other Signals	35

9. MN6617 CIRCUIT OPERATION	36
9-1 Recording System Signal Processing	36
9-2 Reproducing System Signal Processing	41
9-3 LSI and Associated Operations of Peripheral Circuits	44
10. AUDIO CIRCUIT	51
10-1 D-A Converter	52
10-2 Deglitch Circuit	52
10-3 LPF and De-emphasis Circuit	53
11. MUTING CIRCUIT and SYSTEM CONTROL	54
TROUBLESHOOTING	
1. SYMPTOM CHECK	55
2. CHECKING PROCEDURE	56
2-1 Loading Check	56
2-2 Disk Sensor Check	60
2-3 Focus Search Check	62
2-4 Spindle Motor Circuit Check	65
2-5 RF Signal Check, etc.	67
2-6 LCD and LED Check	71
2-7 Audio Circuit Check	73
2-8 Operation Check	76
2-9 Readjustment	78
3. APPENDIX	79
3-1 Connector Voltages	79
3-2 Microcomputer Pin Voltages	83

CIRCUIT OPERATION

1. TRAVERSE MECHANISM and OPTICAL PICKUP

1-1 Traverse Mechanism

Fig. 1-1 shows the traverse mechanism.

The traverse mechanism conveys power from the motor to the wormgear, gear A, gear B, and main gear, and drives the optical pickup with the rack gear.

Gear A and gear B are double clutch gears.

The inner REST switch is closed by the optical pickup base when the optical pickup is returned to the innermost position by the traverse mechanism.

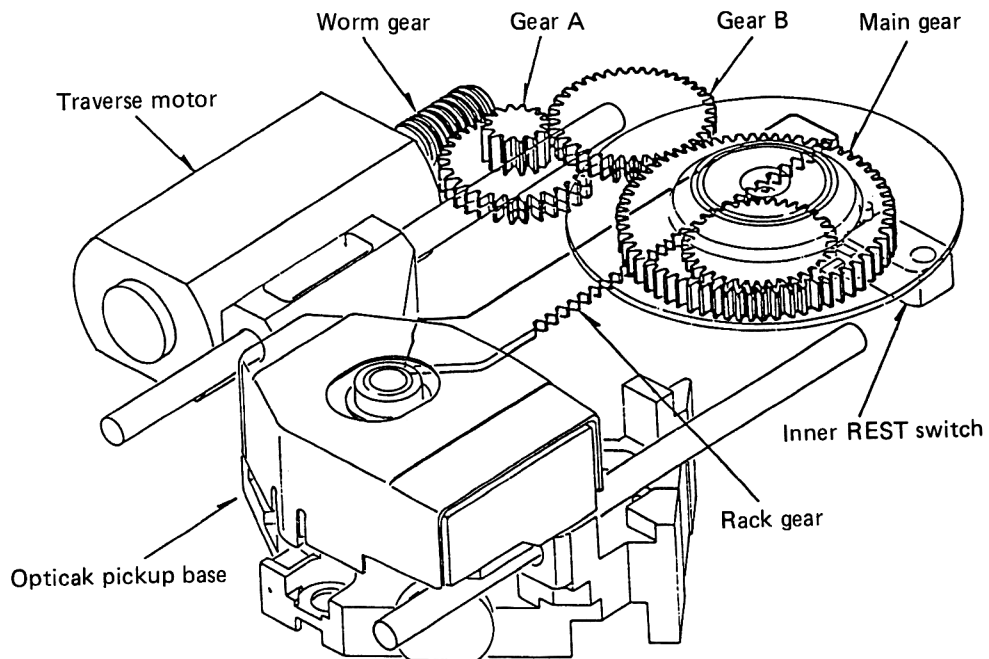


Fig. 1-1 Traverse Mechanism

1-2 3-Beam Type and Single Beam Type

The optical pickup of the SL-XP7 is a single beam type instead of a 3-beam type. The structure of the optical pickup is shown in Fig. 1-2.

The largest difference in optical structure between the 3-beam type and single beam type is that the single beam type has no diffraction grating. Therefore, the single beam type has only beam on the optical path, and its photo-detector is divided into four instead of six (four plus two).

The two sub-beams of the 3-beam type have the function of generating tracking errors. The process of generating tracking errors by the single beam type that is without sub-beams is described in the following. Focus errors are generated by the astigma method as in conventional models.

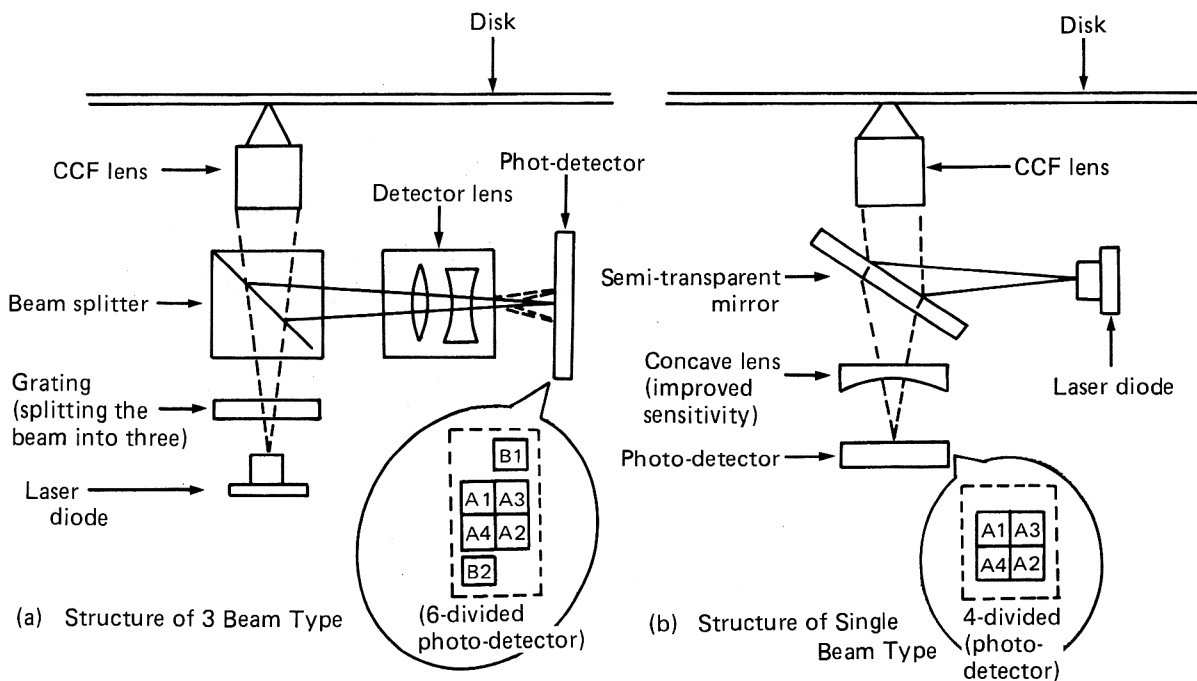


Fig. 1-2 Structure of Optical Pickup

1-3 Tracking Error Generation

Fig. 1-3 shows the basic configuration of the single beam type (phase difference detecting method). The output of the optical pickup is the sum of each diagonal element of the 4-divided photo-detector.

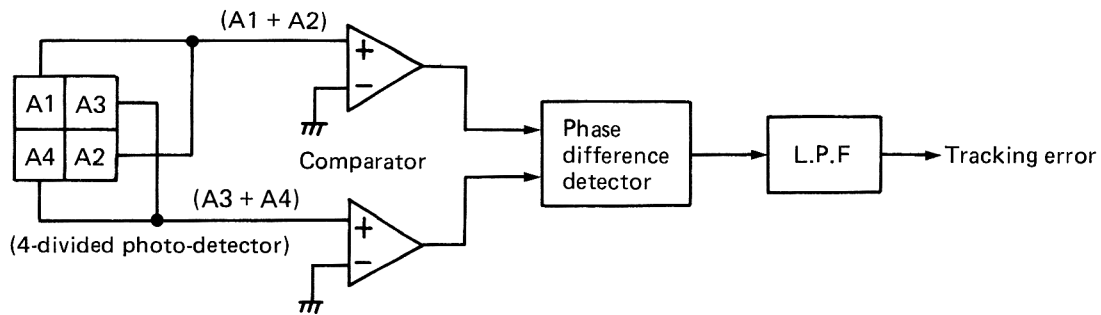
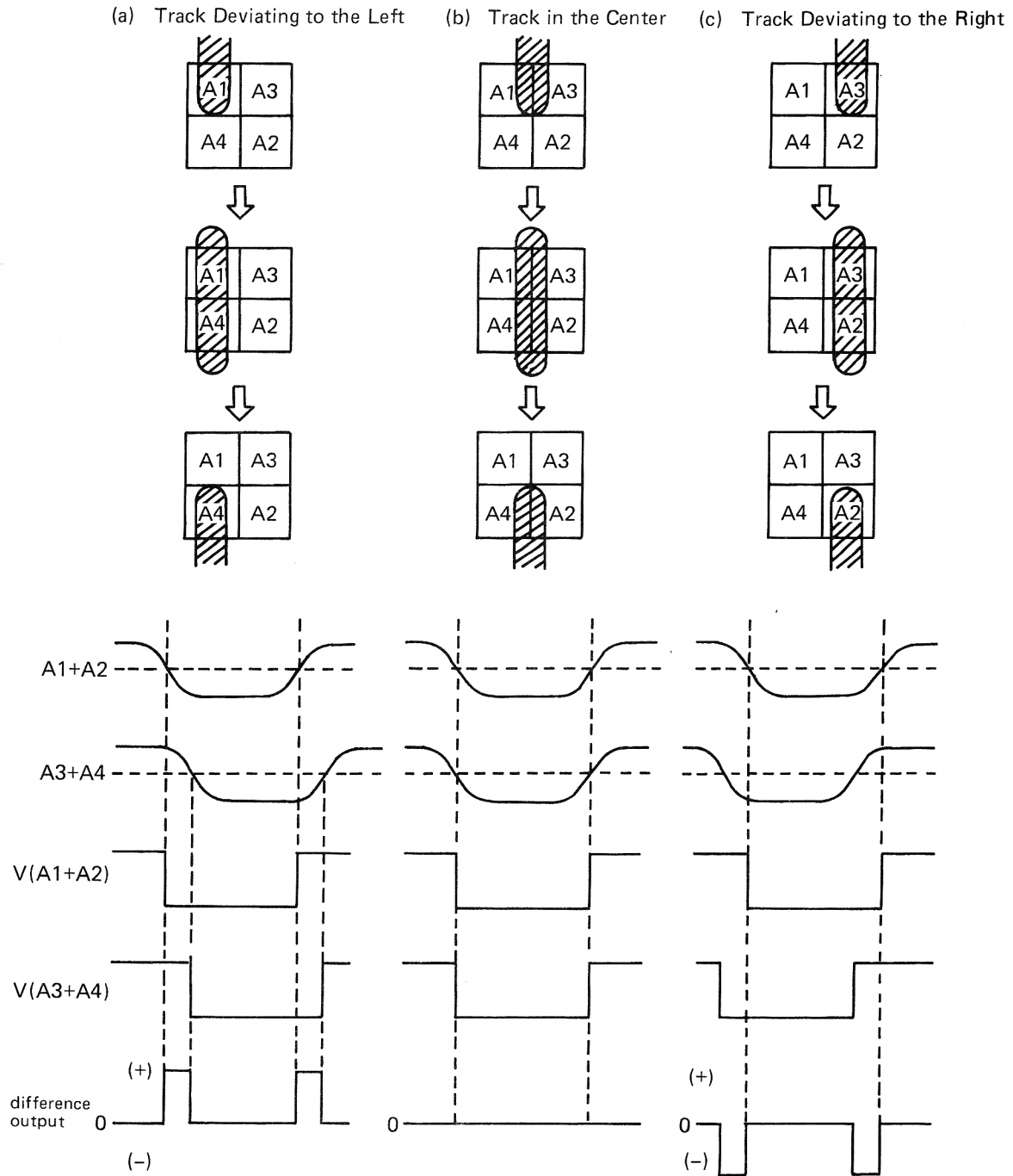


Fig. 1-3 Basic Configuration

If a track pit image is assumed to pass over the 4-divided photo-detector, and deviates to the left or to the right, a phase difference corresponding to the track deviation is generated between output signals $(A1 + A2)$ and $(A3 + A4)$. This phase difference is detected by the phase detector, and passed through the lowpass filter (L.P.F.) to generate a tracking error.

Fig. 1-4 shows a tracking error detection diagram.



- A positive pulse is generated if $V(A1 + A2)$ is earlier than $V(A3 + A4)$, or a negative pulse is generated in the opposite case.

Fig. 1-4 Tracking Error Detection Diagram

2. BLOCK DIAGRAM and CD SYSTEM OPERATION

2-1 Block Diagram

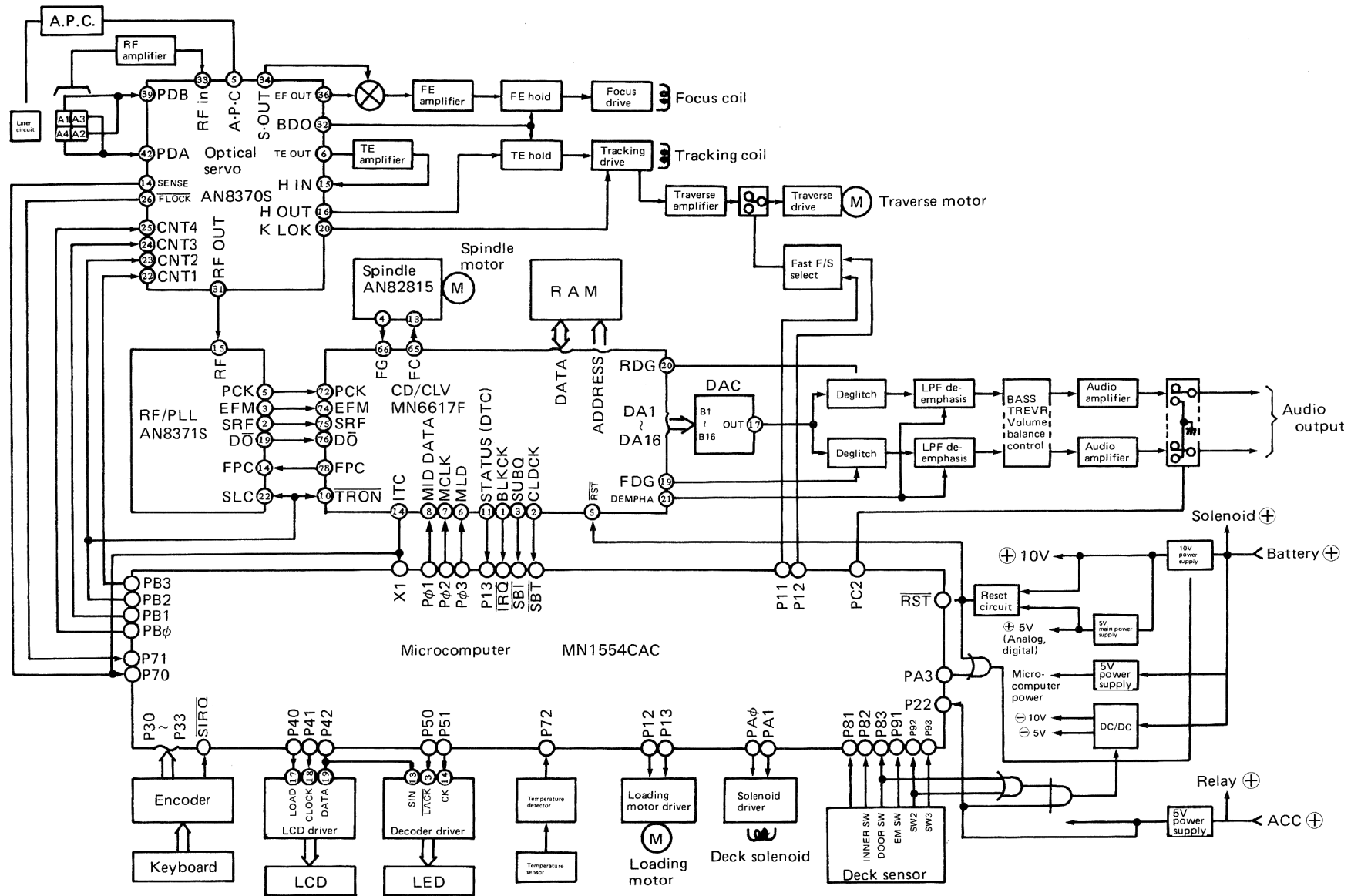
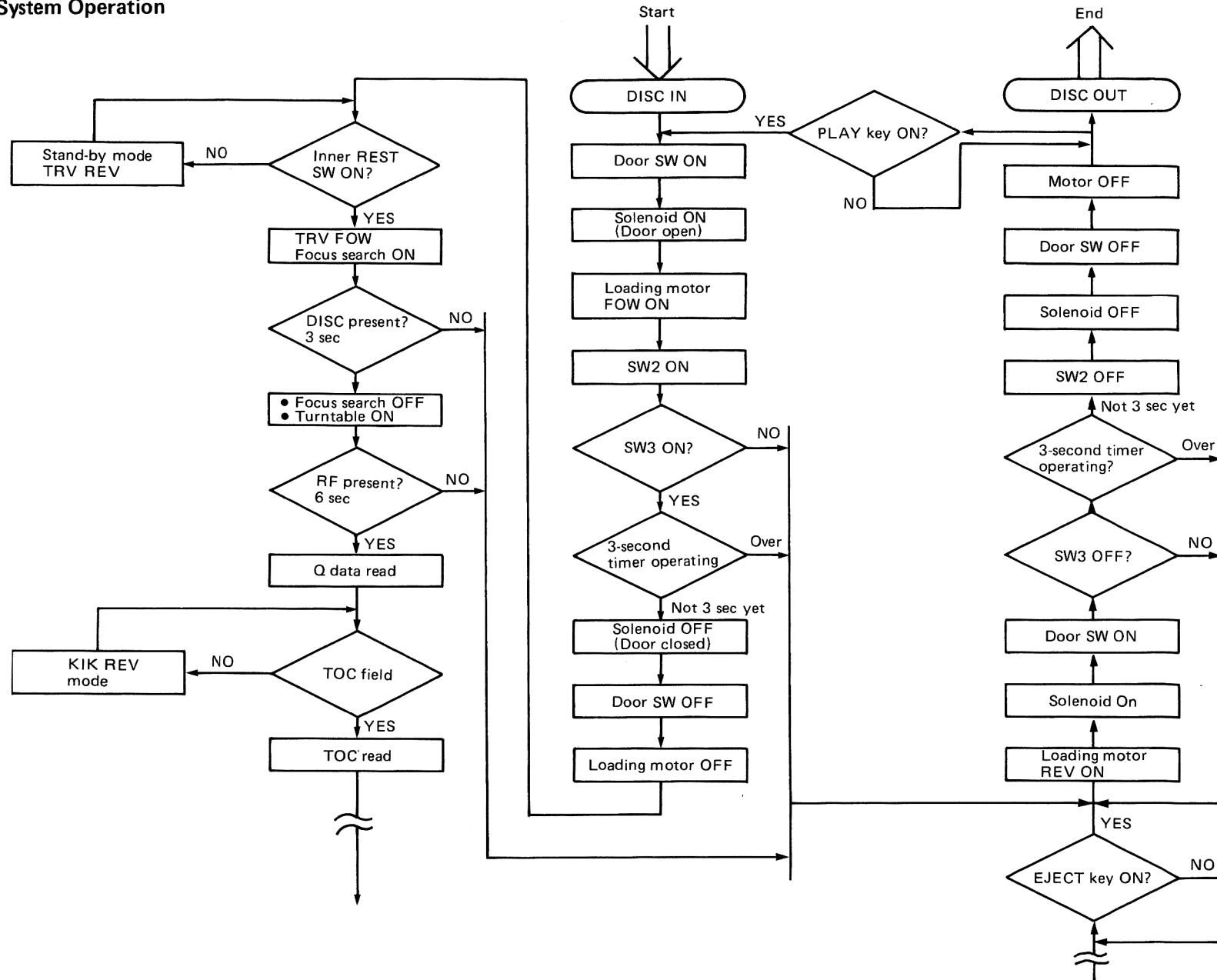
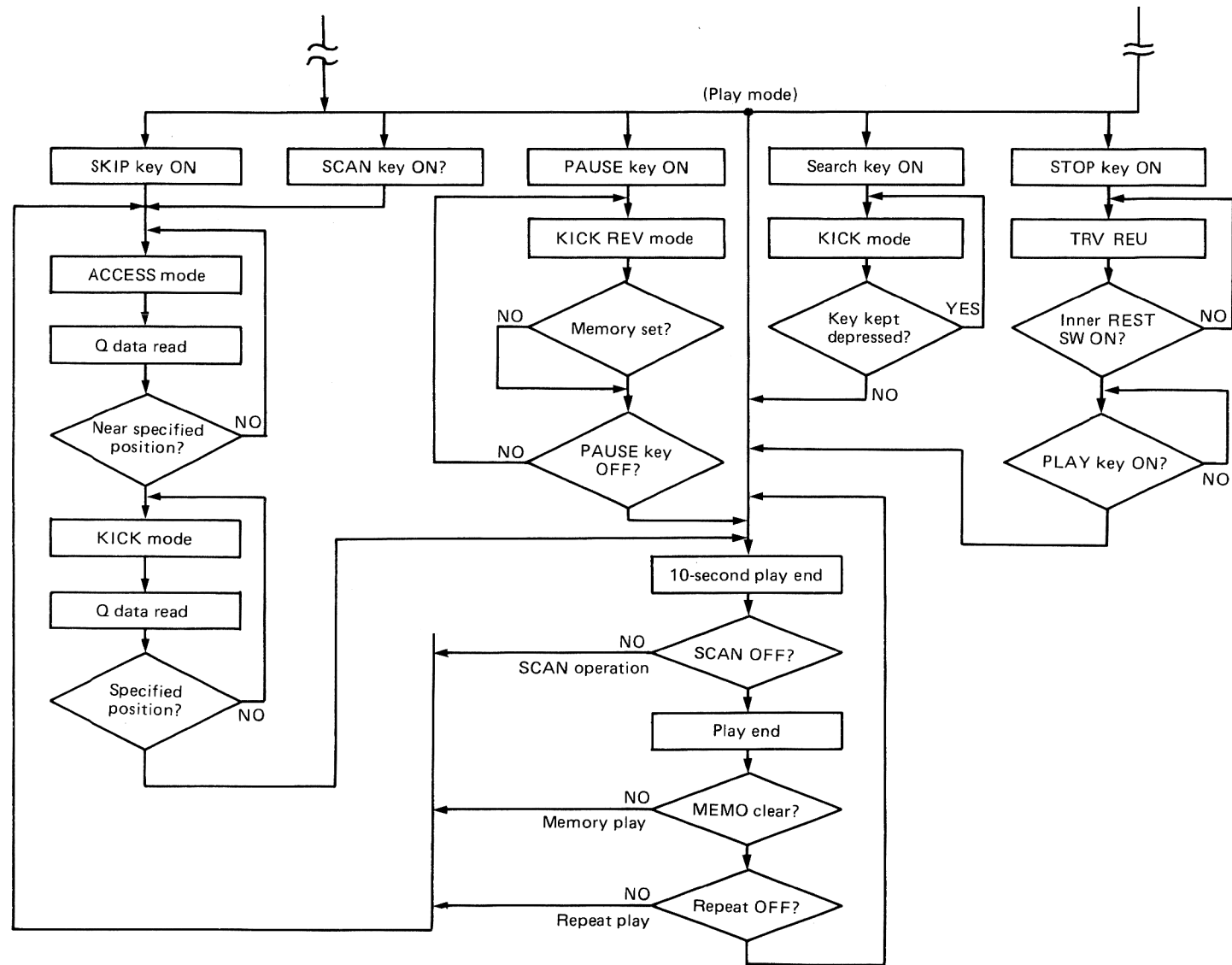


Fig. 2-1 Block Diagram

2-2 CD System Operation





3. LOADING MECHANISM and OPERATION

3-1 Disk Loading Mechanism

The following optical switches are provided as sensors to detect disk position.

- Door switch
- SW2
- SW3

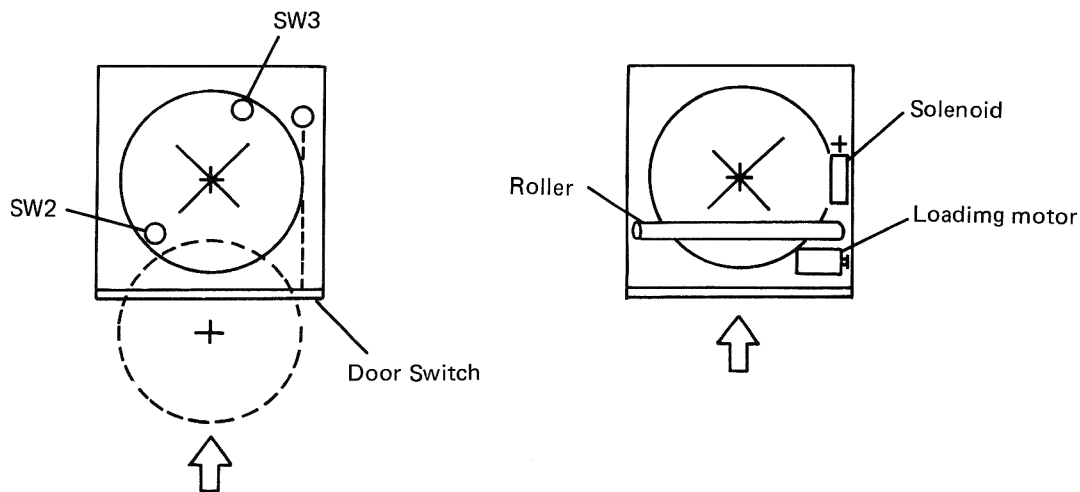


Fig. 3-1 Loading Mechanism Sensor Locations

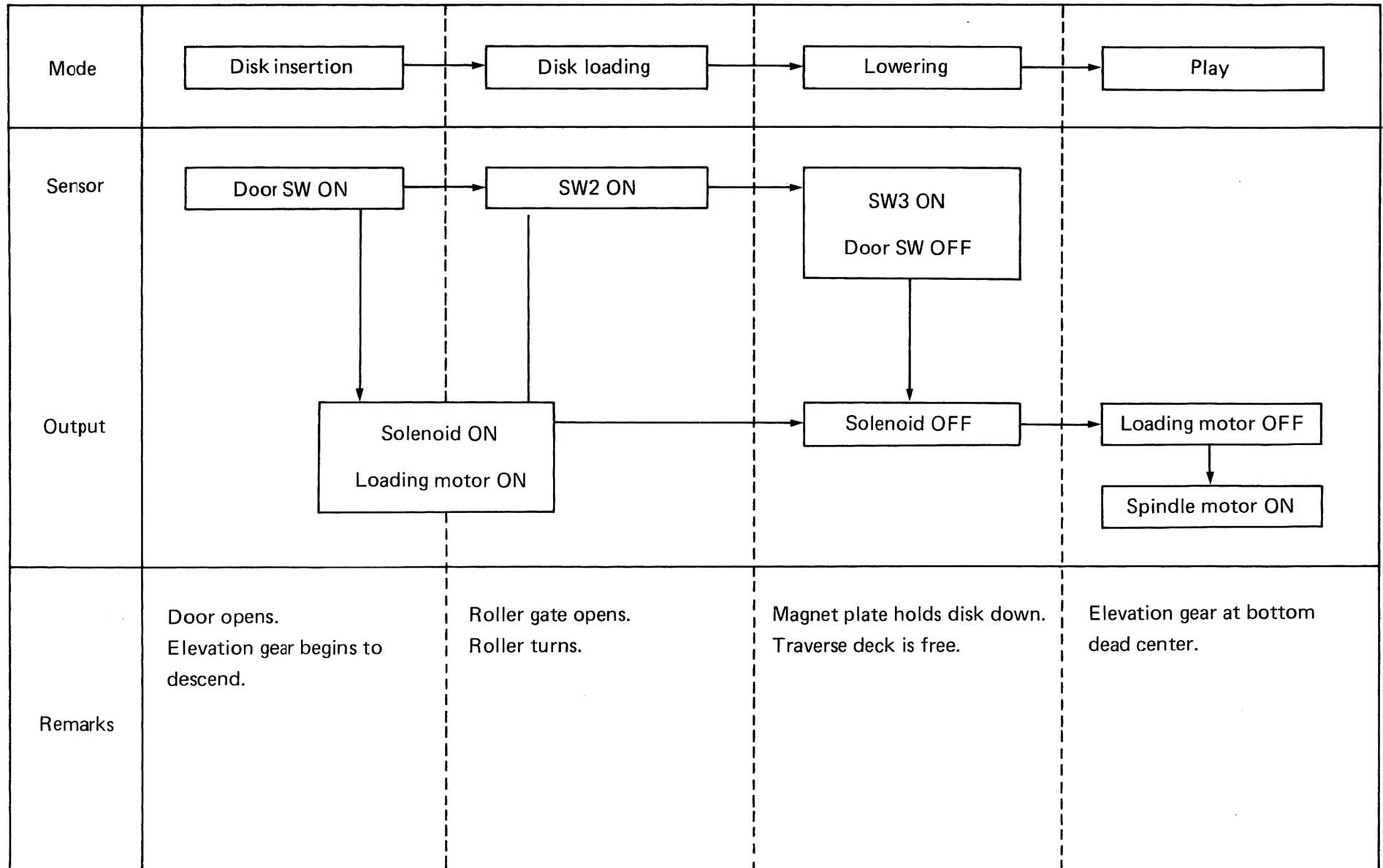
The following are provided as driving devices:

- Loading motor : Turns both ways to load, eject, raise, and lower disks.
- Solenoid : Opens and closes the door, and allows descent on the traverse deck

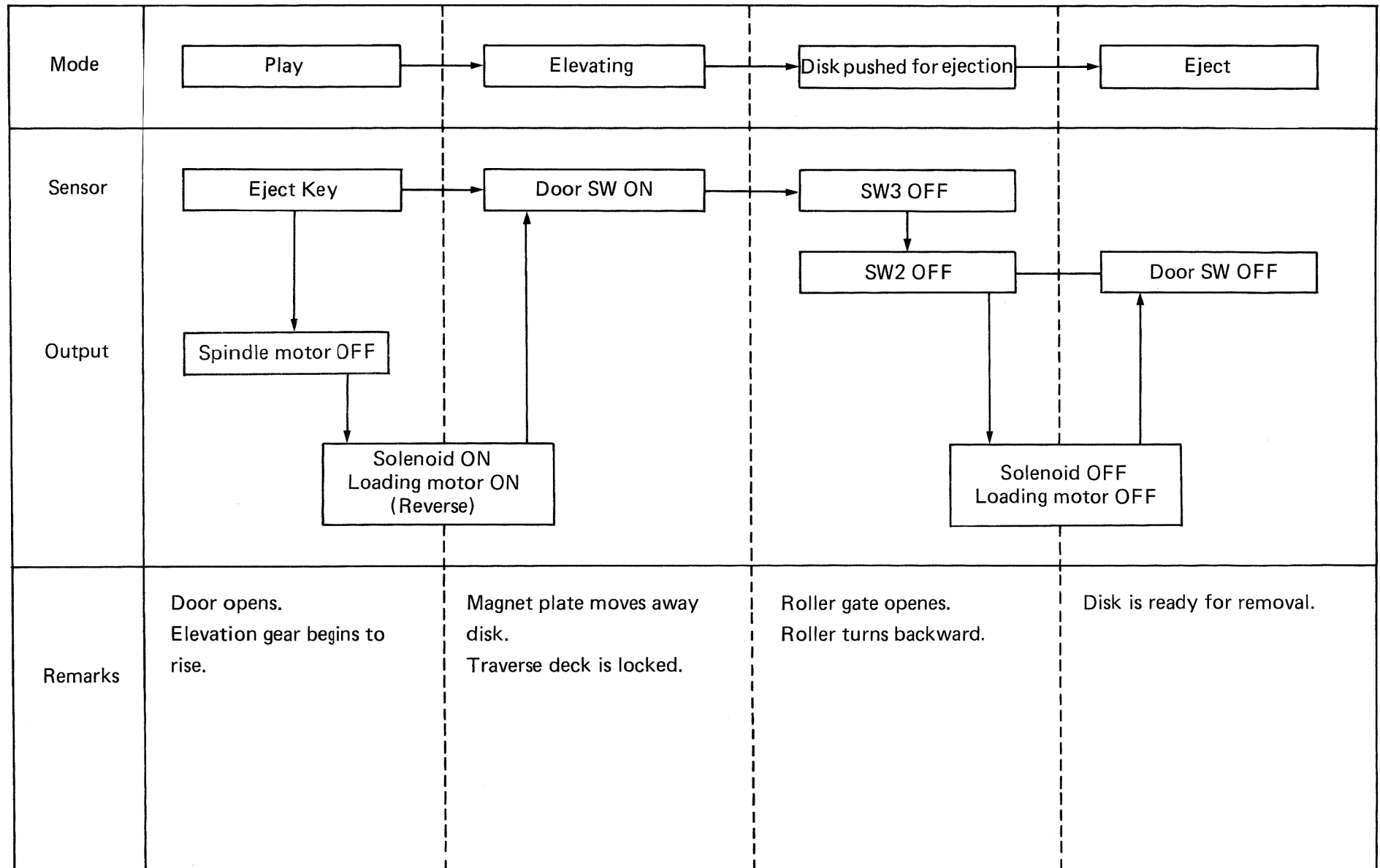
In the play mode, the traverse deck is set on a free cushion base with the disk held down by the magnet plate.

* The drop detection switch is a standby switch that is used to recover the magnet plate by means of EJECT, should it drop when there is no disk loaded.

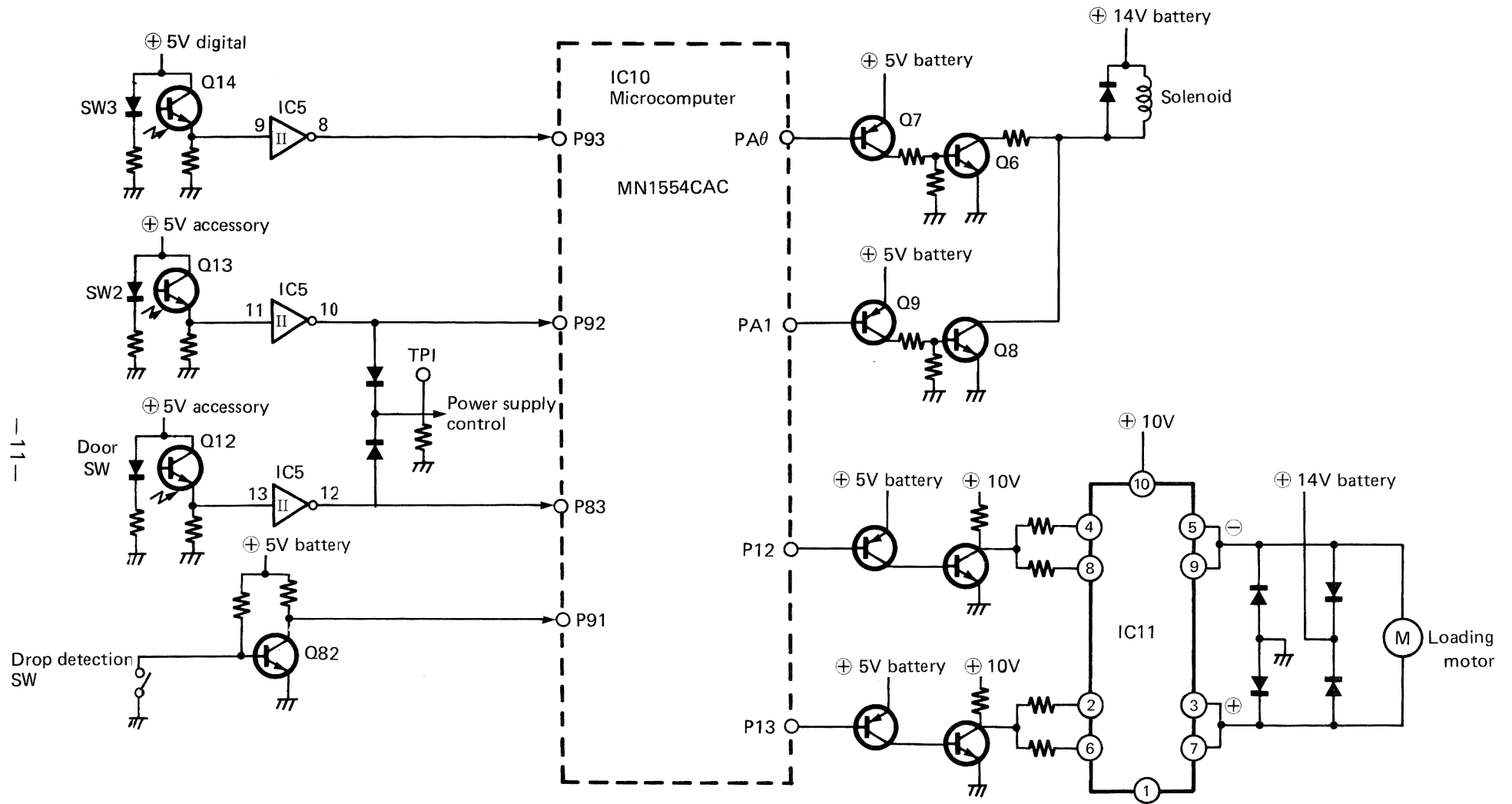
3-2 Loading Operation



3-3 Eject Operation



3-4 Disk Loading Circuit



* Input and output: Low active

Fig. 3-2 Disk Loading Circuit

(1) Sensor input

In disk loading and ejection, the microcomputer controls the loading motor and solenoid with four sensor inputs.

When a disk is set, the sensor information input to the microcomputer is as follows.

SW3	(P93)		0V
SW2	(P92)		0V
Door SW	(P83)		5V
Drop detection SW	(P91)		5V

Loading starts as door SW ON information or SW2 ON information switches on the following:

- Power supply
- Solenoid
- Loading motor

Eject starts as the EJECT SW is pressed.

(2) Solenoid control

The solenoid is controlled by PA ϕ and PA1 outputs.

PA1 is for starting the solenoid.

PA ϕ is for holding the solenoid.

The solenoid operates when the disk goes in or out. (0V during operation)

(3) Loading motor control

There are three loading motor control modes.

- Forward operation for loading
- Reverse operation for eject
- Stop

Pin \ Mode		Loading	Eject	Stop
IC10	P ⑫	5V	0	5V
	P ⑬	0	5V	5V
IC11	P ⑤ ⑨	0	10V	0
	P ③ ⑦	10V	0	0

IC11 is a complementary transistor driver, whose equivalent circuit is shown in Figure 3-3.

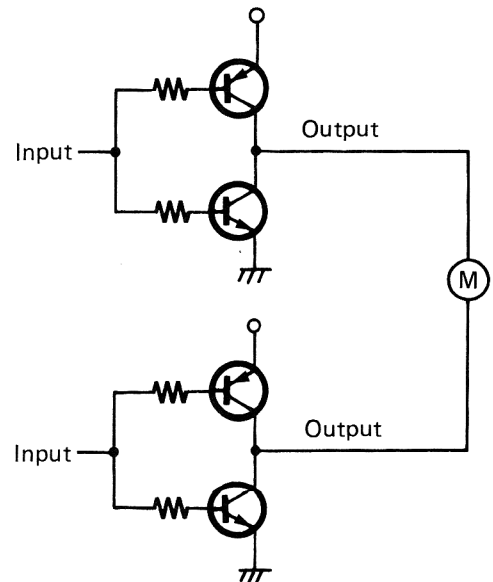


Fig. 3-3 Loading Motor Driver Circuit

4. POWER CIRCUIT

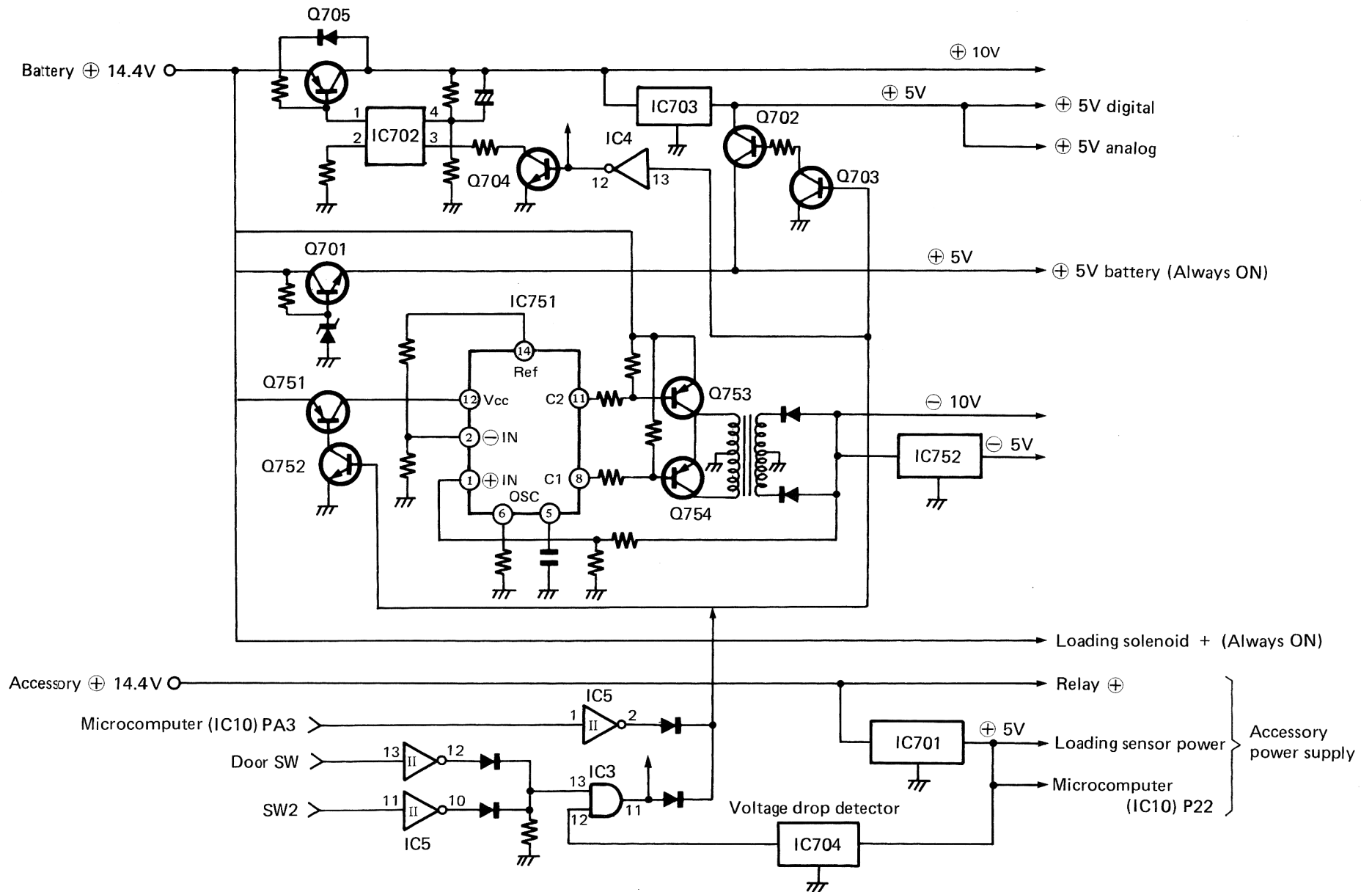


Fig. 4-1 Power Circuit

4-1 Required Power for CD Circuit

The following power inputs independent of the power supply for the system are necessary for the CD circuit.

- 1) Direct input from battery
- 2) Input from accessory power circuit

However, illumination power is supplied from the main amplifier for the system.

The power required for the CD circuit is as follows:

- 1) $\oplus$ 10V, $\ominus$ 10V
- 2) $\oplus$ 5V, $\ominus$ 5V

The $\oplus$ 5V power supply is divided as follows for preventing noise and selection during operation.

- 1) $\oplus$ 5V digital power for digital circuit
- 2) $\oplus$ 5V analog power for analog circuit IC
- 3) $\oplus$ 5V battery power for microcomputer, supplied before disk insertion
- 4) $\oplus$ 5V sensor power from accessory power supply to loading sensor

4-2 Power Supply Control

$\oplus$ 5V battery power is kept supplied. The following voltages, however, are supplied when door SW and SW2 are closed (going low) by disk insertion.

- 1) $\oplus$ 10V
- 2) $\oplus$ 5V digital, $\oplus$ 5V analog
- 3) $\ominus$ 10V, $\ominus$ 5V

Control by signal (IC10) PA3 low from the microcomputer is for supplying power until the disk is ejected in the event of faulty disk loading.

5. MICROCOMPUTER PERIPHERAL CIRCUITS

5-1 Reset Circuit

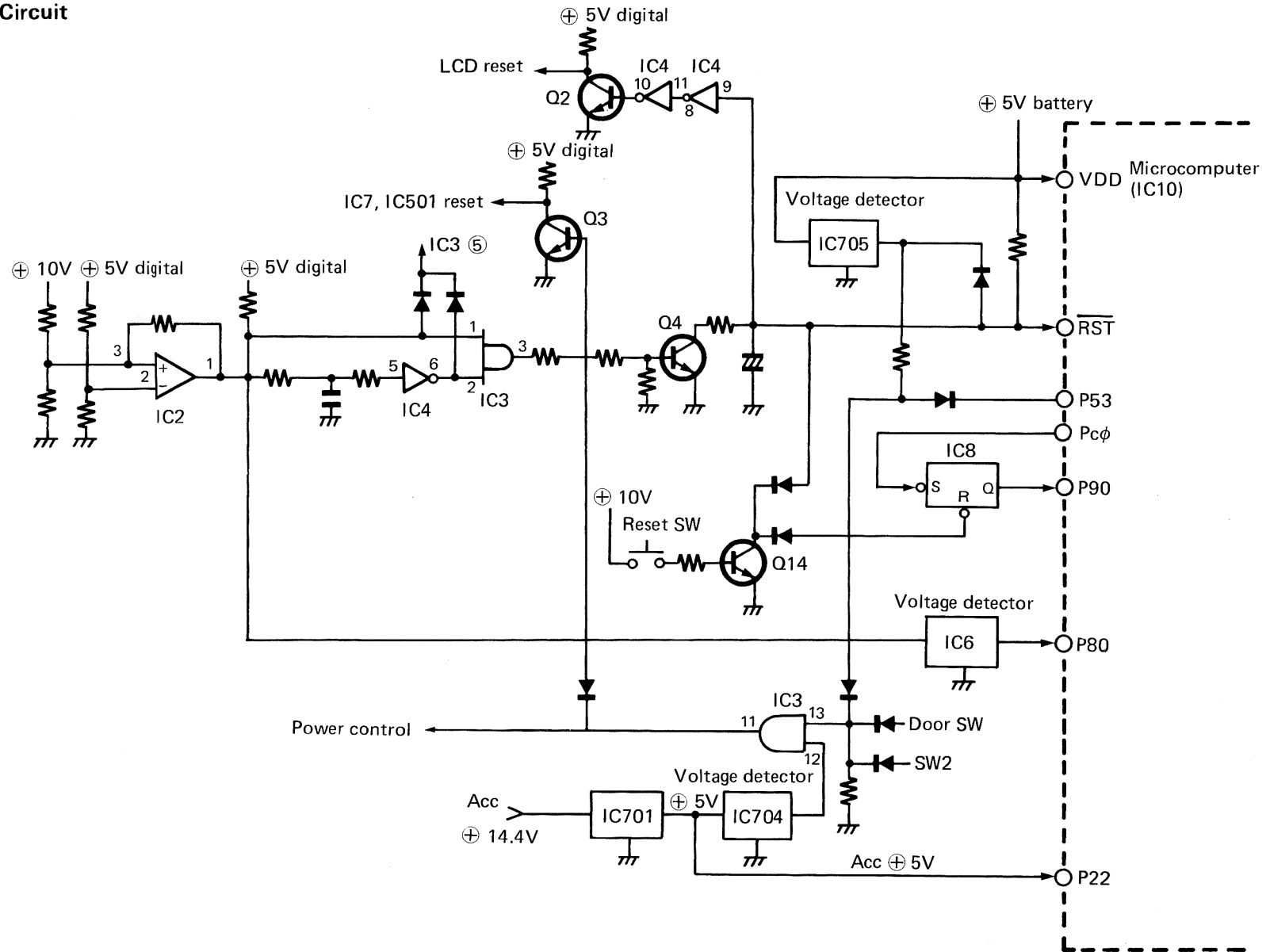


Fig. 5-1 Reset Circuit

The reset circuit is so designed that it will generate a reset pulse when a disk is inserted.

IC2 detects a rising voltage, and IC4 and IC3 generate a reset pulse.

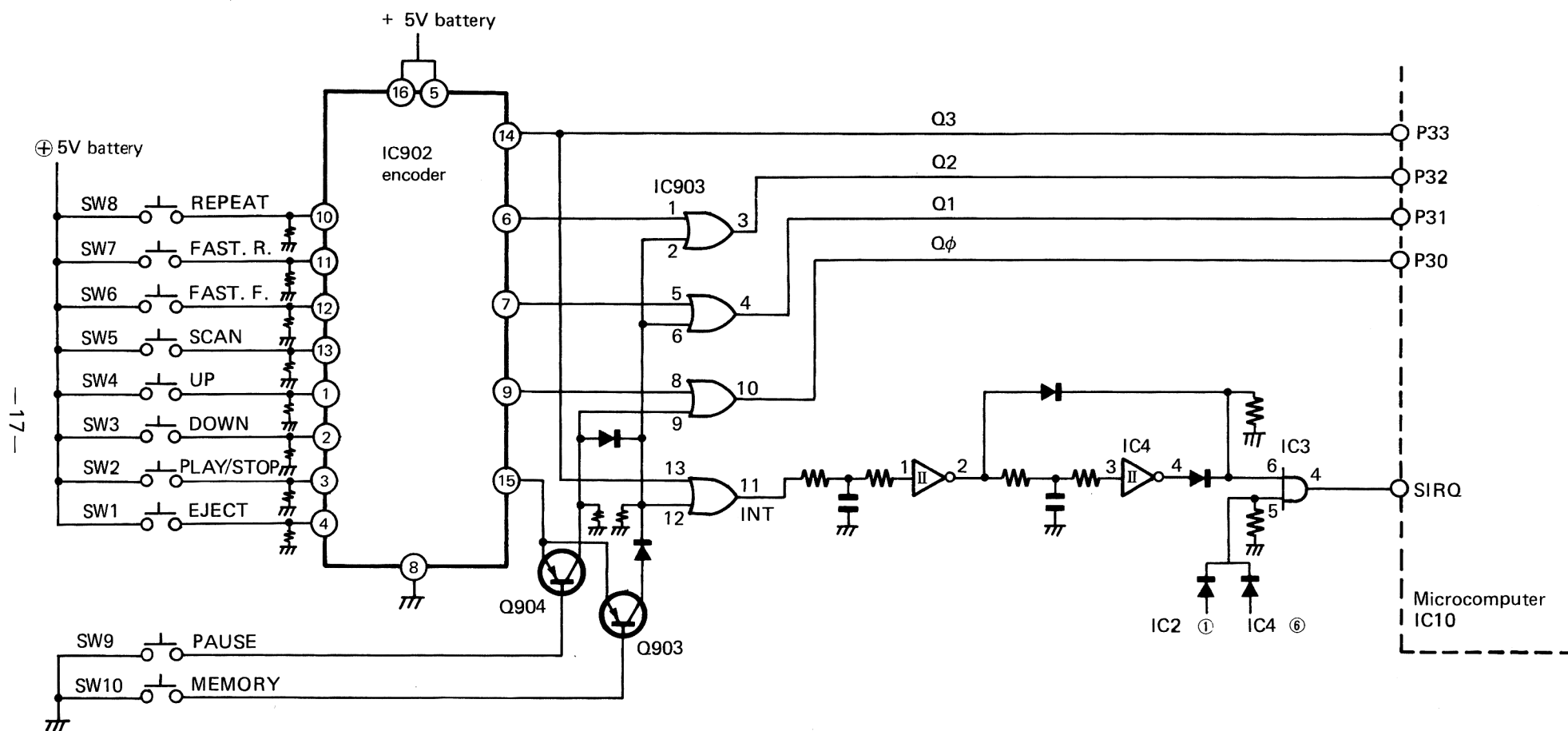
A reset switch is provided for forced resetting.

IC705, IC6, and IC704 (MN1208) are voltage drop detecting ICs. They output input voltages of up to about 4.5V, and are grounded if the input voltage drops below about 4.5V, thus protecting microcomputer operation.

In normal condition, microcomputer board voltages are as follows:

<u>VDD</u>	5V
PST	5V
P53	5V
PC ϕ	5V
P90	0V
P80	5V
P22	5V

5-2 Keyboard Interface Circuit



Ten key inputs are converted to 4-bit codes to give commands to the microcomputer. INT signal is routed through the delay circuit and waveform shaping circuit to supply SIRQ H pulse.

5-2 Keyboard Interface Circuit

5-3 LCD and LED Display Circuit

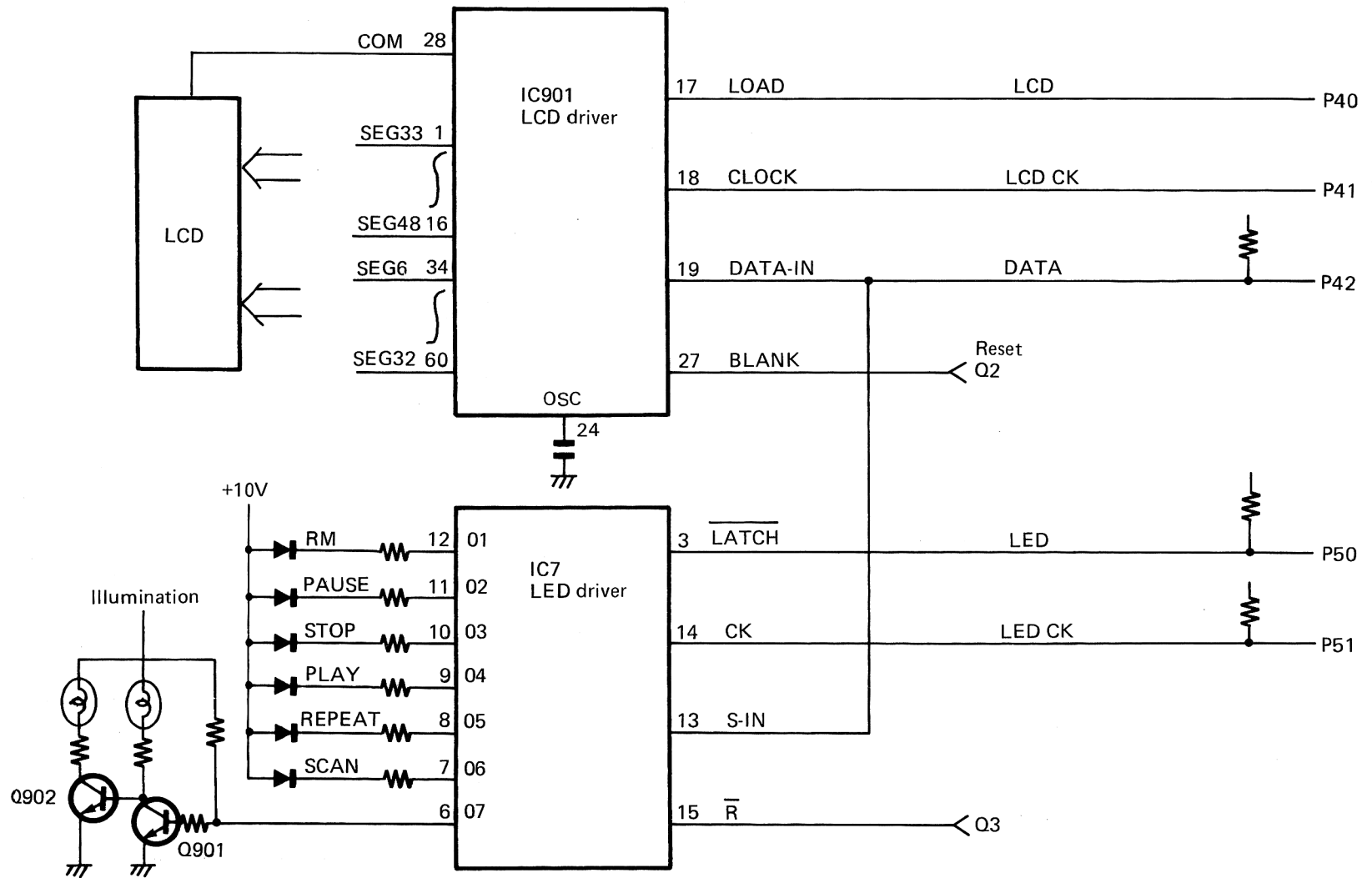


Fig. 5-3 LCD and LED Display Circuit

(1) LCD display

The LCD is used for track and time display.

Serial data synchronized with the clock is input from the microcomputer to the LCD driver, and each time that serial data ends, LOAD signal is added to the data to be sent to the LCD.

OSC is a terminal for generating a COM frequency, which is one-eighth of the OSC frequency. Signal voltage between COM and each SEG are used for LCD display as follows:

SEG output with phase opposite to COM signal: ON

SEG output with same phase to COM signal OFF

(2) LED display

The LED display indicates seven statuses.

Serial data synchronized with the clock is input from the microcomputer to the LED driver, and each time that serial data ends, LATCH signal is added to the data for selecting an LED and giving an indication. The two pilot lamps that are connected to 07 are orange and green and operate as follows:

- 1) Orange pilot lamp lights when no disk is in.
- 2) Green pilot lamp lights when a disk is inserted.
- 3) Orange and green lamps alternately light during loading or ejection.

6. AN8370S CIRCUIT OPERATION

The optical servo reads the signals recorded on a disk using the laser beam.

AN8370S is the main IC for the optical servo, consisting of laser APC, RF signal amplifier, focus servo (including auto focus), tracking servo (including kick), drop output detector, and the control logic that controls the optical servo in this IC.

A block diagram of AN8370S is shown in Fig. 6-1.

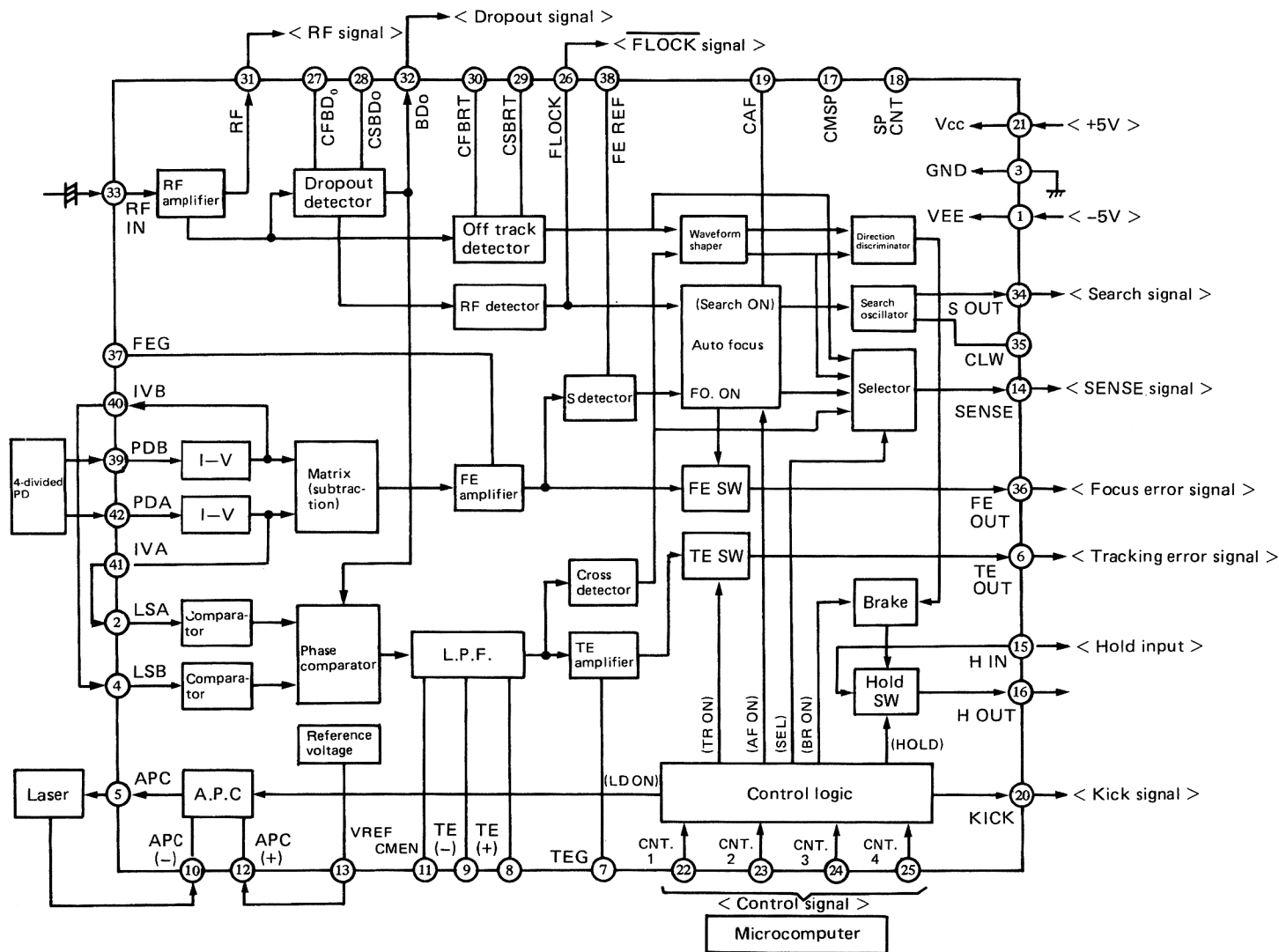


Fig. 6-1 AN8370S Block Diagram

6-1 Control Logic Circuit

Control signals CNT1 to CNT4 are 4-bit parallel output signals from the microcomputer. The logic circuit of AN8370S decodes these control signals and outputs the required control signals to the optical servo system.

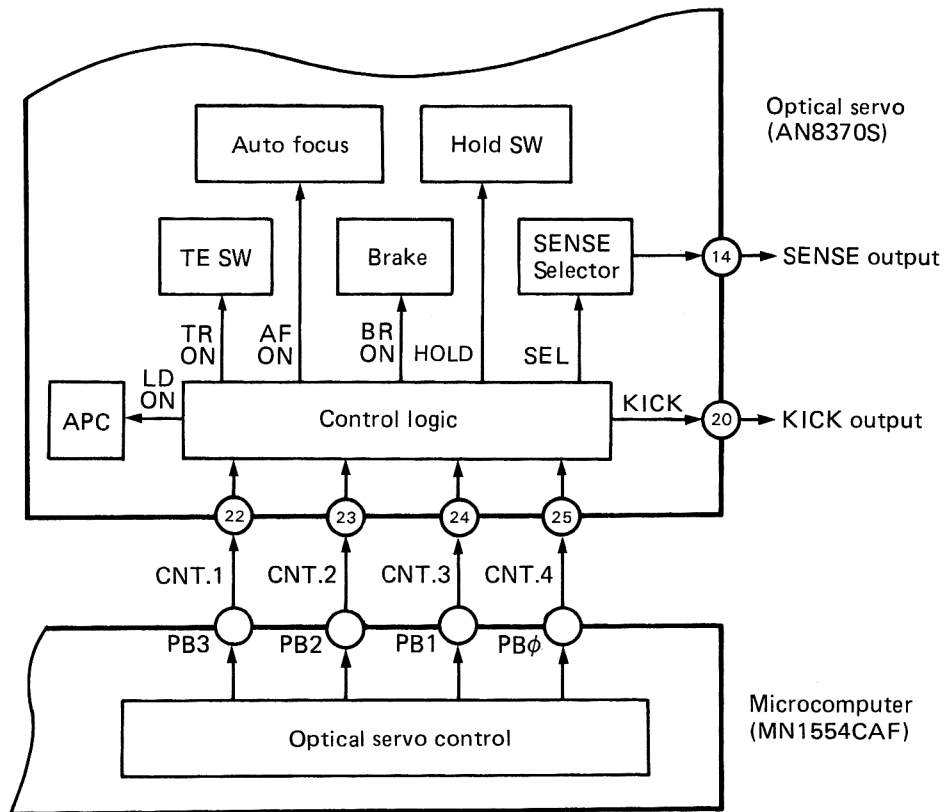


Fig. 6-2 Control Logic Circuit

(1) Control signals and their functions (See Fig. 6-2.)

- 1) LD ON Laser ON command
 - Controls laser APC ON/OFF.
- 2) AF ON Auto focus command
 - Controls the focus servo loop to open or close, and the search OSC switch ON/OFF.
- 3) TR ON Tracking servo command
 - Controls the tracking servo loop switch.
- 4) HOLD Tracking hold command
 - Holds tracking error at preceding value during track kick.
- 5) BR ON Tracking brake command
 - After high-speed feed or kick, brakes the tracking actuator to absorb unnecessary vibration in a short time to facilitate tracking.

- 6) SEL SENSE signal select command
- Selects SENSE signal (that indicates optical servo status) to be sent from AN8370S to the microcomputer.
- SENSE signal changes depending on the system mode.
- 7) KICK Track kick command
- Controls track kick output signals.
- There are KICK FWD (outward) and KICK REV (inward), and three modes of 1 kick, 10 kicks, and 100 kicks according to kick width.
- The logic circuit has 16 ($2^4 = 16$) decoding modes, of which eight are used as shown in Table 6-1.

(1 "H", 0 "L")

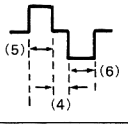
HEX No.	Mode	Logic input				Logic control signal output							Remarks
		CNT1	CNT2	CNT3	CNT4	LD ON	AF ON	TR ON	HOLD	BR ON	KICK	SENSE	
0	PLAY	0	0	0	0	ON	ON	ON	OFF	OFF	OFF	Off track	Play mode
3	BRAKE	0	0	1	1	ON	ON	ON	OFF	ON	OFF	Off track	Brake mode
4	KICK CROSS	0	1	0	0	ON	ON	OFF	ON	OFF	OFF	CROSS	Kick mode 
5	KICK REV	0	1	0	1	ON	ON	OFF	ON	OFF	ON (REV)	Shaping CROSS	
6	KICK FWD	0	1	1	0	ON	ON	OFF	ON	OFF	ON (FWD)	Shaping CROSS	
7	ACCESS	0	1	1	1	ON	ON	OFF	OFF	OFF	OFF	Shaping CROSS	Coarse feed
8	DISK DETECT	1	0	0	0	ON	ON	OFF	OFF	OFF	OFF	S latch	Disk detection
F	STANDBY	1	1	1	1	OFF	OFF	OFF	OFF	OFF	OFF	Shaping CROSS	Forced return

Table 6-1 Control Logic Table

6-2 Laser APC Circuit

Fig. 6-3 shows the laser APC circuit.

When a laser ON mode command is emitted, pin 5 goes high to turn the laser on.

A loop circuit is formed by Q and D (the laser paired with a PIN diode) and the APC in the IC, and laser beam intensity is detected by the PIN diode in APC operation.

If laser beam intensity increases, the PIN diode's photo-electric current increases so that the pin 10 voltage rises (pin 12 voltage is the reference voltage) to decrease laser beam intensity.

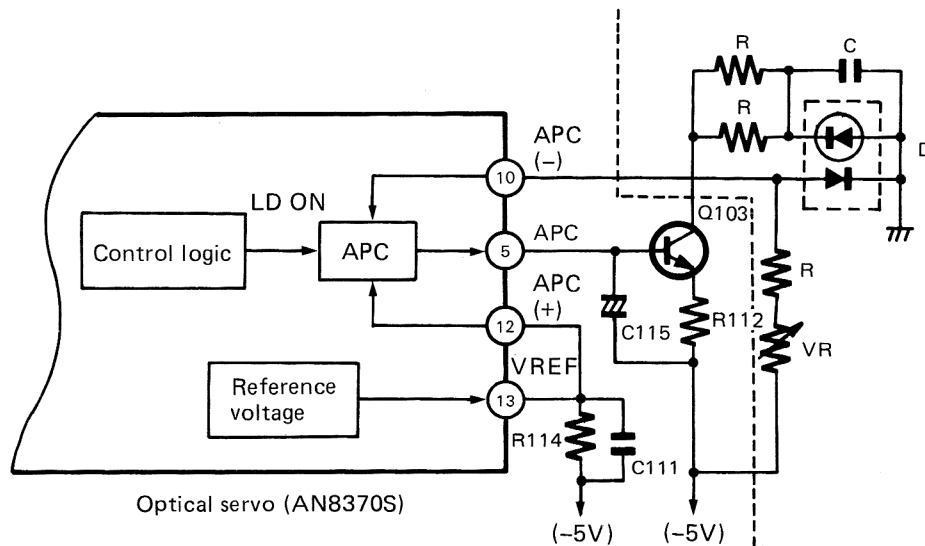


Fig. 6-3 Laser APC Circuit

6-3 RF Amplifier and Dropout Detector Circuits

(1) RF amplifier circuit

RF signal is the total sum of the 4-divided photo-detector ($A1 + A2 + A3 + A4$) of the optical pickup. These are added in the optical pickup, and are input via Q to pin 33. (Q is located in the optical pickup.)

The output is taken out from pin 31, from which it is sent to the next IC for PLL (AN8371S).

The gain between pin 33 and pin 31 is about 34 dB.

(2) Dropout detector circuit

In dropout detection, a dropout on the disk is detected from the above-mentioned RF signal, the tracking servo is stabilized, and its detection signal is output to pin 32.

The detecting process is shown in Fig. 6-5. The level goes high if a dropout is detected.

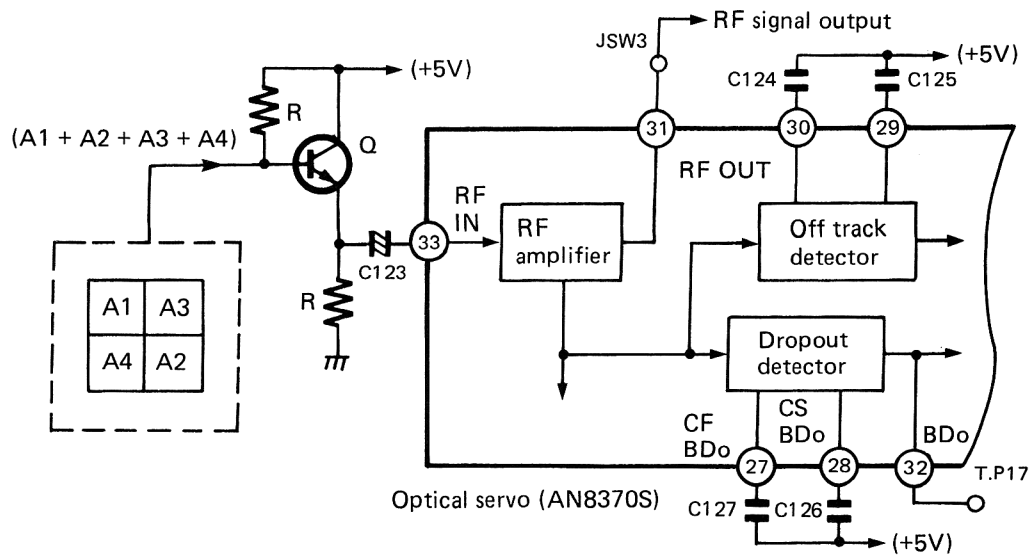


Fig. 6-4 RF Amplifier and Dropout Detection

(3) Off track detector

Off track detection refers to a signal resulting from detection of the beam between tracks.

When the tracking servo is disconnected, the beam crosses a track. In this case, the relationship between RF signal and off track detection signal is as shown in Fig. 6-6.

This detection signal is used as a control signal in the IC and also as SENSE signal to be sent to the microcomputer.

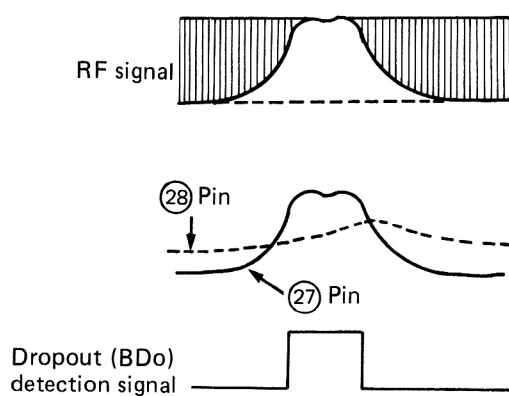


Fig. 6-5 Dropout Detection

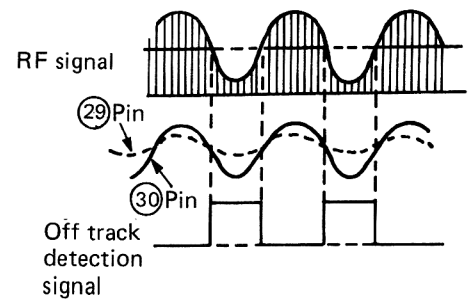


Fig. 6-6 Off Track Detection

6-4 Auto Focus Circuit and Focus Error Circuit

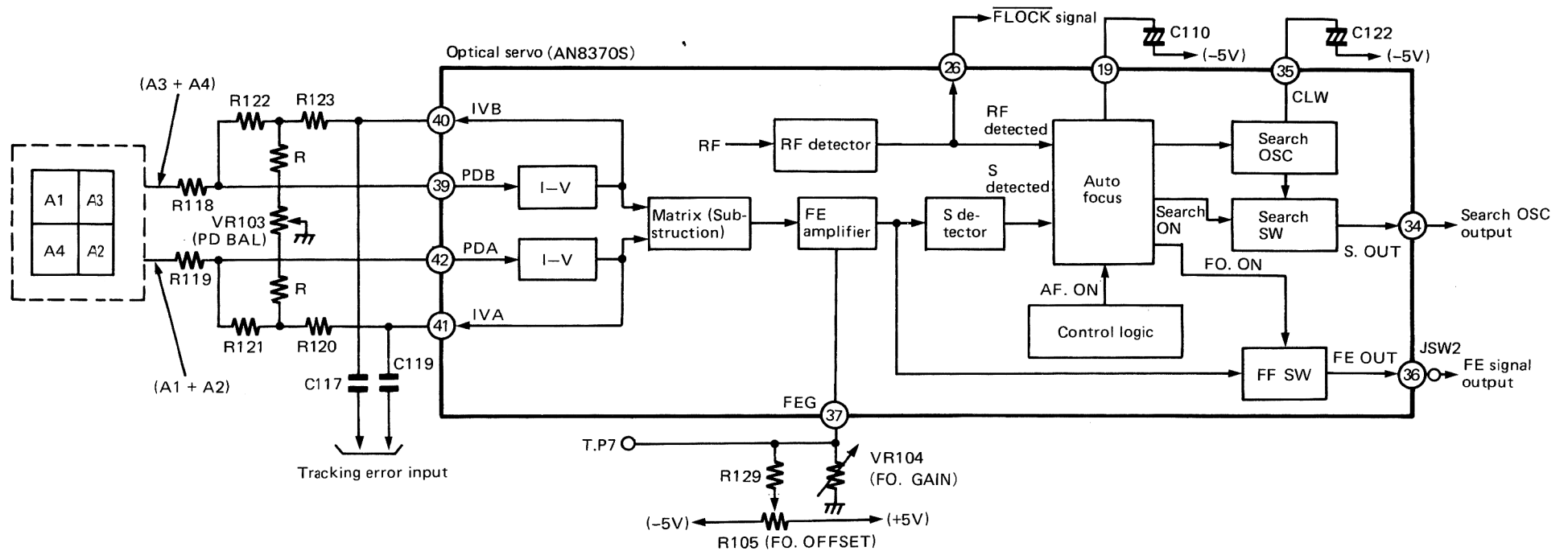


Fig. 6-7 Focus Error Circuit and Auto Focus Circuit

(1) Auto focus circuit

The auto focus uses control logic output AF ON for its operation. It has two inputs, RF detection and S detection, and controls search SW and FE SW.

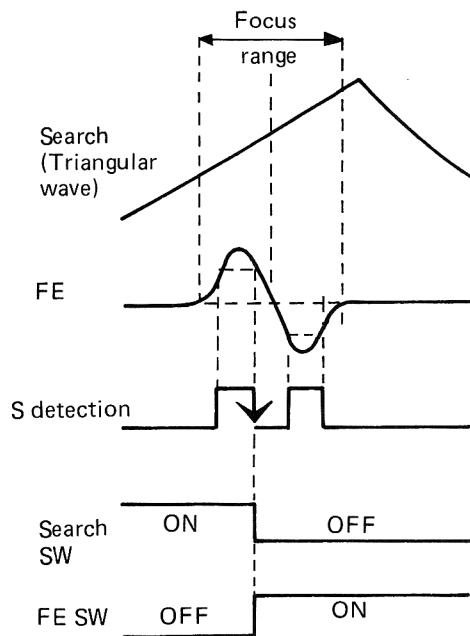


Fig. 6-8 Fo ON Timing

1) Disk detection

AF ON turns search SW on and FE SW off.

In this state, the search signal (triangular wave) drives the focus coil (raising or lowering the lens).

When the distance between the disk and lens is brought close to the focal point by focus search, the FE signal develops an S-curve, which is detected to turn search SW off and FE SW on, thus closing the focus serve loop.

2) Re-focusing

The auto focus discriminates the state of the servo by RF detection while the focus servo loop is closed. When RF detection is no longer possible due to defocusing, search SW is turned to start focus search.

This is input to the microcomputer, and whether the focus is correct can be checked at pin 26 (F LOCK). (Low when the focus is correct.)

(2) Focus error circuit

The diagonal sums $(A1 + A2)$ and $(A3 + A4)$ of the 4-divided photo-detector are input to pins 39 and 42. I-V converts the photo-detector output currents to voltages. An external resistor is used for gain setting. VR101 is a photo-detector balance (PD BAL) adjuster, used to absorb the sensitivity difference of the photo-detector.

Subtraction of the two converted signals, that is, $(A1 + A2) - (A3 + A4)$, produces a focus error signal (FE).

This FE signal is routed via the FE amplifier and FE SW to pin 36 (FE OUT).

VR104 is the focus gain (FO GAIN) control, and VR105 the focus offset (FO OFFSET) control.

6-5 Tracking Error Circuit and Kick, Hold Circuits

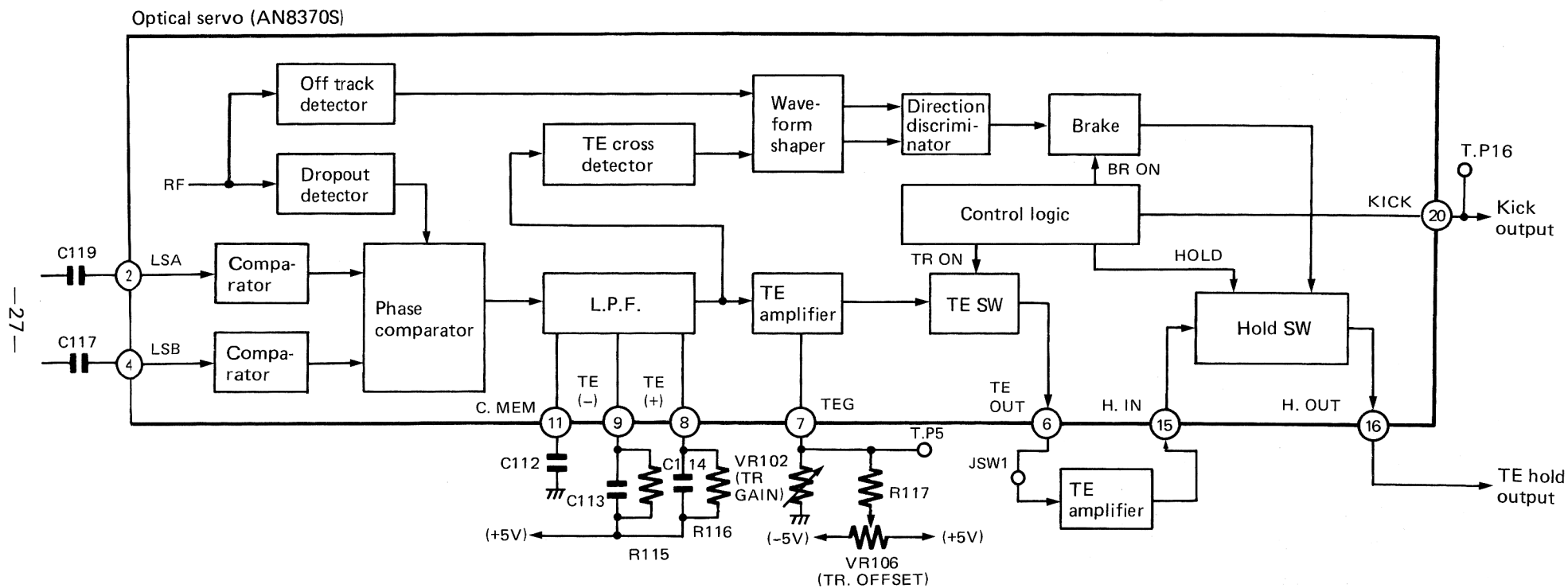


Fig. 6-9 Tracking Error Circuit and Kick, Hold Circuits

(1) Tracking error circuit

The tracking error circuit inputs the I-V converted signals to pins 2 and 4 via C119 and C117.

The comparators shape the waveforms.

In the tracking error process of the single beam type, the phase difference between the two signals is detected and a pulse corresponding to the phase difference is generated as described in 2-3. Deviating direction is judged from the polarity of the pulse. The generated pulse is smoothed by the LPF into an analog tracking error (TE) signal. This is how the phase comparator and LPF operate.

Then, the TE signal is output from pin 6 (TE OUT) via the TE amplifier and TE SW.

VR102 is the tracking gain (TR GAIN) control, and VR106 the tracking offset (TR OFFSET) control.

(2) Kick and hold circuits

Kick means jumping the beam from one track to another during a pause or manual search. Kick timing is set by a control signal from the microcomputer, and a kick signal is output to pin 20. The waveform of each kick signal is shown in Fig. 6-10.

The hold circuit holds the preceding value of a TE signal in a kick section. TE SW is kept off during a kick. Signal waveforms during a pause kick are shown in Fig. 6-11.

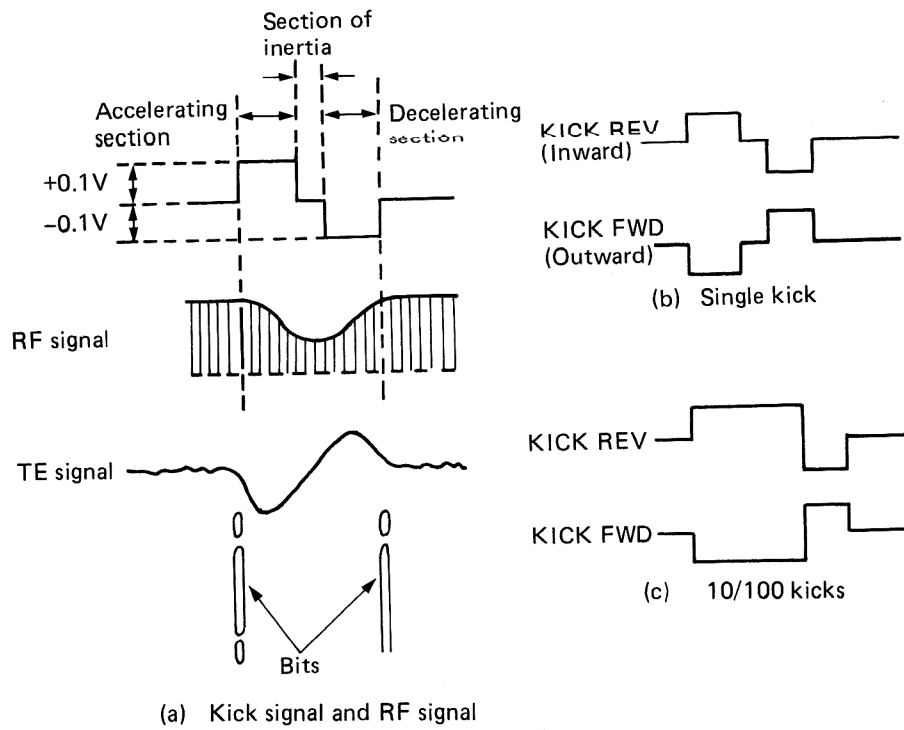


Fig. 6-10 Kick Signal Waveforms

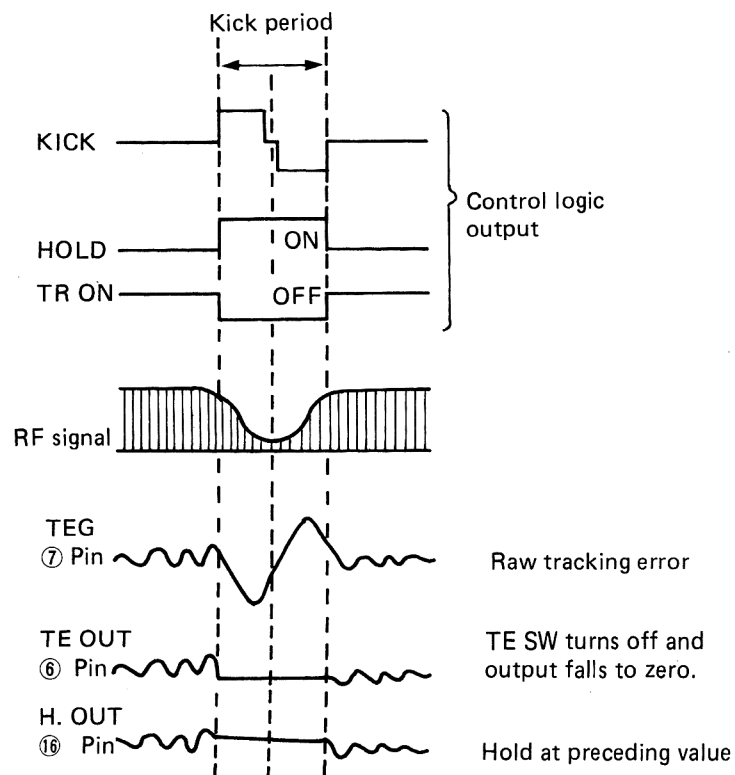


Fig. 6-11 Pause Kick

(3) Brake circuit

The brake circuit brakes the tracking actuator to quickly retract the tracking servo after rapid feed or kick (10/100).

The process is shown in Fig. 6-12.

The brake circuit applies a single limiter (indicated by the dotted line in the figure) to the TE signal to absorb the energy after a kick.

The figure shows the case of KICK REV. In the case of KICK FWD, the brake is applied with the single limiter inverted in polarity. Therefore, the brake circuit requires a kick direction discriminating signal, which is generated from a tracking cross signal and off track signal as shown in Fig. 6-13.

The cross signal has an advanced phase in KICK FWD, and a delayed phase in KICK REV. Braking direction is determined by detecting this phase difference.

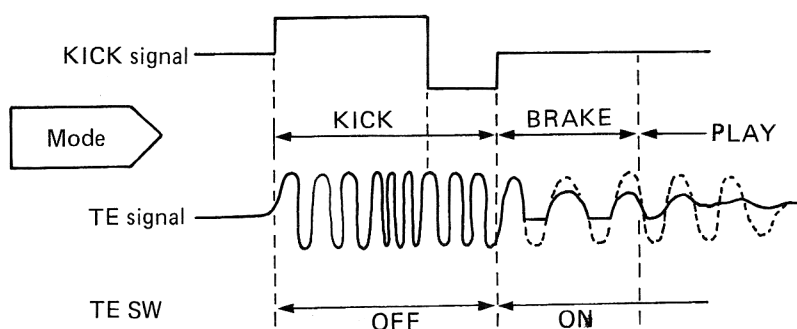


Fig. 6-12 Brake Timing

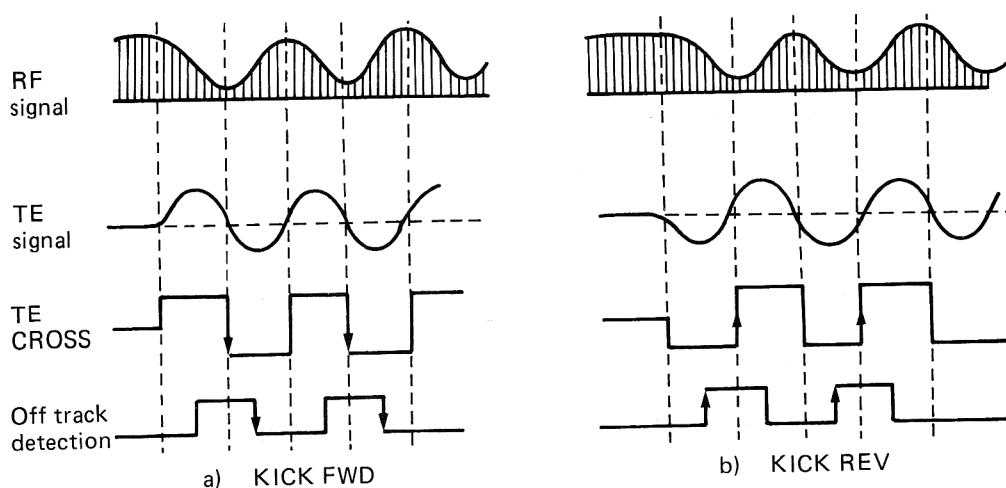


Fig. 6-13 Kick Direction Discrimination

6-6 Focus Tracking Drive Circuit

Fig. 6-14 shows the focus tracking servo drive circuit.

Both servos have the same drive circuit except that the focus servo has a search circuit and that the tracking servo has a kick hold circuit.

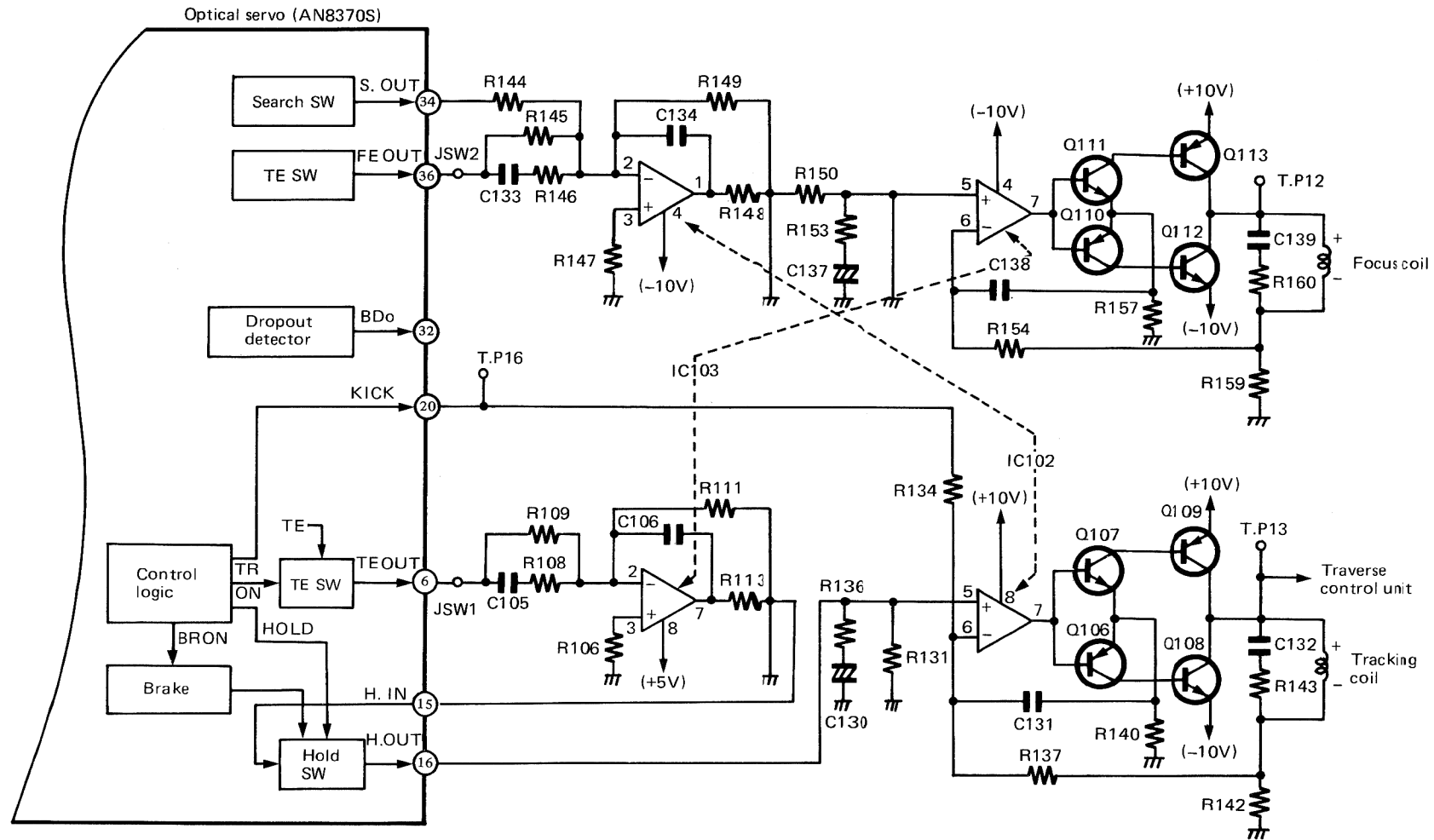


Fig. 6-14 Focus Tracking Drive Circuit

7. TRAVERSE SERVO CIRCUIT OPERATION

The traverse servo detects the inclination of the objective lens, moves the optical pickup, and thus keeps the objective lens within the range of tracking servo.

The traverse servo circuit is shown in Fig. 7-1. It consists of a traverse detector circuit, mode select circuit, and traverse drive circuit.

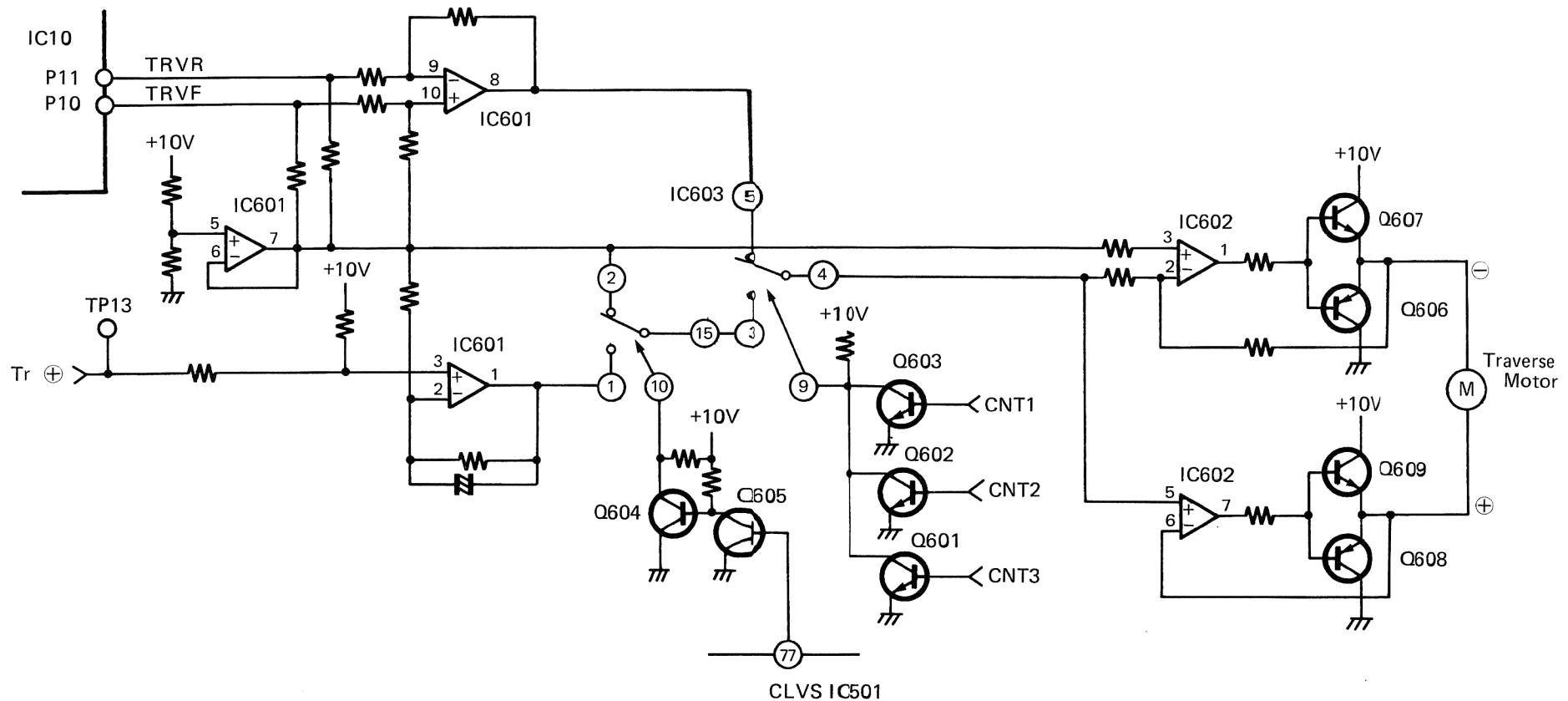


Fig. 7-1 Traverse Servo Circuit

7-1 Traverse Detector Circuit

The tracking error signal corresponds to the eccentric component of the disk as viewed within the range of a single disk revolution. In a little wider range, however, a DC voltage (servo error voltage) is generated as shown in Fig. 7-2. This DC voltage drives the traverse mechanism.

The traverse detector circuit referred to here covers from the tracking coil output to the lowpass filter (pin 1 of IC601) output. Pin 8 of IC601 outputs forced drive signals, that is, forced forward H and reverse L, by micro-computer commands.

Pin 7 of IC601 is set to a reference voltage of +5V.

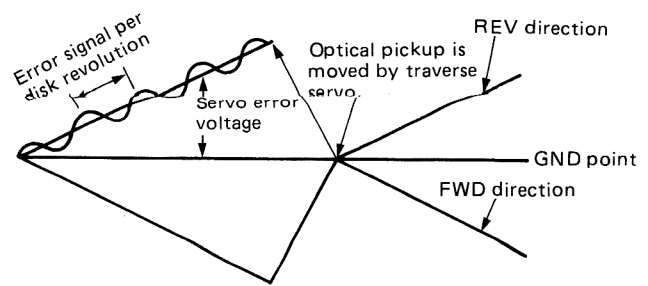


Fig. 7-2 Traverse Servo Voltage

7-2 Mode Select Circuit

The multiplexer (IC603) is used for selecting three inputs in mode selection.

1) Servo error voltage selection

The output of pin 1, IC601 is applied to the drive circuit when pin 10 and pin 9 of IC 603 are high.

2) Brake selection by reference voltage

The output of pin 7, IC601 is applied to the drive circuit when pin 10 of IC603 is low and pin 9 of IC603 is high.

3) Forced forward/reverse selection

The output of pin 8, IC601 is applied to the drive circuit when pin 9 of IC603 is high.

The input to pin 9 of IC603 is controlled by CNT1 to CNT3 of the microcomputer via the NOR circuit. Thus, pin 9 goes high when all of CNT1 to CNT3 are low (during disk play), or goes low when one of CNT1 to CNT3 is high (other than during disk play).

The input to pin 10 of IC603 is controlled by CLVS signal 77 of LSI (IC501). Thus, pin 10 goes high when T11 is OK.

7-3 Traverse Drive Circuit

The traverse drive circuit consisting of IC602, Q606 to Q609 drives the traverse motor using the reference voltage applied to pin 3 of IC602 according to the polarity corresponding to the voltage difference applied to pin 2 and the voltage difference.

8. AN8371S CIRCUIT OPERATION

AN8371S receives an RF signal input from the preceding optical servo IC (AN8370S), and generates signals (EFM signal and PCK signal) from it to be input to the subsequent digital signal processing LSI (MN6617).

An internal block diagram is shown in Fig. 8-1. AN8371S consists of an AGC circuit, data slice circuit, PLL circuit, and dropout detector circuit.

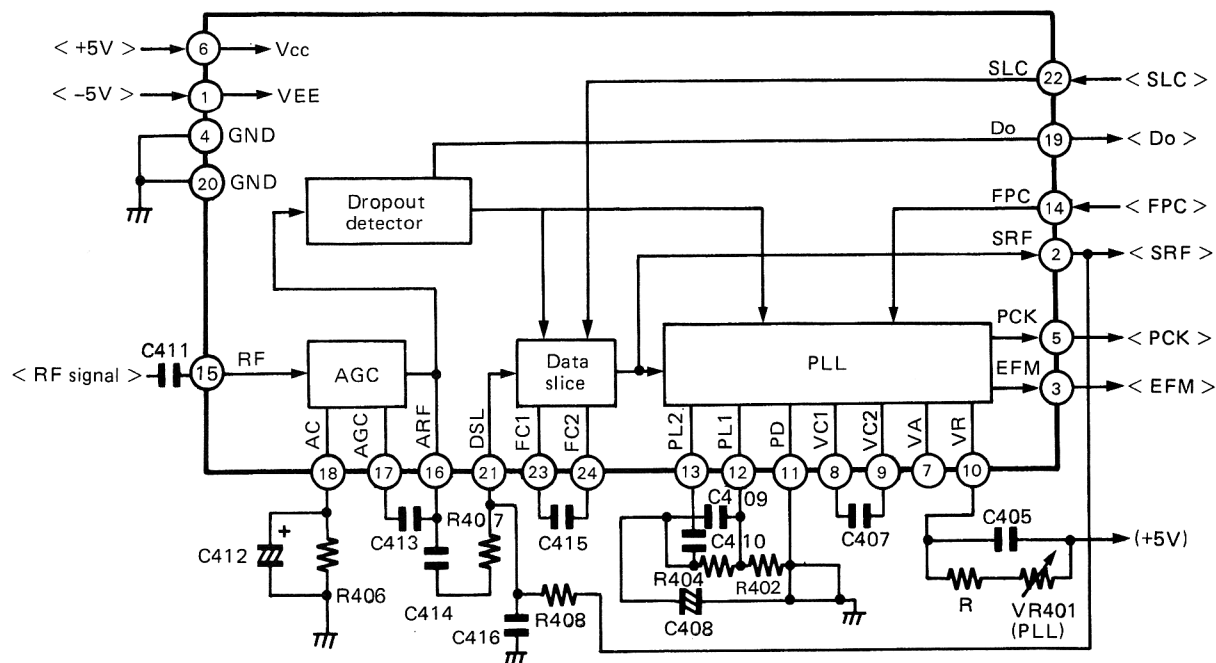


Fig. 8-1 AN8371S Block Diagram

8-1 EFM Signal

RF signal is a continuous signal as shown in Fig. 8-2.

Therefore, it cannot be processed as a digital signal unless it is shaped into a digital signal. This waveform shaping is performed by the data slice circuit, whose output signal is EFM signal.

8-2 PCK Signal

PCK signal is a clock signal synchronized with EFM signal.

EFM signal has some jitter components until it is written into the memory. Therefore, a clock signal synchronized with EFM signal is necessary in order for the digital signal processing circuit to accurately receive data.

PCK signal is generated from an EFM signal input by the PLL circuit. Fig. 8-3 shows a timing chart of EFM signal and PCK signal.

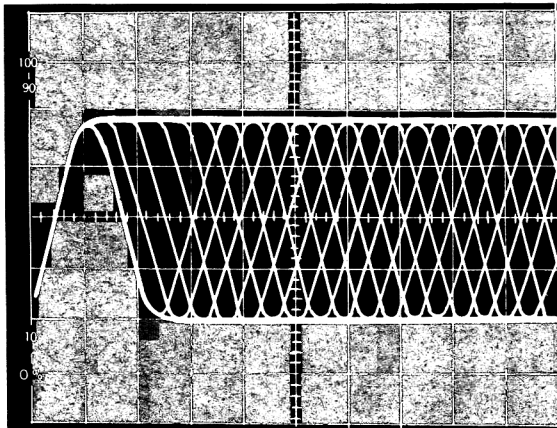


Fig. 8-2 RF Signal

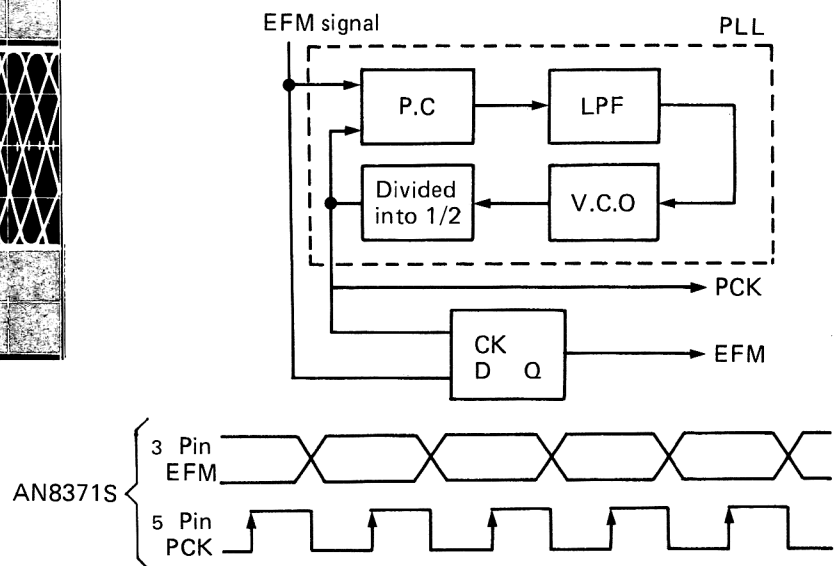


Fig. 8-3 EFM Signal and PCK Signal

8-3 Other Signals

A part from EFM signal and PCK signal, AN8371S outputs SRF signal and D0 signal, and receives input signals of FPC and SLC.

SRF signal is generated by data-slicing RF signal.

D0 signal is a dropout detecting signal for RF signal (high at dropout). Both are input to the CLV control in LSI (MN6617).

FPC signal is a VCO control signal for the PLL circuit that is fed back from the LSI, and SLC signal is a slice level control signal from the microcomputer (MN1554CAC).

Their functions are described in 9-3.

9. MN6617 CIRCUIT OPERATION

Before beginning a circuit operation description, the signal processing of the recording system is explained.

9-1 Recording System Signal Processing

(1) Encoder

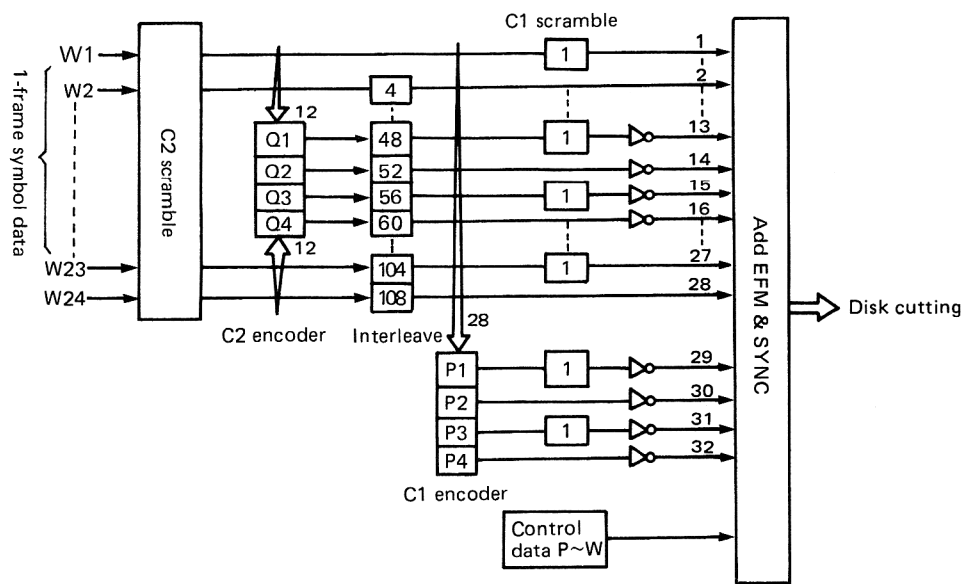


Fig. 9-1 Recording System Signal Processing

Fig. 9-1 shows a block diagram of CD recording system signal processing. A brief description of CD recording system signal processing is given in the following.

- 1) The CD sampling frequency is 44.1 kHz, and quantizing involves A—D conversion to 16 bits per sample. The 16 bits are divided into two parts, upper 8 bits and lower 8 bits.

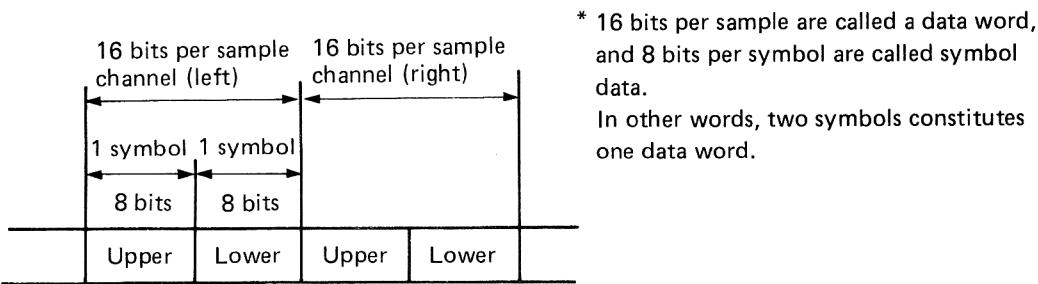


Fig. 9-2 Symbol Data and Other Than Data

- 2) 12 samples of data (six each for the right and left) are taken as a frame for signal processing. In other words, 24-symbol data constitutes a frame. W1 to W24 shown in Fig. 9-1 correspond to it.
- 3) Data W1 to W24 are first scrambled by C2. Scramble may be considered to be a small interleave. In C2 scramble, the data are delayed over two frames alternately in units of sample data.
- 4) After the C2 scramble, the data enter the C2 encoder, which generates 4-symbol parity signals Q1 to Q4 from the 24-symbol data using reed solomon code, and arranges them in the frame center.

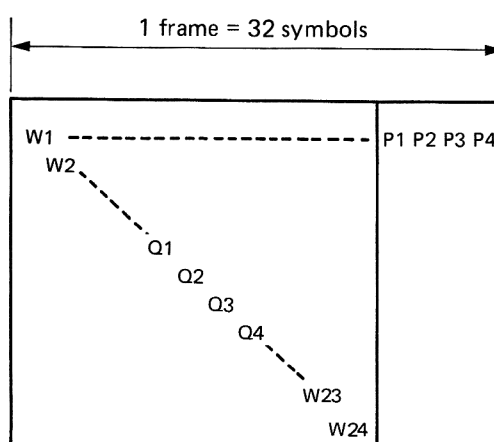
The symbol data count in one frame is 28 in total.

- 5) Next is the interleave.

As shown in the figure, the interleave delays the 28-symbol data, including the Q parity signals, over the range of 4 frames minimum to 108 frames maximum.

- 6) The C1 encoder generates 4-symbol parity signals P1 to P4 from the 28-symbol data, which have been rearranged by interleaving, using reed solomon code, and arranges them at the frame end.

Here, the symbol data count in a frame is 32 in total. (See Fig. 9-3.)



* Symbol parties P1 to P4 are added to the 28 symbols arranged horizontal in a frame, and four symbol parities are added to the 24 symbols arranged diagonally.

Fig. 9-3 CIRC Code Composition

- 7) The C1 scramble delays only the even-numbered symbols of the 32-symbol data in a frame to the next frame.
- 8) Control data P to W are otherwise called subcodes or users bits, containing time information for random access or display and consisting of a single symbol (8 bits).
These signals are arranged at the head of a frame.
Entry of these control signals increases the symbol data count in a frame to 33 in total.
- 9) Finally, the above 1-frame data are EFM-modulated, a frame synchronizing signal is added, and the data are recorded on the disk.

EFM modulation is described in detail in the next paragraph.

All these are intended for improving the correction capacity.

This system is called CIRC (Cross Interleave Reed Solomon Code).

(2) EFM modulation

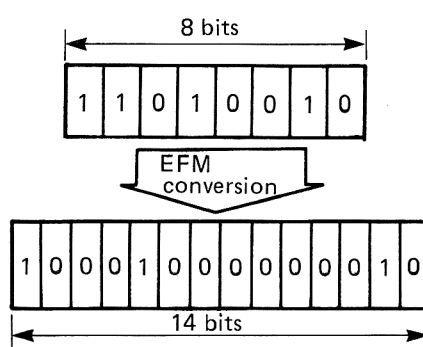


Fig. 9-4 EFM Modulation

EFM is a new modulation system for CD, and stands for Eight to Fourteenth Modulation meaning modulating 8 symbol data bits to 14 channel bits.

EFM is necessary because the data bit arrangement of 1's and 0's cannot be defined if symbol data remains as such.

If 1's or 0's continue so long, data will be a direct current so that self-clocks cannot be generated by the tracking servo (using pits) and reproducing system.

Conversely, if 1's and 0's are frequently reversed, it will be necessary to reduce the laser beam to a very small diameter.

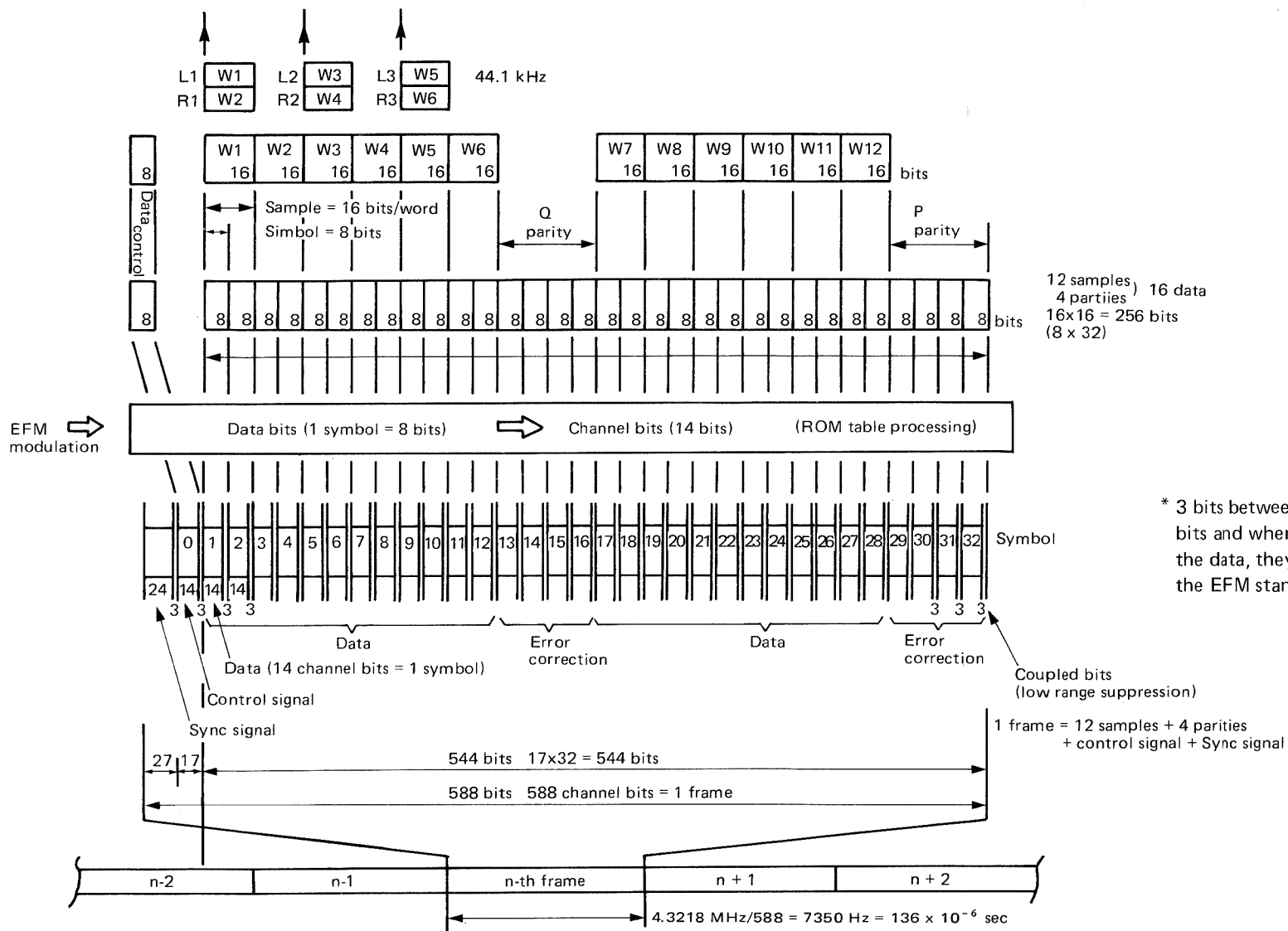
Therefore, signals to be recorded must be such that 1's and 0's are reversed at appropriate intervals and that the DC component is as small as possible.

8 bits has 256 combinations (2^8), while 14 bits has 16,384 combinations (2^{14}).

EFM selects bit patterns, in which 2 to 10 0's are present between two 1's, from among 2^{14} combinations for correspondence to 2^8 data patterns.

Therefore, signals to be recorded have definitions with 3T to 11T (T meaning 1, 1 clock after EFM = 4.32 MHz).

Fig. 9-5 shows the process of up to the recording signal format.



* 3 bits between 14 bits call the coupled bits and when they are observed through the data, they serve as connectors to maintain the EFM standards.

Fig. 9-5 CD Signal Format

(3) Control data

Control data is recorded in one symbol (8 bits) in one frame.

The 8 bits are named P, Q, R, S, T, U, V, and W in this order. Actually, however, only bits P and Q of the 8 bits are used.

Subcode channel No.	P	Q	R	S	T	U	V	W
ϕ	Sync signal							
1	Sync signal							
2								
96								
97								

1/75 second
(13.3 ms)

Fig. 9-6 Control Data

Control P signal

This signal discriminates program signals (music signals) or intervals between melodies on the read-in track, and indicates the program end on the read-out track using a 2 Hz repetitive pulse.

Control Q signal

Details of melodies recorded on the read-in track, program track, and read-out track of the disk are recorded for use in various kinds of display and control.

Read-in track

Music No.	Point (Melody No.)	Melody time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
-----------	--------------------	---------------------------------------	------	--

Program track and read-out track

Control emphasis	ID No.	Music No.	Music No. index	Melody time (Minutes, seconds, frame)	Zero	Overall time (Minutes, seconds, frame)
------------------	--------	-----------	-----------------	---------------------------------------	------	--

Fig. 9-7 Control Q Details

9-2 Reproducing System Signal Processing

(1) Decoder

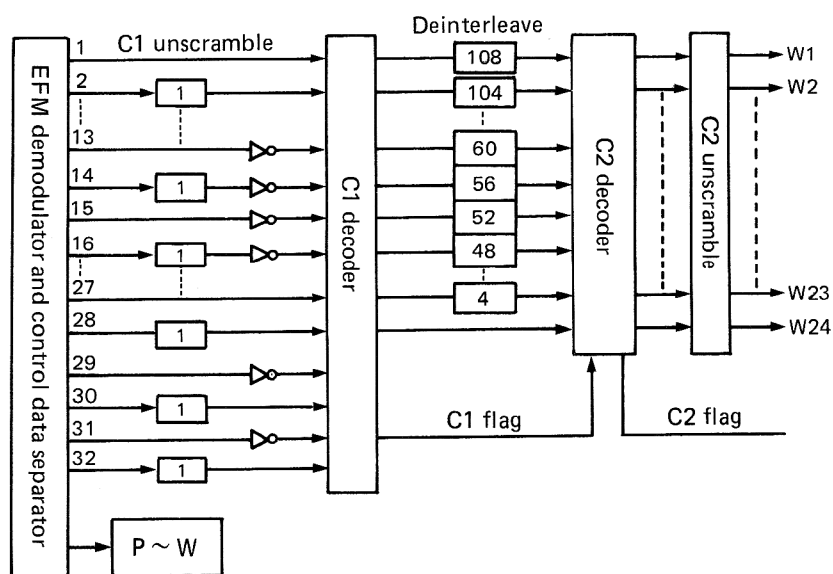


Fig. 9-8 Reproducing System Signal Processing

A reproducing system signal processing block diagram is shown in Fig. 9-8.

The flow of reproducing system signal processing is just opposite to the flow of recording system signal processing. Here, decoders are error detection and correction are used in place of the C1 and C2 encoders that are used for parity generation.

Both the C1 and C2 decoders are capable of error detection and correction. If data is incorrecable in the C1 decoder, C1 flag is attached to that data, and after deinterleaving, the data is sent to the C2 decoder for accurate correction.

If data is incorrecable in the C2 decoder, an average value is obtained from the preceding and next data after C2 unscramble, followed by interpolation processing.

(2) LSI signal processing

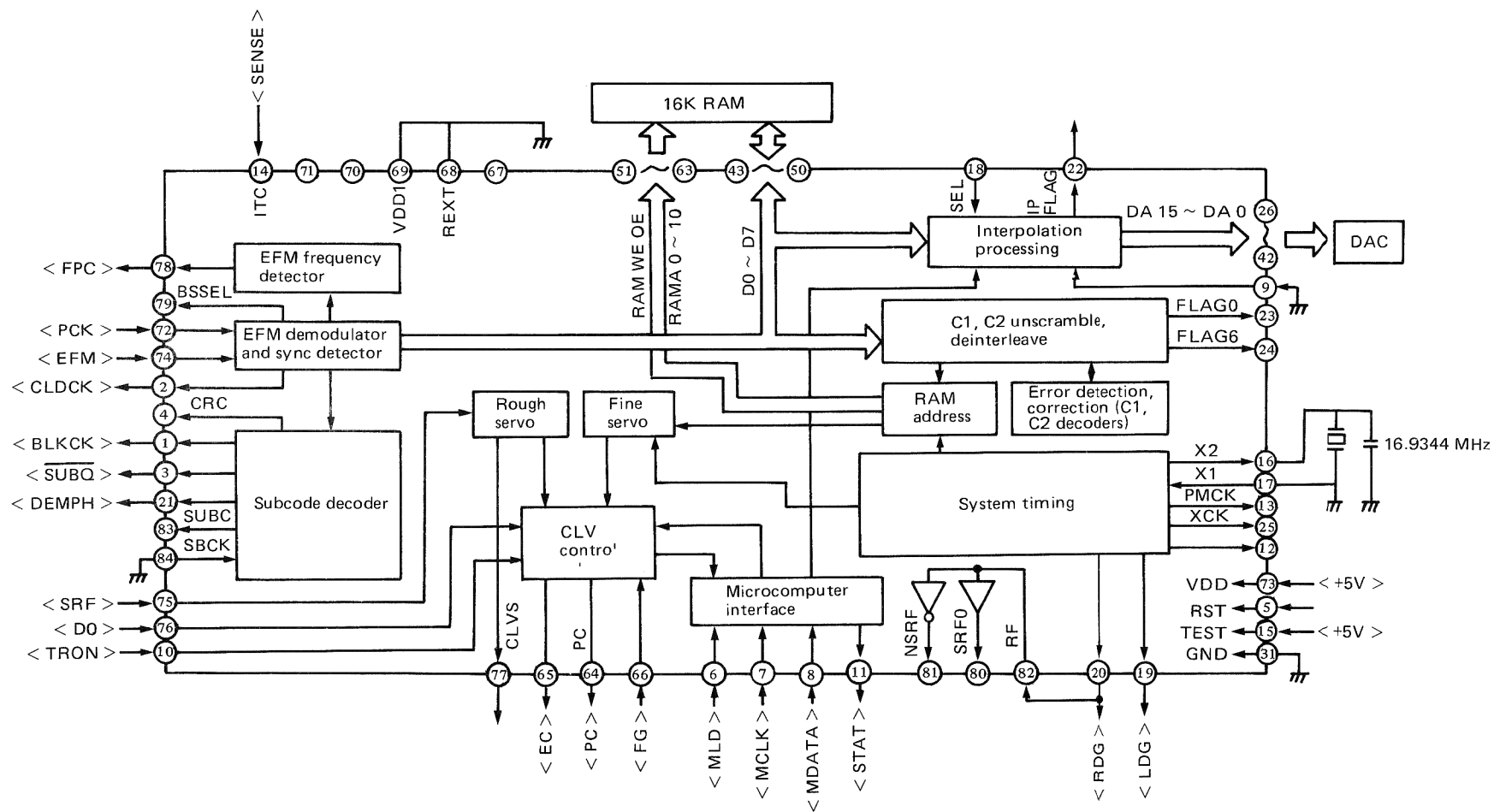


Fig. 9-9 LSI Block Diagram

Fig. 9-9 shows an LSI block diagram.

1) EFM demodulation, subcode demodulation

In EFM demodulation, an extraction clock PCK is used to latch EFM signal every 14 bits to return it to the original 8-bit data.

The frame clock of 7.35 kHz in EFM signal is output from pin ② (CLDCK) by synchronized detection.

In subcode demodulation, the synchronized signal in the 14 bits preceding EFM demodulation is detected, and subcode Q data is output to pin ③ (SUBQ) and emphasis/de-emphasis signal to pin ④ (DEMPHA) (high in case of emphasis signal).

Also, a subcode Q block clock of 75 Hz is output from pin 1 (BLKCK).

2) C1, C2 unscramble, deinterleave

Time base processing (data rearrangement) is used during recording to deal with various burst errors and random errors that occur through the recording and reproducing systems.

The data thus rearranged is returned to the original state by an operation opposite to that of recording, using an external RAM (16K).

(a) Pins 43 to 50 (D0 to D7) : Data signals to RAM

(b) Pin ⑤① (RAM WE) : Command signal to write in RAM

(c) Pin ⑤② (RAM OE) : Command signal to read from RAM

(d) Pins ⑤③ to ⑤⑩ (RAMA0 to RAMA10) : RAM address signals

3) Error detection, correction (C1, C2 decoders)

The C1 decoder detects errors in data subsequent to C1 unscramble and corrects them, using parity P signal. The C2 decoder detects errors in data subsequent to deinterleave and corrects them, using parity Q signal.

4) System timing

Xtal signal (16.9344 MHz) that is connected to pin ①⑥ (X2) and pin ①⑦ (X1) is divided to generate a clock for signal processing in the LSI and a timing clock signal for external circuits.

(a) Pin ①② (SMCK)

Signal output of 4.23 MHz

(b) Pin 19 (L DG), pin 20 (R DG)

Signal output of 44.1 kHz used as a deglitch signal for audio circuit

5) CLV servo

The CLV servo circuit is described in 9-3 (3).

9-3 LSI and Associated Operations of Peripheral Circuits

(1) Operations associated with AN8371S

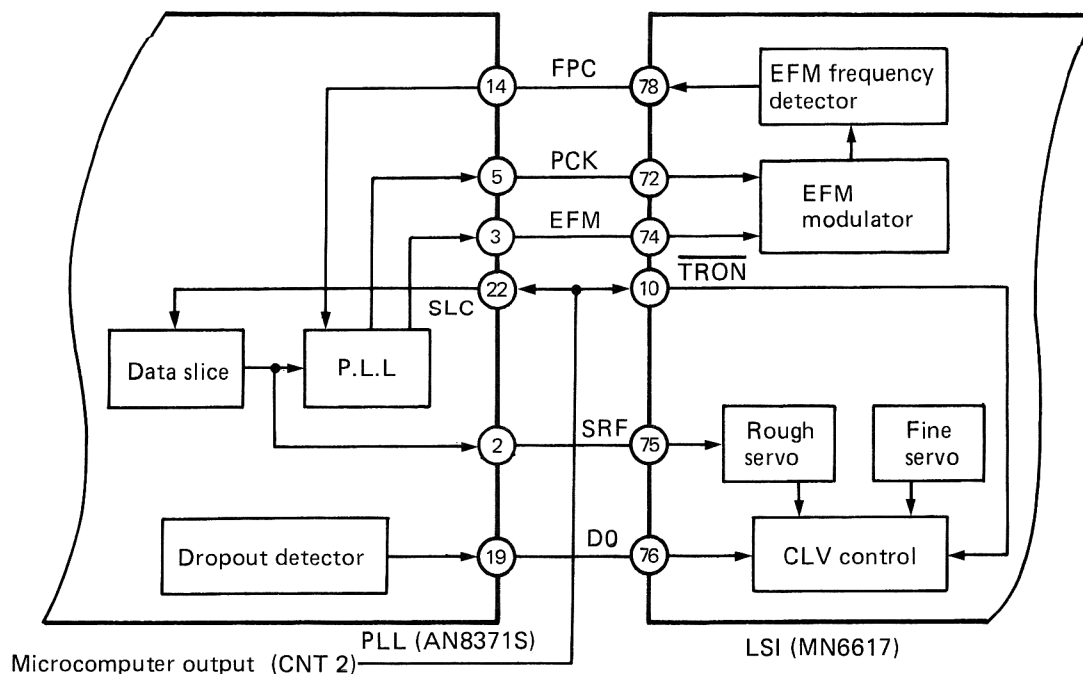


Fig. 9-10 AN8371S and LSI Connection Diagram

Fig. 9-10 shows an LSI (MN6617) and PLL IC (AN8371S) connection diagram.

There are four inputs and one output as viewed from the LSI, and the output (CNT2) from the microcomputer is input to both ICs.

1) PCK, EFM, FPC signals

PCK and EFM signals are as described in Section 6.

FPC signal results from measuring the pulse width of EFM signal (3T to 11T) with PCK signal (1T).

If EFM signal is a signal of 2T or less, a feedback operation takes place to raise the VCO frequency of the PLL. If EFM signal is 12T or over, the VCO frequency is lowered.

During play (PLL has completed operating), pin 78 of the LSI is constant at 2.5V. If the PLL is still operating, a control pulse of $2.5V \pm 2.5V$ is output.

2) SRF, D0 signals

SRF signal is generated by data-slicing RF signal (data slice output of AN8371S), and is used as an input signal for the CLV rough servo (11T servo) in the LSI. The CLV servo is described later.

D0 signal is a dropout detection signal (dropout detection output of AN8371S) in AN8371S, and is used as a control signal for CLV control in the LSI. The control method is such that, when a dropout is detected, the 11T servo is turned off to allow free rotation. Fig. 9-12 shows SRF signal and D0 signal.

3) CNT2 (SLC or TR ON) signal

CNT2 signal is one of the 4-bit control logic signals of the optical servo IC (AN8370S). This signal remains high until the tracking servo is in the OFF mode (See Table 6-1).

This is used to control both ICs as follows.

- (a) The CLV control in the LSI is kept under speed control even during rapid feed by changing the rough servo to the traverse tracking servo mode, and when the tracking servo turns on, the CLV servo is immediately set ready for operation.
- (b) The data slice circuit in AN8371S is so controlled that the slice level is forcibly raised to permit accurate generation of SRF signal as an input signal to the rough servo.

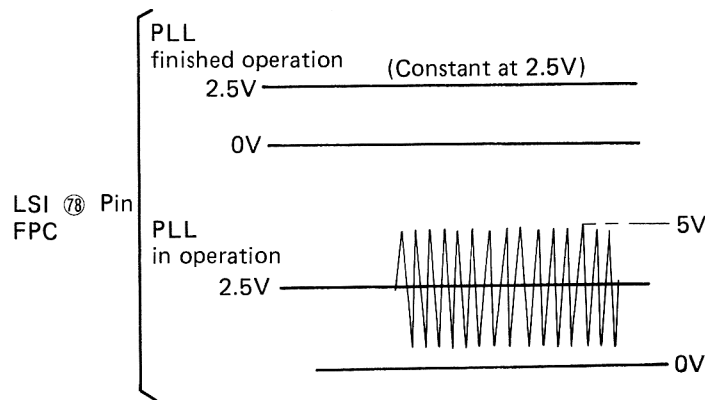
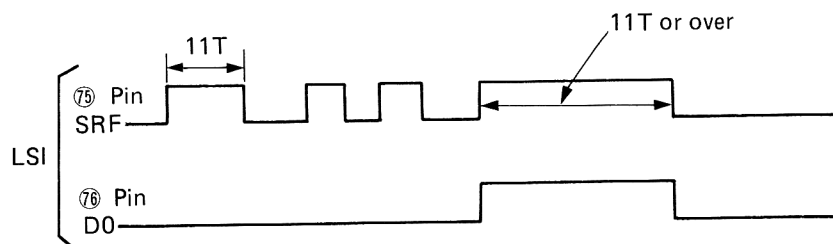


Fig. 9-11 FPC Signal



- SRF signal has a period of 11T or longer when a dropout occurs, which is interpreted as slow rotation, causing the spindle motor to run uncontrolled. Thus, D0 signal makes rotation free.

Fig. 9-12 SRF Signal and C0 Signal

(2) Operations associated with microcomputer

Fig. 9-13 shows an LSI (MN6617) and microcomputer (MN1554CAF) connection diagram. There are four inputs and five outputs as viewed from the LSI.

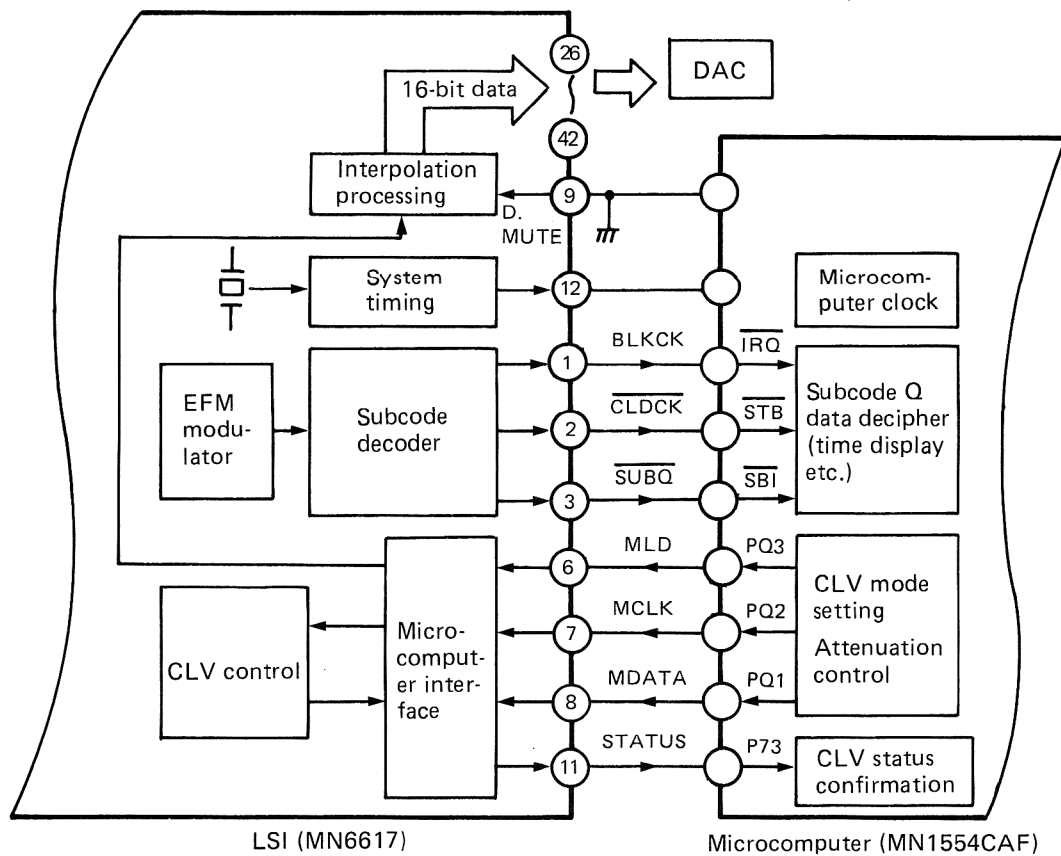


Fig. 9-13 LSI and Microcomputer Connection Diagram

1) BLKCK, CLDCK, SUB Q signals

These three signals transfer the subcode Q data, which has been demodulated from the EFM data in the LSI, to the microcomputer. The microcomputer deciphers the subcode Q data for time base control and display.

BLKCK is a clock signal of one block (98 frames) extracted from the EFM data; CLDCK is a clock signal of one frame similarly extracted, and SUB Q becomes subcode Q data.

Data transfer is serial.

Fig. 9-14 shows a timing chart of the three signals.

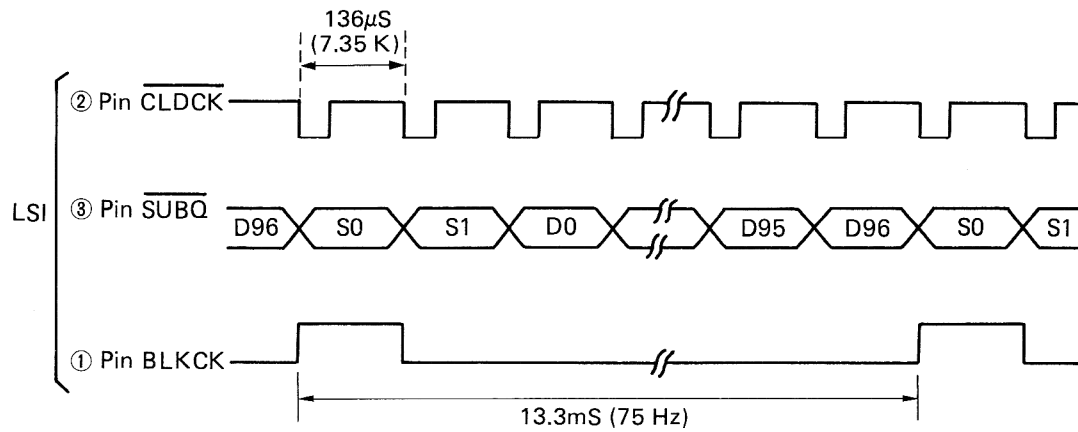


Fig. 9-14 Subcode Q data Transfer (LSI to Microcomputer)

2) MLD, MCLK, MDATA signals

These three signals transfer data from the microcomputer to the LSI.

- (a) MLD : Data loading signal
- (b) MCLK : Data transfer clock signal
- (c) MDATA : Transfer data

Data is transferred to set the modes of the CLV control and interpolation circuit in the LSI.

Fig. 9-15 shows MDATA details.

No.	8 bits								Remarks
	A4	A3	A2	A1	D4	D3	D2	D1	
①	0	0	0	1	1 (with FG)	1 (TT ON)	0	0	CLV mode
							(STOP)		
					0 (without FG)	0 (TT OFF)	0	1	
							(ACC)		
②	0	0	1	0	—	—	1	0	Interpolation circuit
							(BRAKE)		
							1	1	
							(PLAY)		
③	0	1	1	1	—	—	1	1	Status output switching
							(ATT-12dB)	(MUTE ON)	
							0	0	
							(ATT OFF)	(MUTE OFF)	
④	0	1	1	1	—	—	0	0	Status output switching
							(CRC)		
							1	0	
							(CLVOK)		
⑤	0	1	1	1	—	—	1	1	Status output switching
							(TT STOP)		

Fig. 9-15 MDATA Details

MDATA is composed of 8 bits, and is serially transferred.

Of the 8 bits, the upper 4 bits are for partitioning transfer data and the lower 4 bits are the data.

Fig. 9-16 shows a timing chart of the three signals.

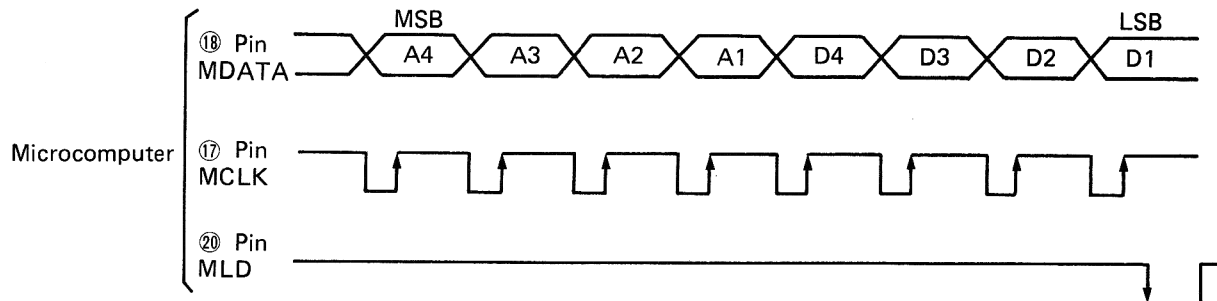


Fig. 9-16 MDATA Transfer (Microcomputer to LSI)

STATUS signal is data to be transferred from the LSI to the microcomputer, and its contents are confirmed as necessary by the microcomputer according to the output switching mentioned in Item 3), Fig. 9-15.

(3) CLV servo

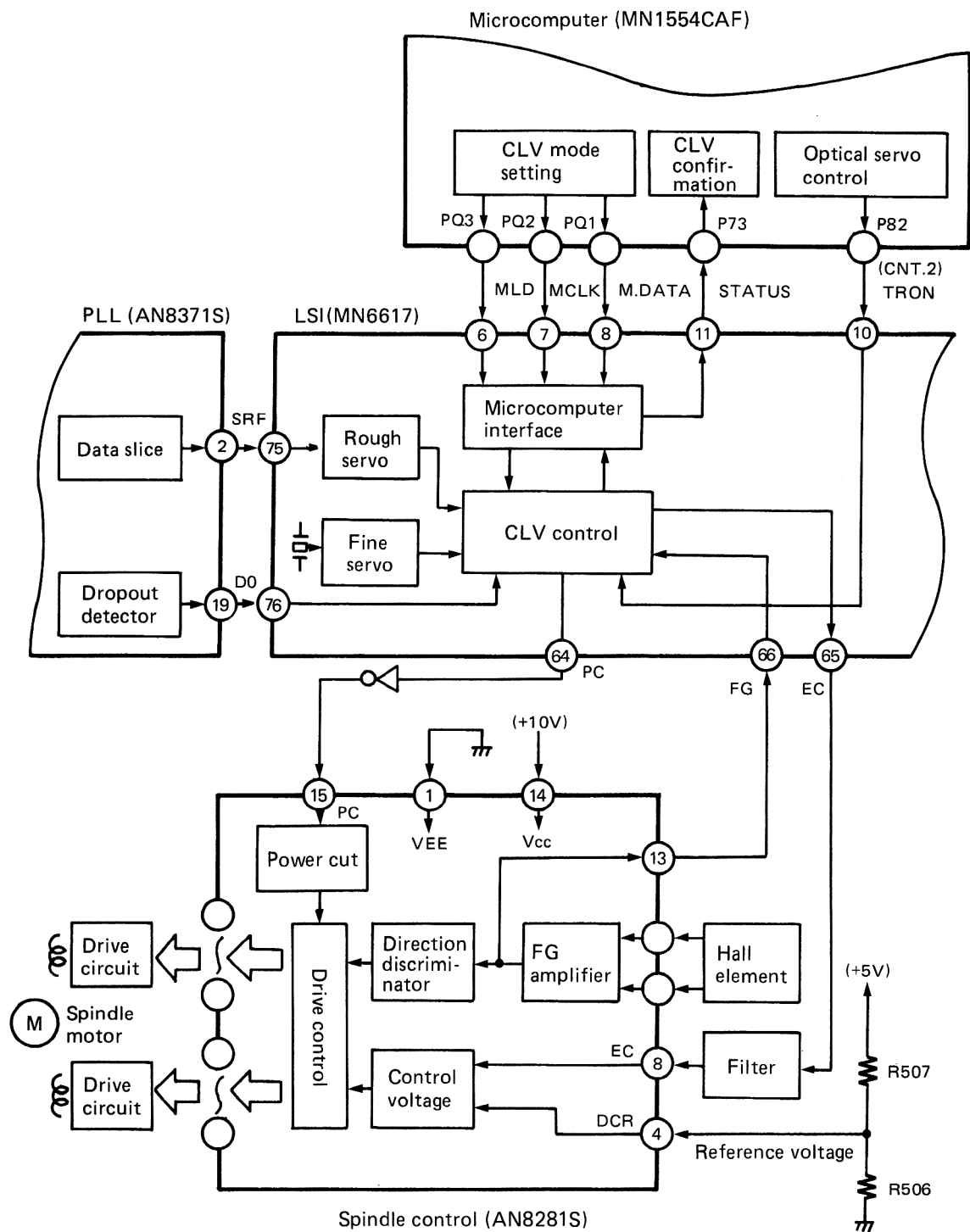


Fig. 9-17 CLV Servo Circuit

The CLV servo is a rotary servo which keeps disk linear velocity constant.

The servo is divided into three parts, that is 11T servo, CLV servo, and traverse tracking servo. The traverse tracking servo and 11T servo are rough servos, and the CLV servo is a fine servo. The CLV control which is the nucleus of the circuit is in the LSI. The CLV servo circuit consists of micro-computers AN8371S and AN8281S.

Figure 9-17 shows the CLV servo circuit.

1) AN8281S circuit operation

AN8281S is the spindle motor drive IC. Control voltage EC is applied from the CLV control to pin 8, and the reference voltage divided by R506 and R507 (DCR) is applied to pin 4.

FG and rotating direction are detected by the Hall element, and FG signal is fed back to the CLV control circuit.

The ON/OFF function of this IC (pin 15) controls starts and stops.

Fig. 9-18 shows EC signal versus the motor.

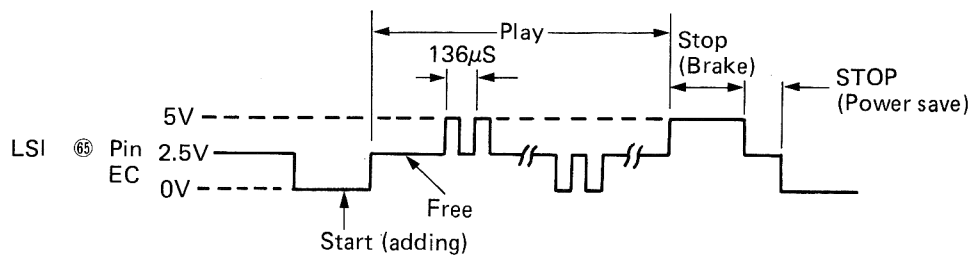


Fig. 9-18 CLV Control Output

2) 11T servo

The 11T servo brings disk rotation close to constant linear velocity.

This requires a signal extracted from the disk, that is, SRF signal. (SRF signal is generated by data-slicing RF signal.)

The 11T servo operates in such a way that the longest period of SRF signal will be 11 times the period of the reference clock 1T (4.3 MHz).

When the 11T servo operates, the spindle motor comes close to the rotating speed of the fine servo.

3) CLV servo

The CLV servo is started by the 11T servo when digital signals are ready for processing in the LSI.

For digital signal processing, data is once written in an external RAM and read out again. A RAM address generator circuit is provided in the LSI for this purpose. Because a crystal clock is used for reading data and a clock synchronized with reproducing signals (corresponding to PCK) is used for writing data, however, address locations may be unsteady relative to RAM capacity. The fine servo detects address location signals and operates to keep the read/write pace constant.

4) Traverse tracking servo

The traverse tracking servo tracks the traverse position to keep linear velocity constant when the traverse mechanism is in rapid feed.

This servo operates in such a way that two consecutive shortest period of SRF will be 6 times the period of the reference clock 1T (4.3 MHz).

TRON is used for switching the traverse tracking servo to the 11T servo or vice versa. That is, the traverse tracking servo is selected when TRON is high; the 11T servo is selected when TRON is low.

10- AUDIO CIRCUIT

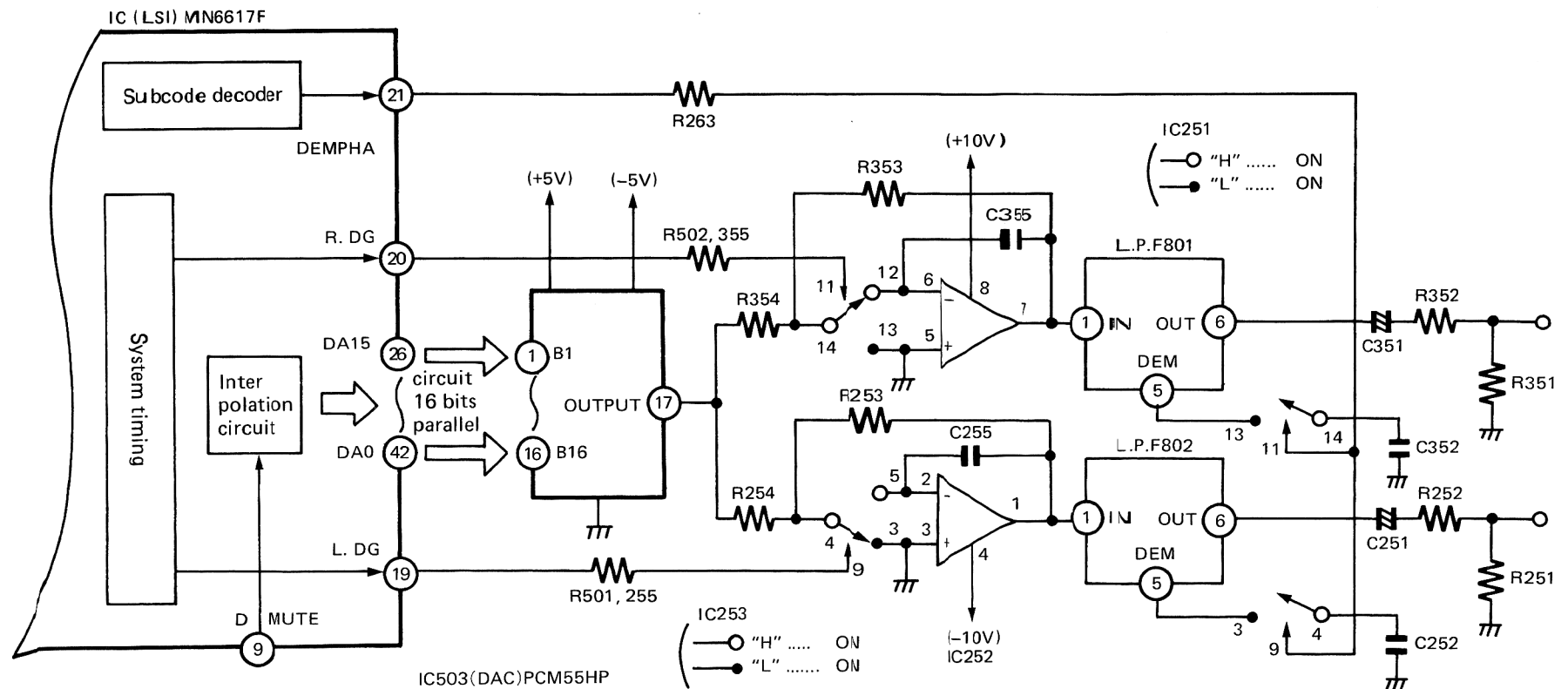
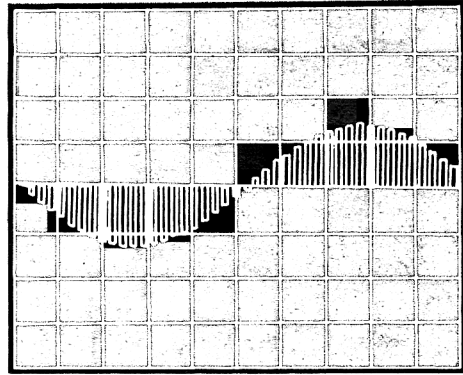


Fig. 10-1 Audio Circuit

10-1 D—A Converter

IC801 is a D—A converter, which converts digital signals to analog signals. Digital signals are input to pins 1 to 16 of IC801, 16 bits parallel. Pin 1 (B1) is MSB, and pin 16 (B16) is LSB. PAM signal is output from pin 17 (OUT).



(Lch = 1 kHz, Rch = 0)

Fig. 10-2 PAM Signal

10-2 Deglitch Circuit

The deglitch circuit consists of IC253 (analog switch), IC252 (operational amplifier), C255 and C355 (for hold).

Pin 19 (L DG) of IC601 (LSI) is the deglitch signal output for the left channel, and pin 20 (R DG) of IC601 is the deglitch signal output for the right channel.

Fig. 10-3 shows a demodulation timing chart of the deglitch circuit.

When L DG signal is low, the left channel circuit takes signals in, while the right channel circuit outputs the signals that have been taken in (and held by C255).

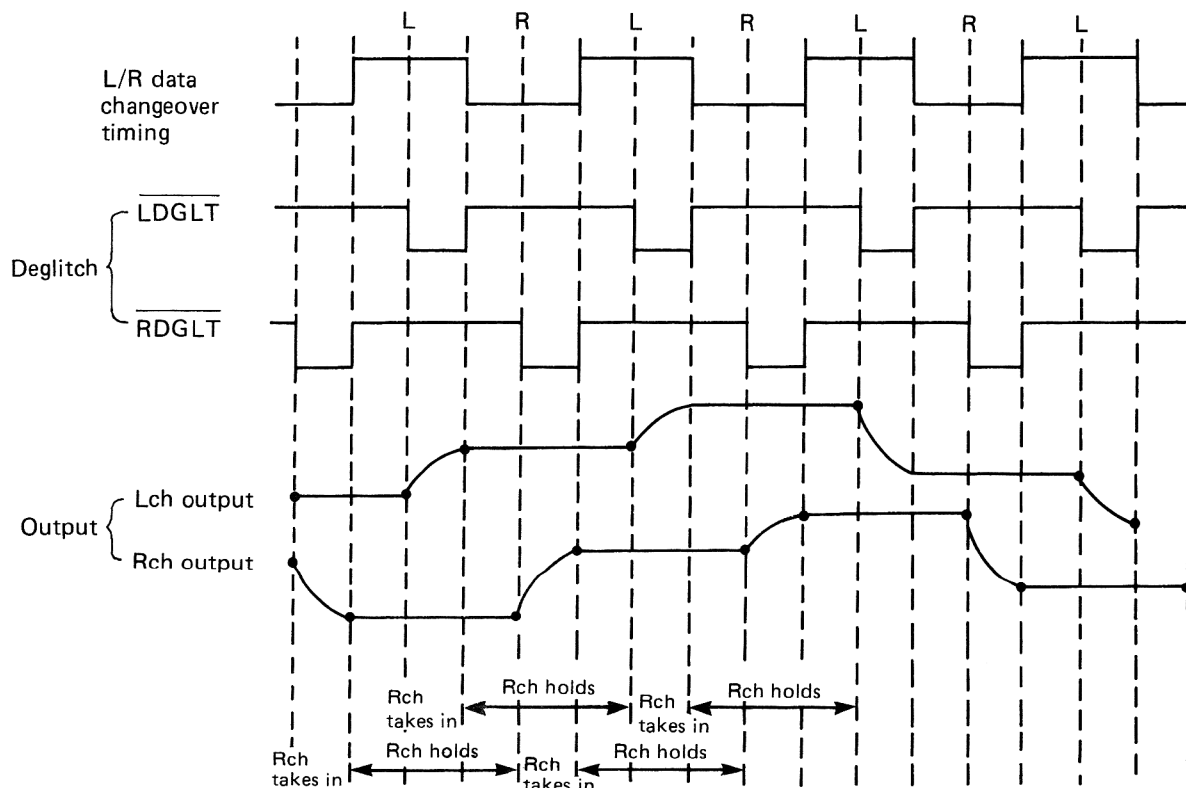


Fig. 10-3 Delitch Timing

10-3 LPF and De-emphasis Circuit

The lowpass filter (LPF) restores the signals containing harmonics (staircase waves) received from the deglitch circuit to complete audio signals.

The LPF also has a de-emphasis function.

It reads emphasis or no emphasis from subcode (Q channel), and outputs a control signal from pin 21 (DEMPHA) of IC601 (LSI). De-emphasis is applied when the output signal is high.

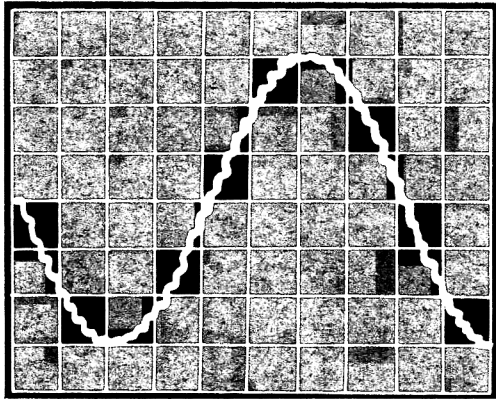


Fig. 10-4 LPF Input

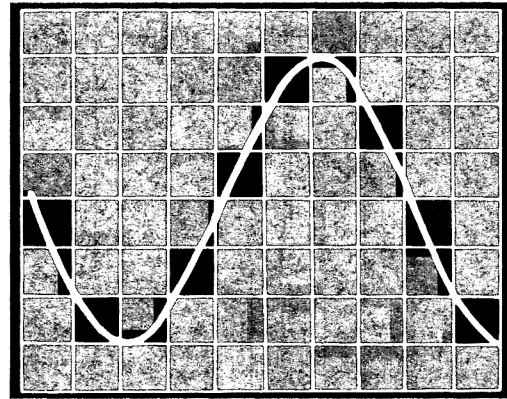


Fig. 10-5 LPF Output

11. MUTING CIRCUIT and SYSTEM CONTROL

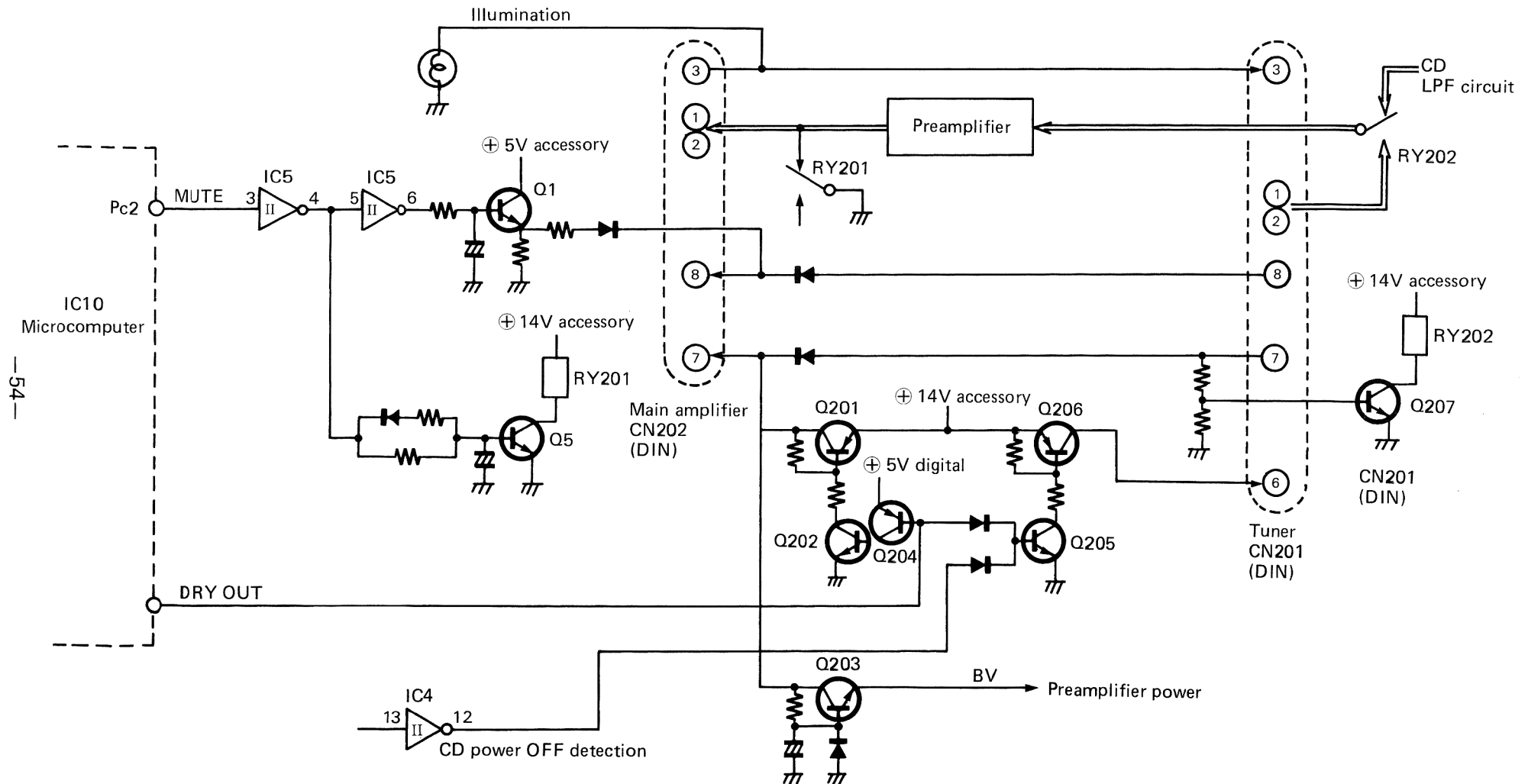


Fig. 11-1 Muting and System Control Circuit

TROUBLESHOOTING



TROUBLESHOOTING

If trouble has occurred, check the symptoms according to the flowchart shown in Section 1.

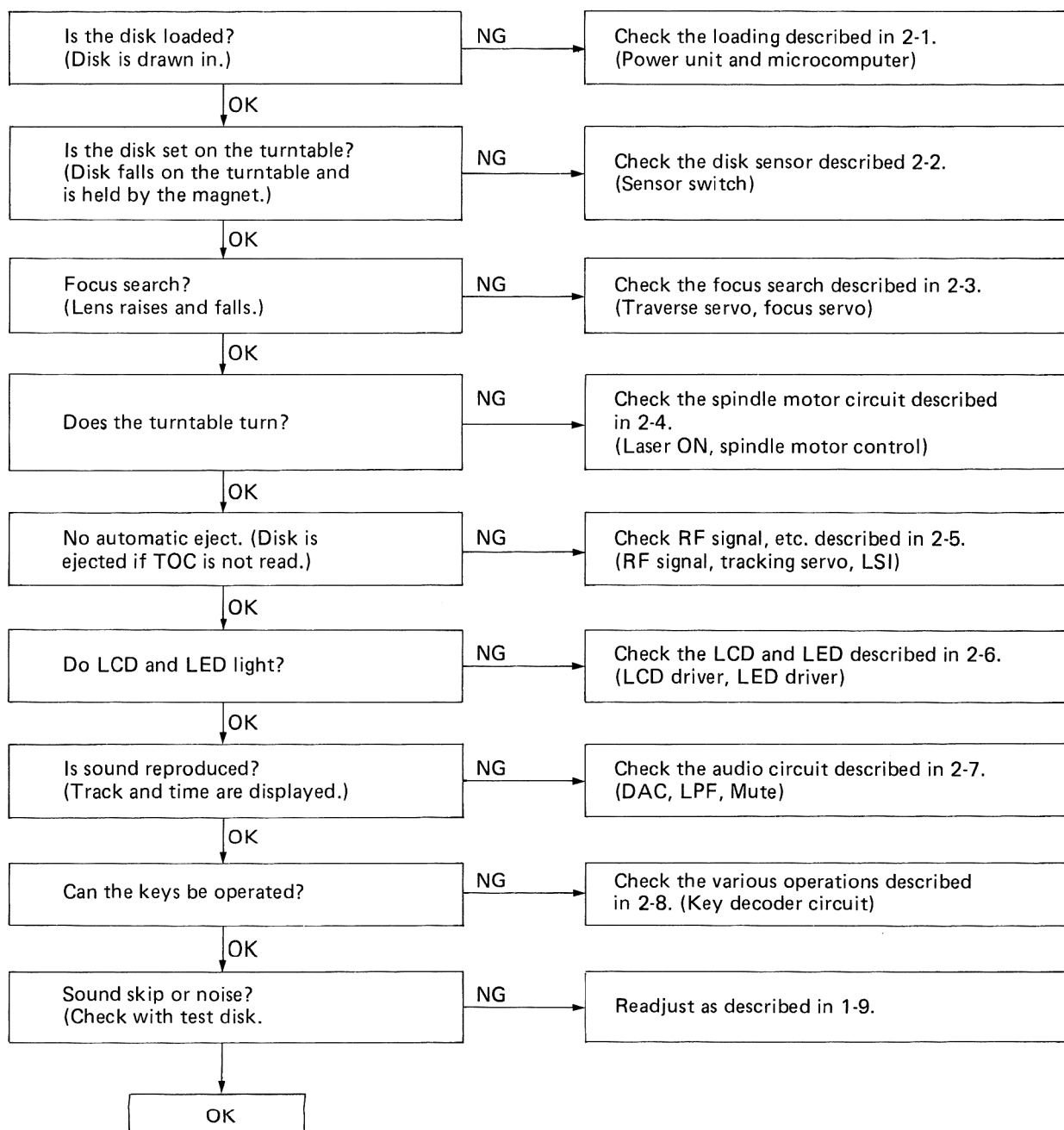
Note: Do not skip steps in the flow of symptom confirmation.

Keep power on and, if the set is disassembled for troubleshooting, make sure that the symptoms are the same as before disassembling the set.

Note: When checking the set with the top cover open, cover the top of the deck to prevent the optical sensor from wrong operation.

1. SYMPTOM CHECK

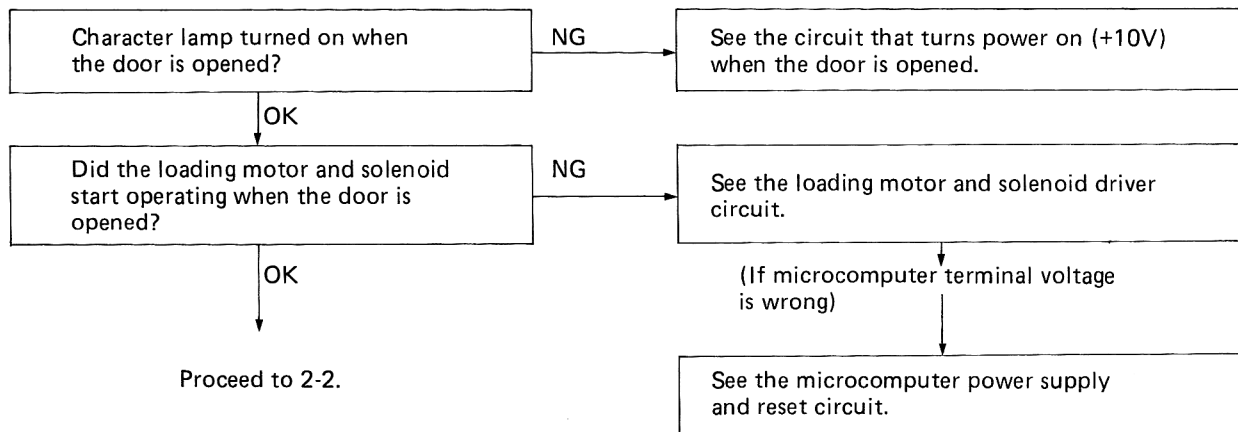
(Insert a disk.)



2. CHECKING PROCEDURE

2-1 Loading Check (Power unit and microcomputer)

For a disk to be drawn in, the door sensor must detect it to turn power on and the loading motor and solenoid must operate under microcomputer control.



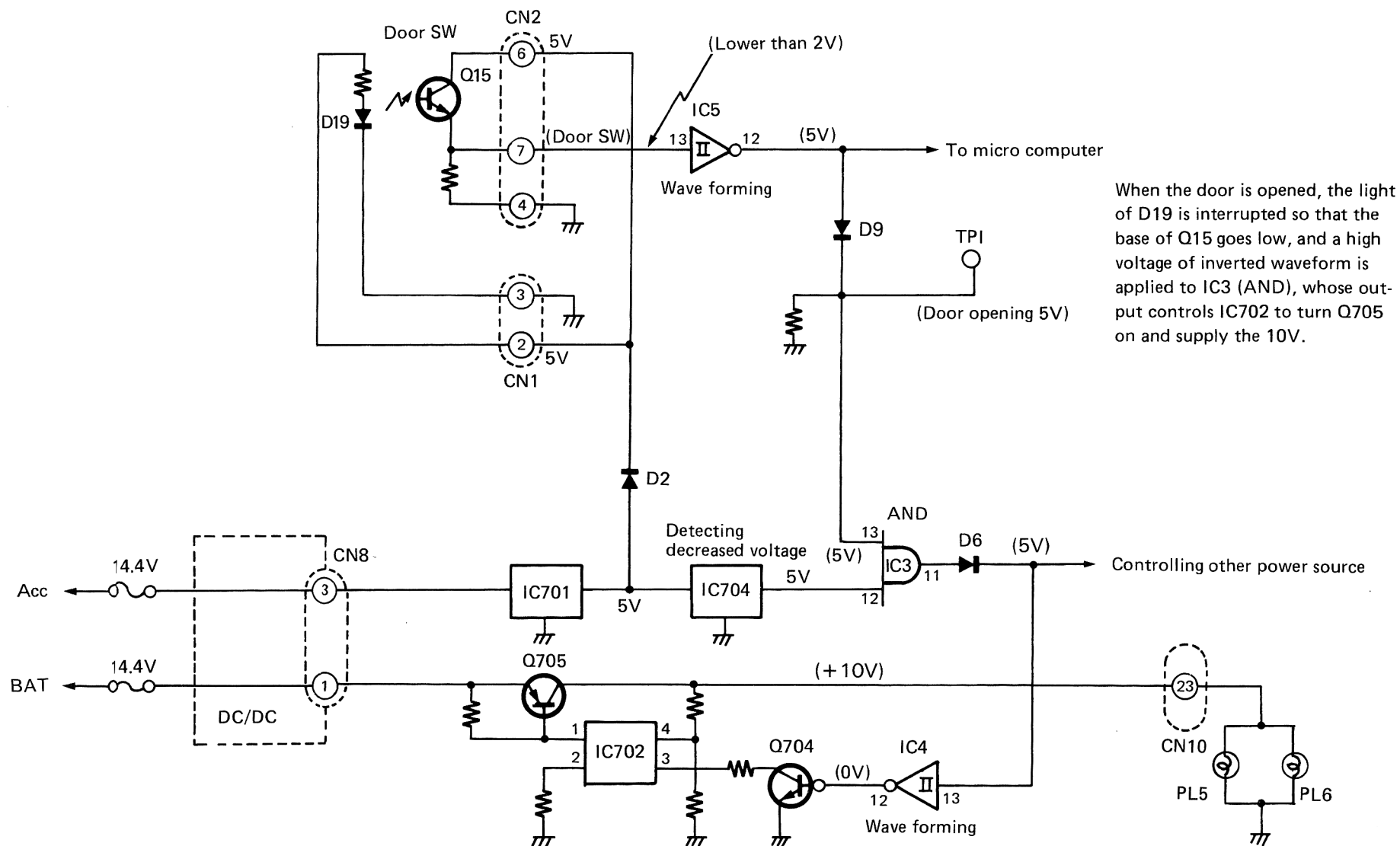


Fig. 2-1-1 Circuit That Turns Power ON (+10V) When Door is Opened
(Circuit voltages apply where door is open)

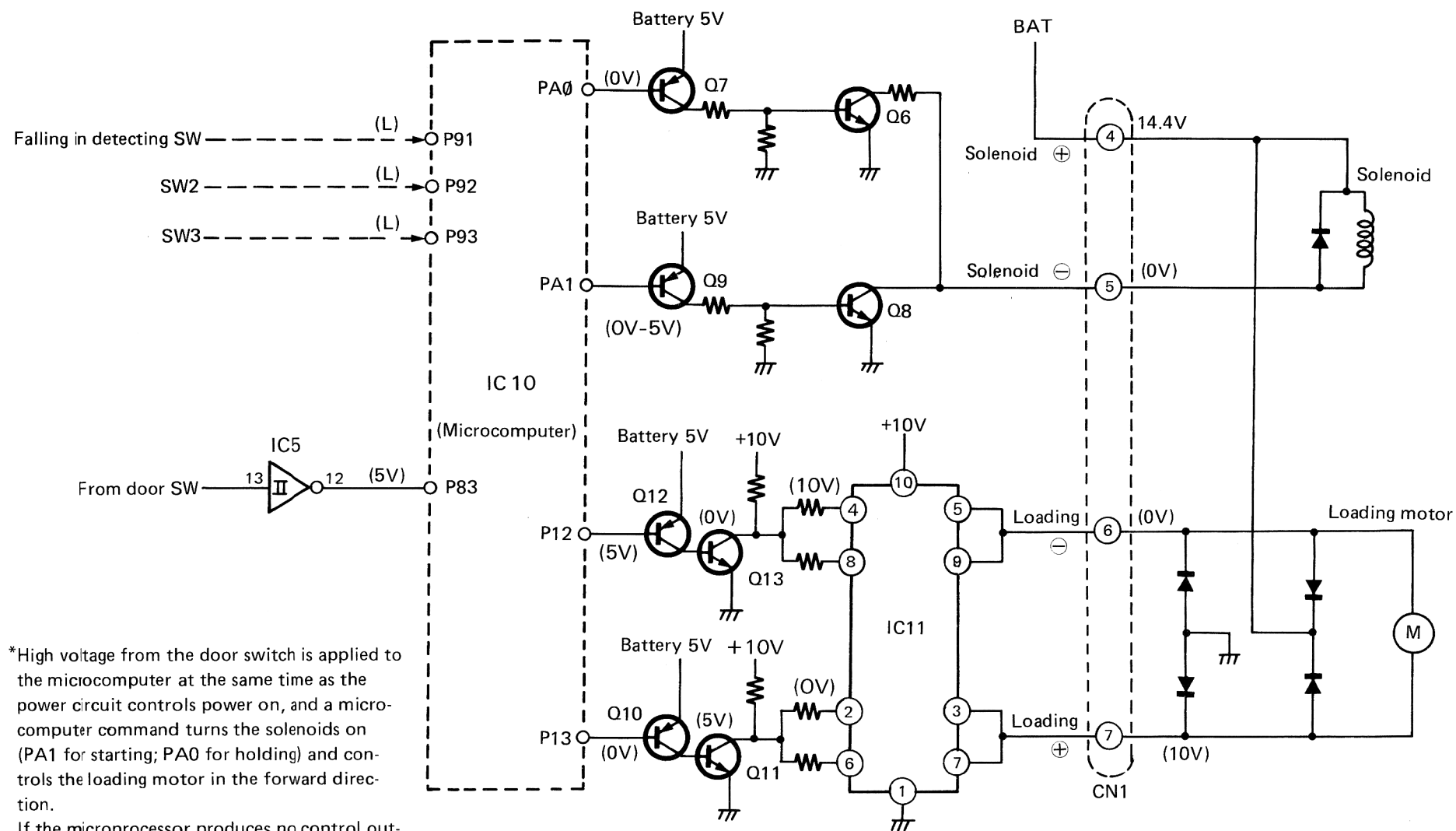
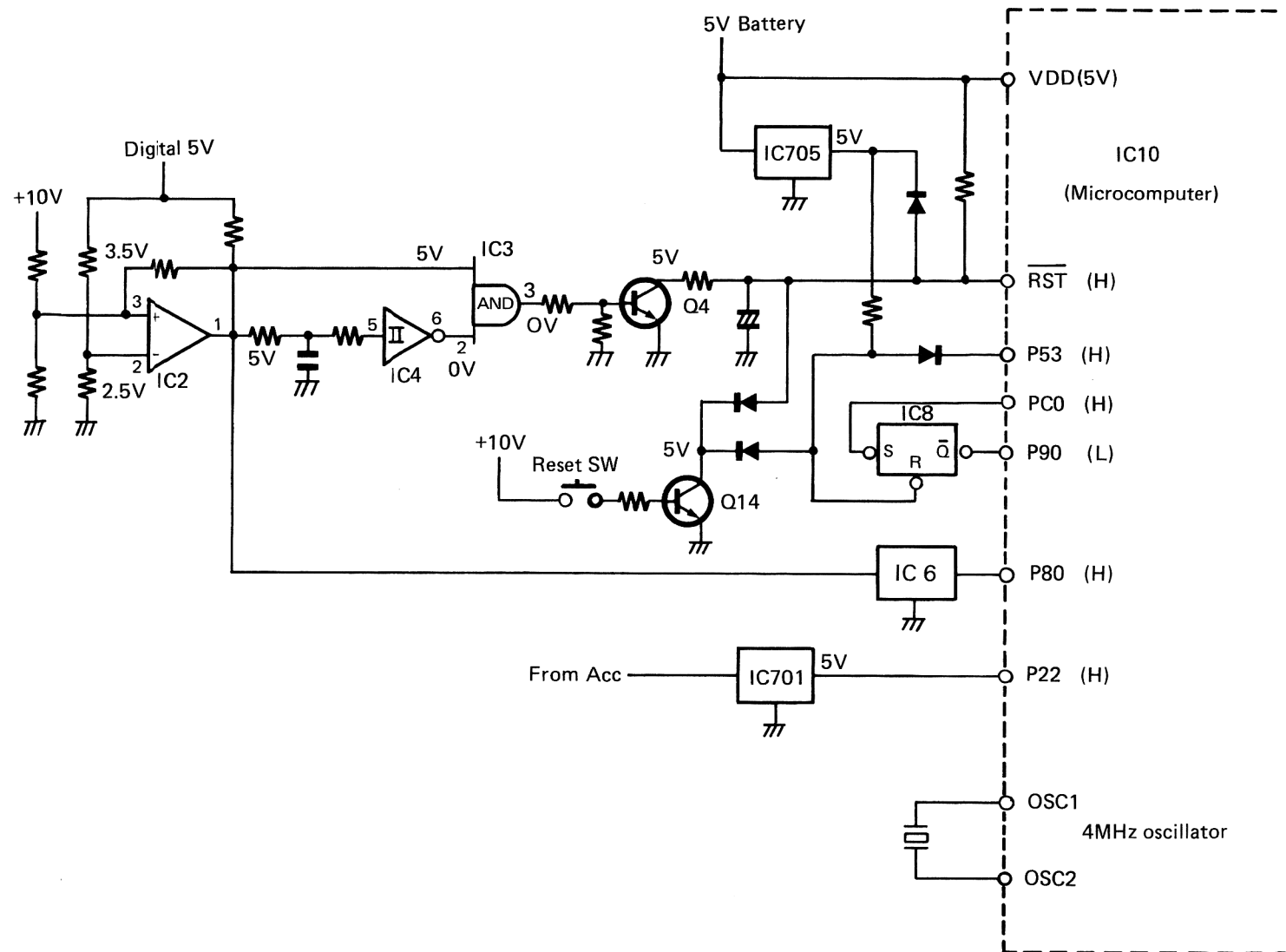


Fig. 2-1-2 Loading Motor and Solenoid Driver Circuit
(Circuit voltages apply where door is open and loading is under way.)

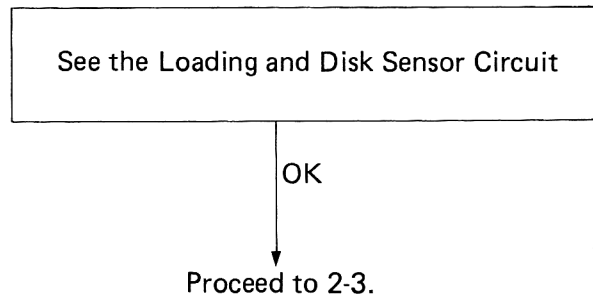


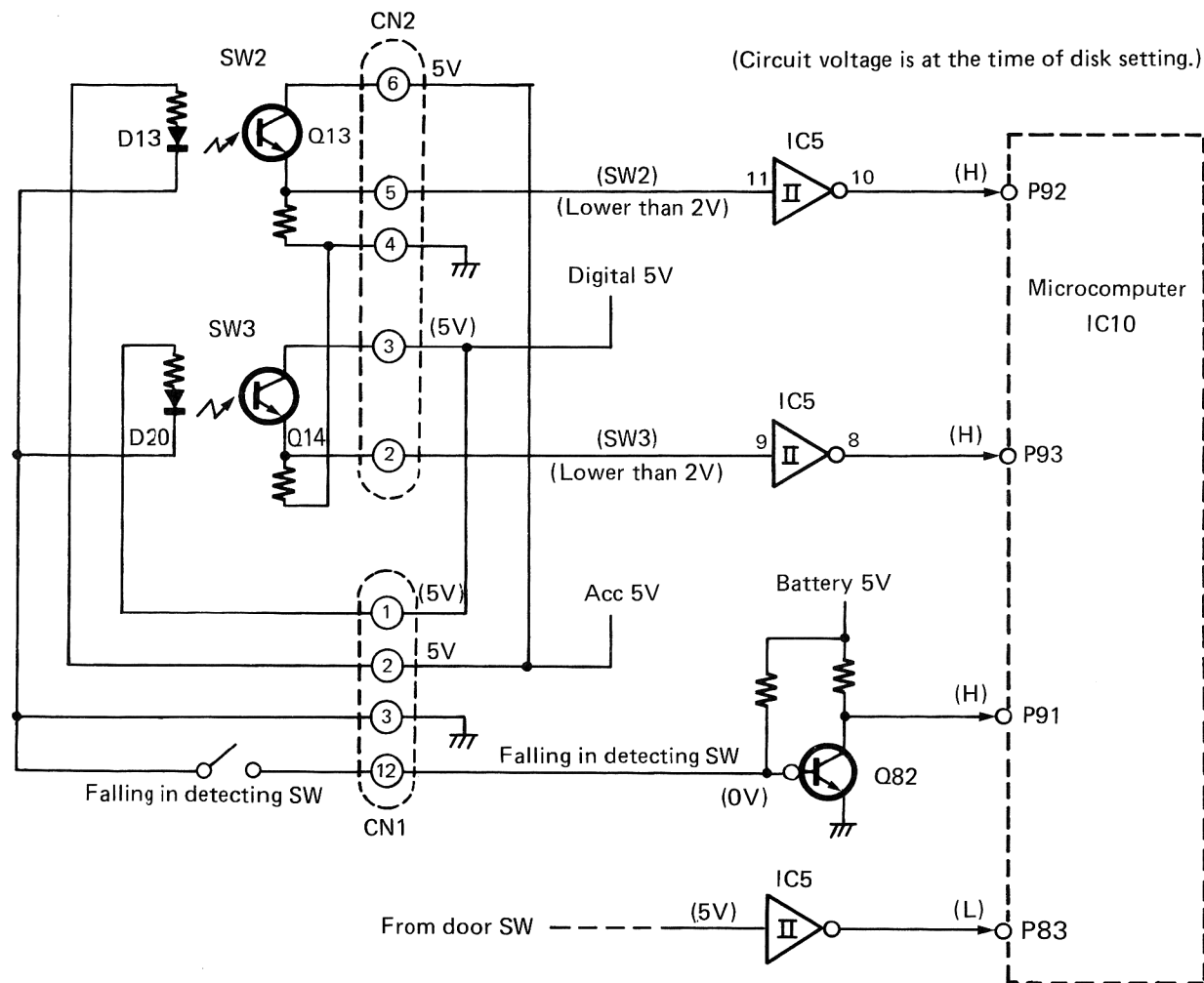
*This circuit diagram shows the power supply for operating microcomputer. Check each microcomputer port.

Fig. 2-1-3 Microcomputer Power and Reset Circuit
(Circuit voltages apply where a disk is in.)

2-2 Disk Sensor Check (Sensor Switch)

For a disk to be set on the turntable, the sensor must detect that the disk has been properly inserted so that the disk may be placed on the turntable and held down by the magnet. Check the loading sensor for its detecting operation.





Mode	Disk off	Door open	Door closed	Disk set
Micro-computer pin				
P92	L	L	H	H
P93	H	L	H	H
P91	L	L	L	H
P83	L	H	L	L

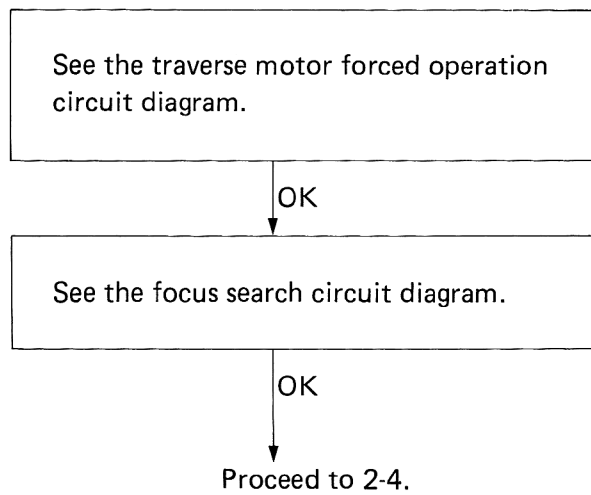
*When a disk is loaded, SW2 turns on the turn SW3 on and turn the solenoid off. Thus, the disk lowers and is held in position on the turntable by the magnet. If detecting circuit is not in good order, it becomes EJECT MODE.

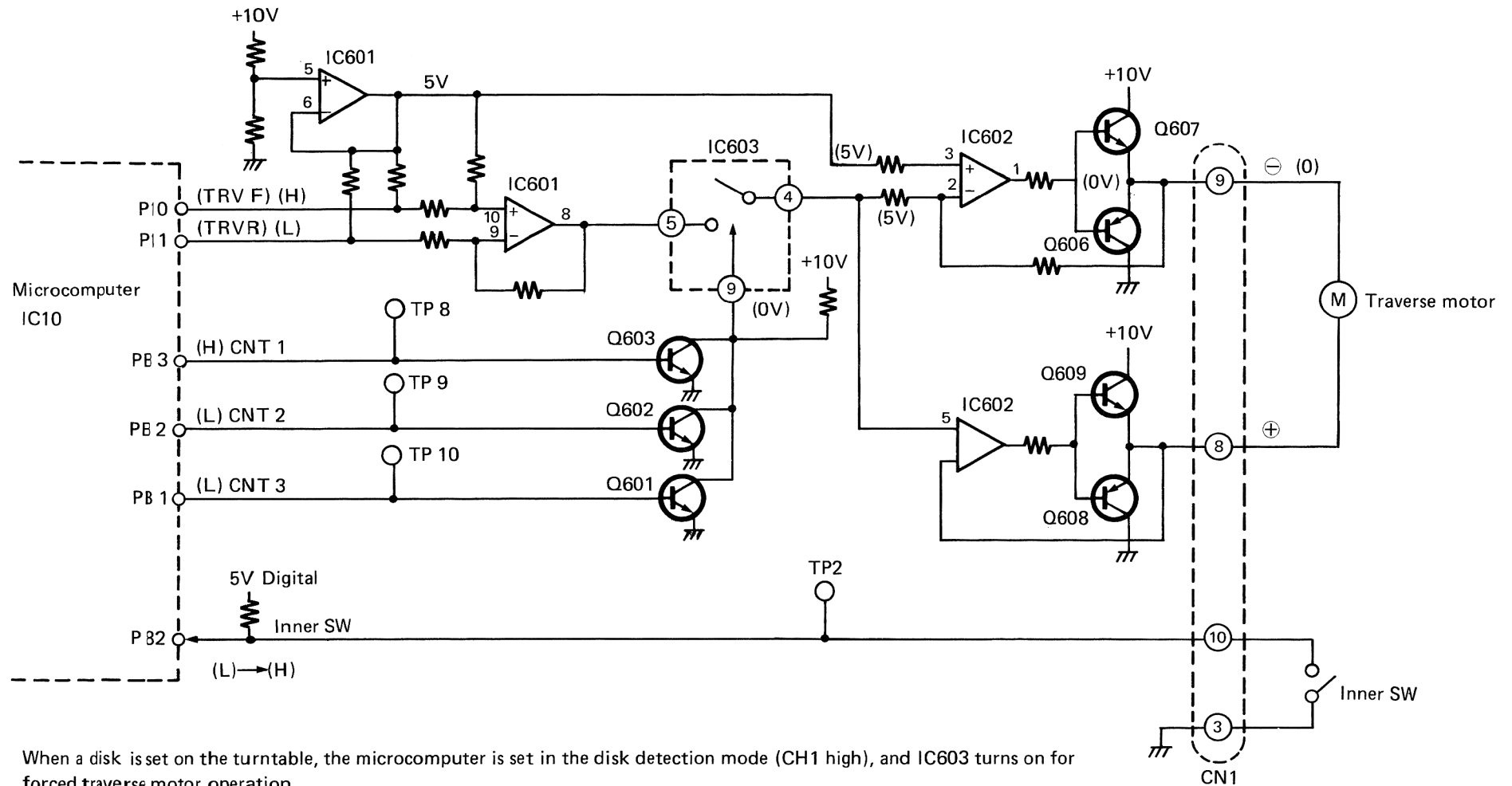
Fig. 2-2-1 Loading and Disk Sensor Circuit (Circuit voltages apply where a disk is set.)

2-3 Focus Search Check (Traverse Servo, Focus Servo)

When a disk is set on the turntable, the traverse motor runs forward, and the focus starts searching.
Proceed as follows:

- (1) Traverse motor forward run check
- (2) Focus search check



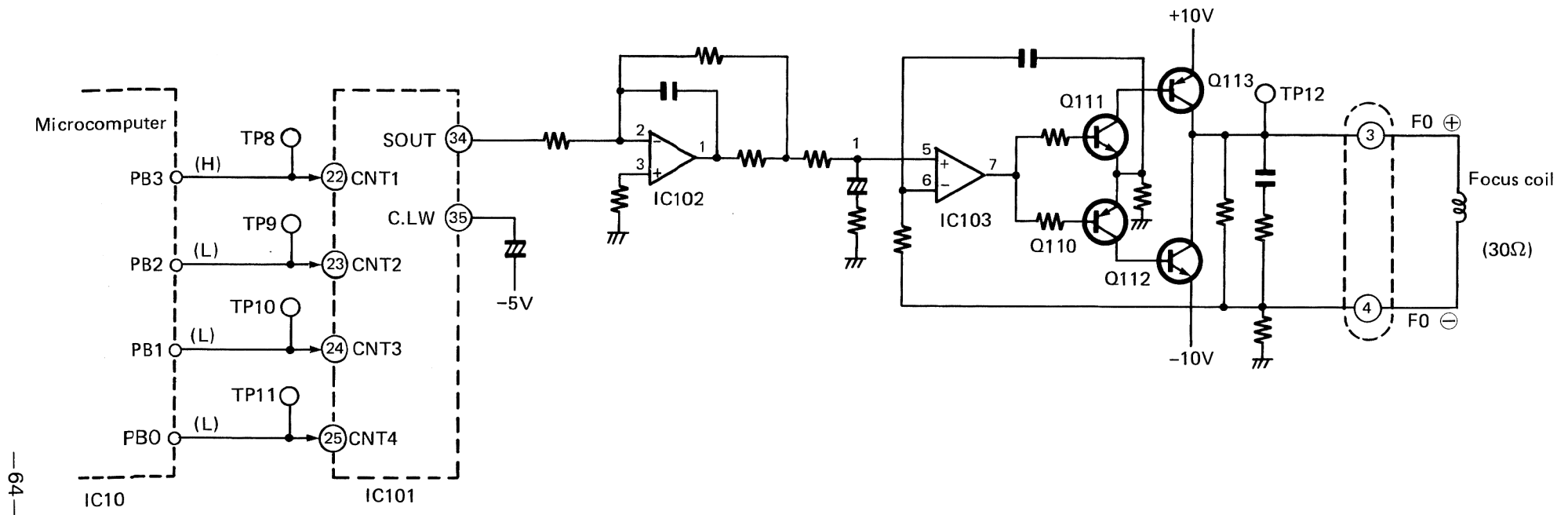


When a disk is set on the turntable, the microcomputer is set in the disk detection mode (CH1 high), and IC603 turns on for forced traverse motor operation.

Internal switch state detection by P82 on the microcomputer controls the traverse motor to run forward or backward (P10, P11), and the forced mode ends when the optical pickup reaches the prescribed position.

1. Check microcomputer PB3 (CNT1) that it is high.
2. Check if microcomputer P82 changes from low to high.
(Normally, operation starts in the internal switch ON state. The switch turns off when the traverse motor starts running forward.)
3. If P82 shows no change, check the circuit from P10 and P11 to the traverse motor.

Fig. 2-3-1 Traverse Motor Forced Operation Circuit (Circuit voltages apply where the traverse motor is running forward.)



Forcus search takes place for disk detection. CNT1 is high due to a command (disk detection) from the microcomputer, IC101 ③4 outputs a search signal (triangular wave) to drive the focus coil.

Fig. 2-3-2 Focus Search Circuit (Circuit voltages apply where forcus search is under way.)

2-4 Spindle Motor Circuit Check (Laser ON, spindle motor control)

When the focus is adjusted by focus search, a spindle motor start command is sent to start the spindle motor.

First, check the laser that it is ON and then the spindle motor circuit.

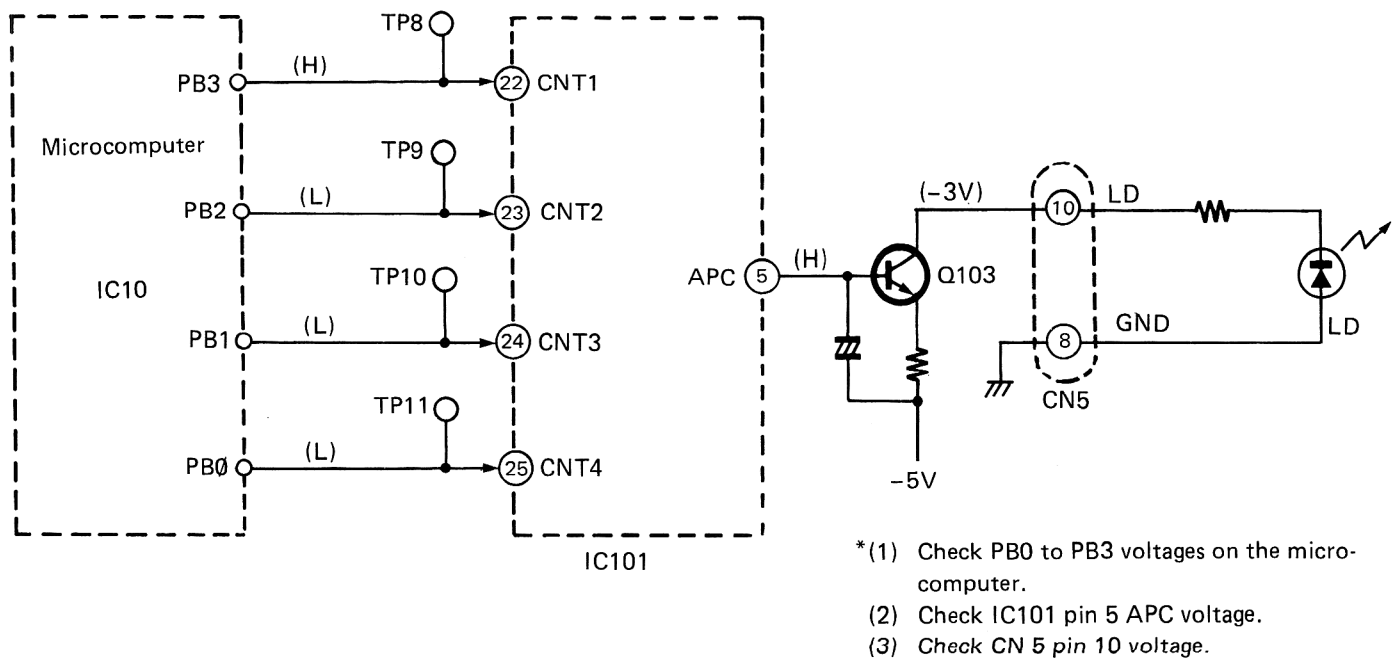
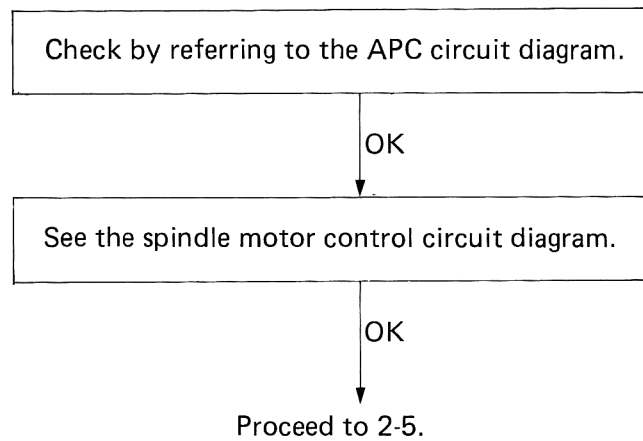
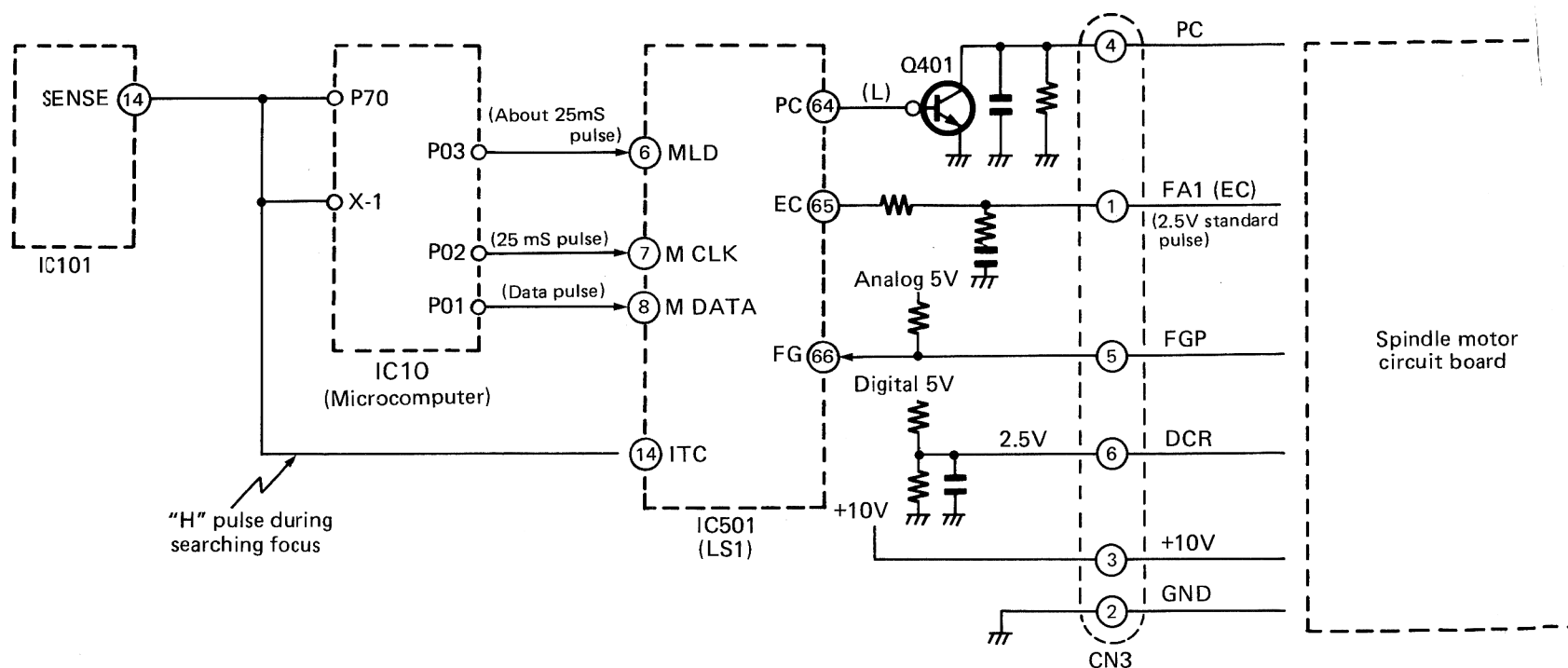


Fig. 2-4-1 APC Circuit (Circuit voltages apply where the disk detection mode is in effect.)



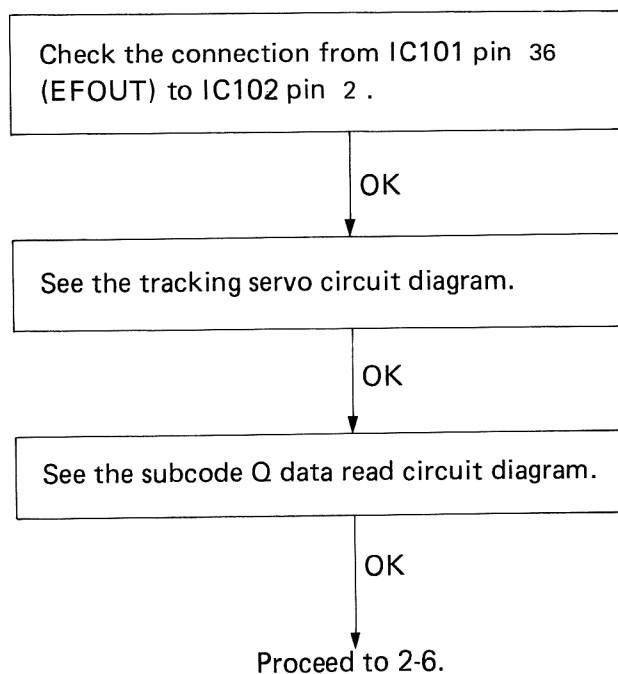
The turntable turns if focus search is under way and the laser is ON.
 High pulse (disk detection) is applied from SENSE ⑭ of IC101 to the microcomputer during search,
 and an LCV control signal is applied from the microcomputer to the LSI.
 The LSI generates a spindle motor ON signal (PC) and spindle motor drive pulse (EC).
 The traverse deck has the spindle motor drive IC, so the final check point is up to CN3.

Fig. 2-4-2 Spindle Motor Control Circuit (Circuit voltages apply where the play mode is in effect.)

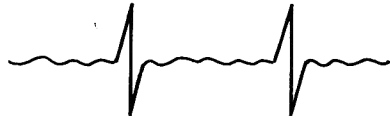
2-5 RF Signal Check (RF Signal, Tracking Servo, LSI)

- 1) When an RF signal is detected, the focus circuit changes from the search mode to the focus servo mode.
- 2) In this case, a focus lock signal is applied to the microcomputer, which sends a tracking ON, command to start the tracking servo circuit.
- 3) Then, the RF signal is converted to an EFM signal by the PLL IC, and the converted EFM signal is applied to the LSI. The LSI demodulates the subcode Q data and supplies it to the microcomputer, which deciphers the TOC and selects the play mode.

—If a fault is detected anywhere up to this point, the eject mode follows.—



1) RF waveform when the focus servo is faulty.



Check points

- Microcomputer input and output ports
- Connection from IC101 pin ③⑥ FEOUT to IC102 pin 2
- Focus offset readjustment
- Focus gain readjustment

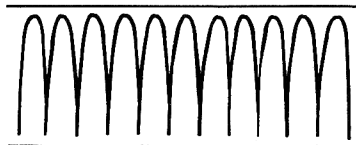
2) RF waveform when the tracking servo is faulty.

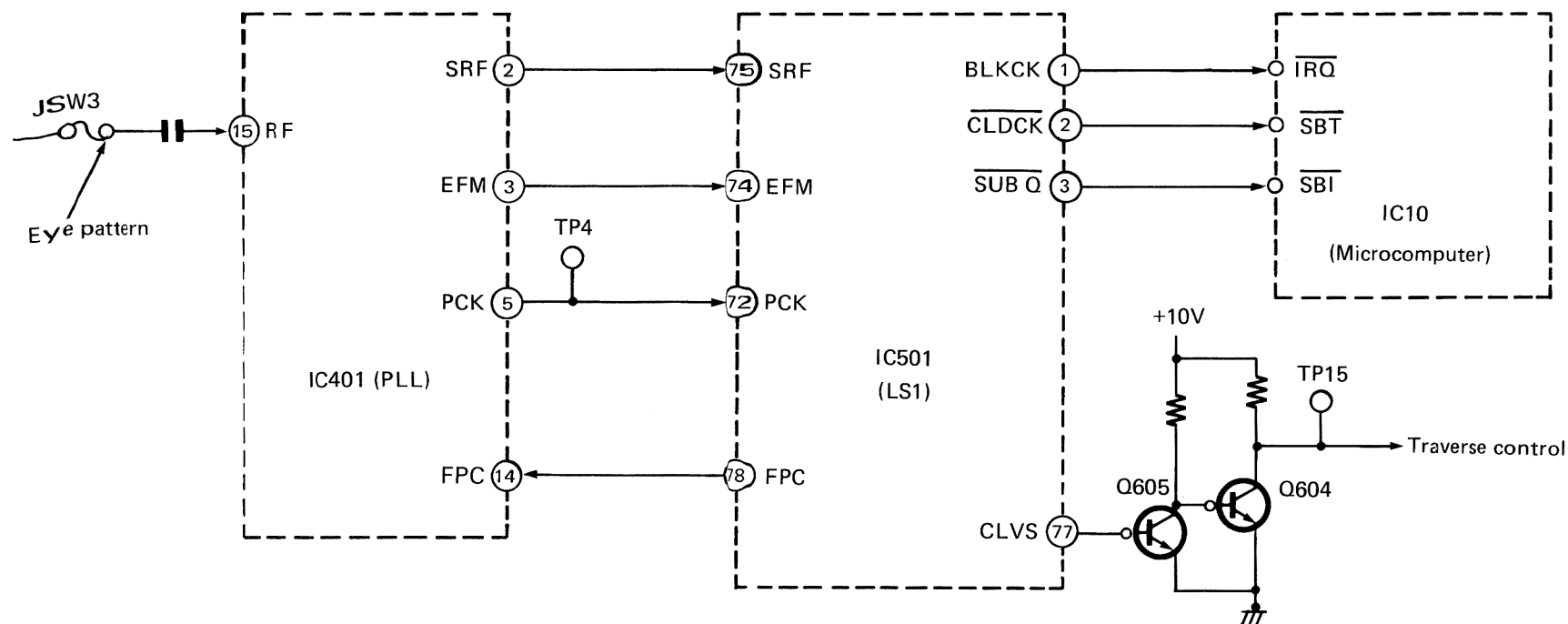


Check points

- Tracking servo circuit from IC101 pin ⑥ TEOUT to pin ⑮ HIN, pin ⑯ HOUT, CN4 pins ① and ②
- Tracking gain readjustment

3) Normal waveform





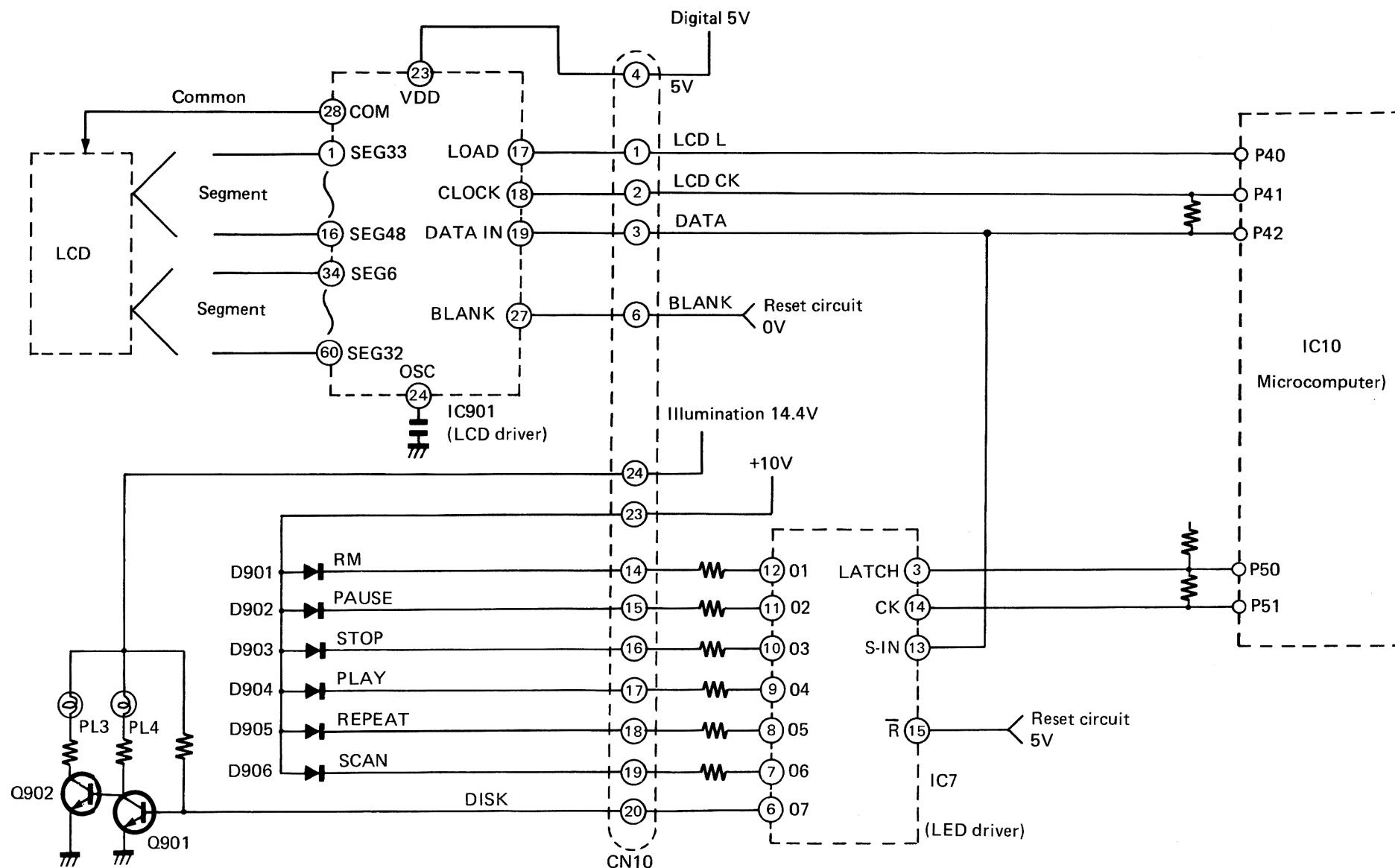
1. RF signal (JSW3) amplitude and eye pattern opening are not sufficient. Readjust the eye pattern.
2. Check PLL IC pin 3 EFM and pin 5 PCK.
3. Check LSI decision output pin 78 FPC and pin 77 CLVS.
 - ① CLVS Normal, FPC NG → PLL IC is faulty.
 - ② LLVS NG, FPC NG → Traverse deck (spindle circuit board) is faulty.
4. Check ① BLKCK, ② CLDCK, and ③ SUBQ from LSI.

Fig. 2-5-2 Subcode Q Data Read Circuit (Circuit voltages apply where the play mode is in effect.)

2-6 LCD and LED Check (LCD driver, LED driver)

When TOC is deciphered, data is fed from the microcomputer via the drivers to select the ON mode. The LCD flashes until the play mode starts.

See the LCD and LED circuit diagram.



LCD circuit check

1. Check the LCDL, LCDCK, and DATA signals from the microcomputer in the play mode.
2. Check CN10 pin 6 BLANK.
3. Check pulse waves at COM and SEG pins.
4. Check OSC and VDD.

LED circuit check

1. Check the LATCH, CK, and SIN signals from the microcomputer in the play mode.
2. Check IC7 pin 15 $\bar{R}$.
3. Check IC7 pin 9 04 that it is 0V in the play mode.
4. PL3 and PL4 light only when main amplifier illumination power is applied.

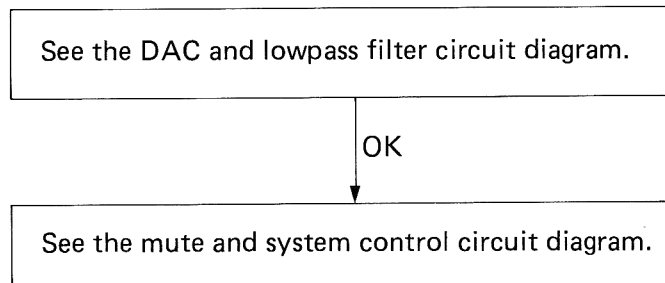
Fig. 2-6-1 LCD and LED Circuits (Circuits voltages apply where the play mode is in effect.)

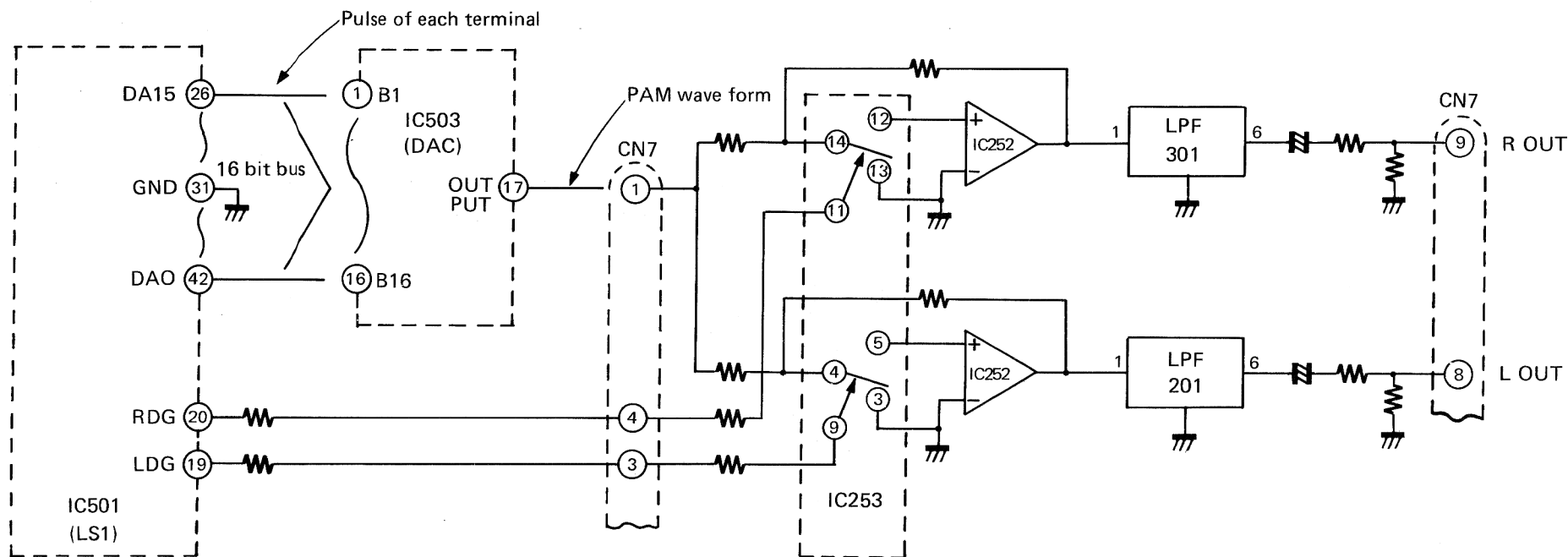
2-7 Audio Circuit Check (DAC, LPF, Mute)

If no sound is reproduced when the LCD is counting correctly in the play mode, check the audio circuit and mute circuit from the LSI to DIN pin output.

The 16-bit signals from the LSI are converted to analog signals by the DAC, and branched to right and left signals by deglitch. The branched signals are routed via the LPFs to be audio signals. Check the route. (In case of DP3, the signals are then routed via the preamplifier to the main amplifier.)

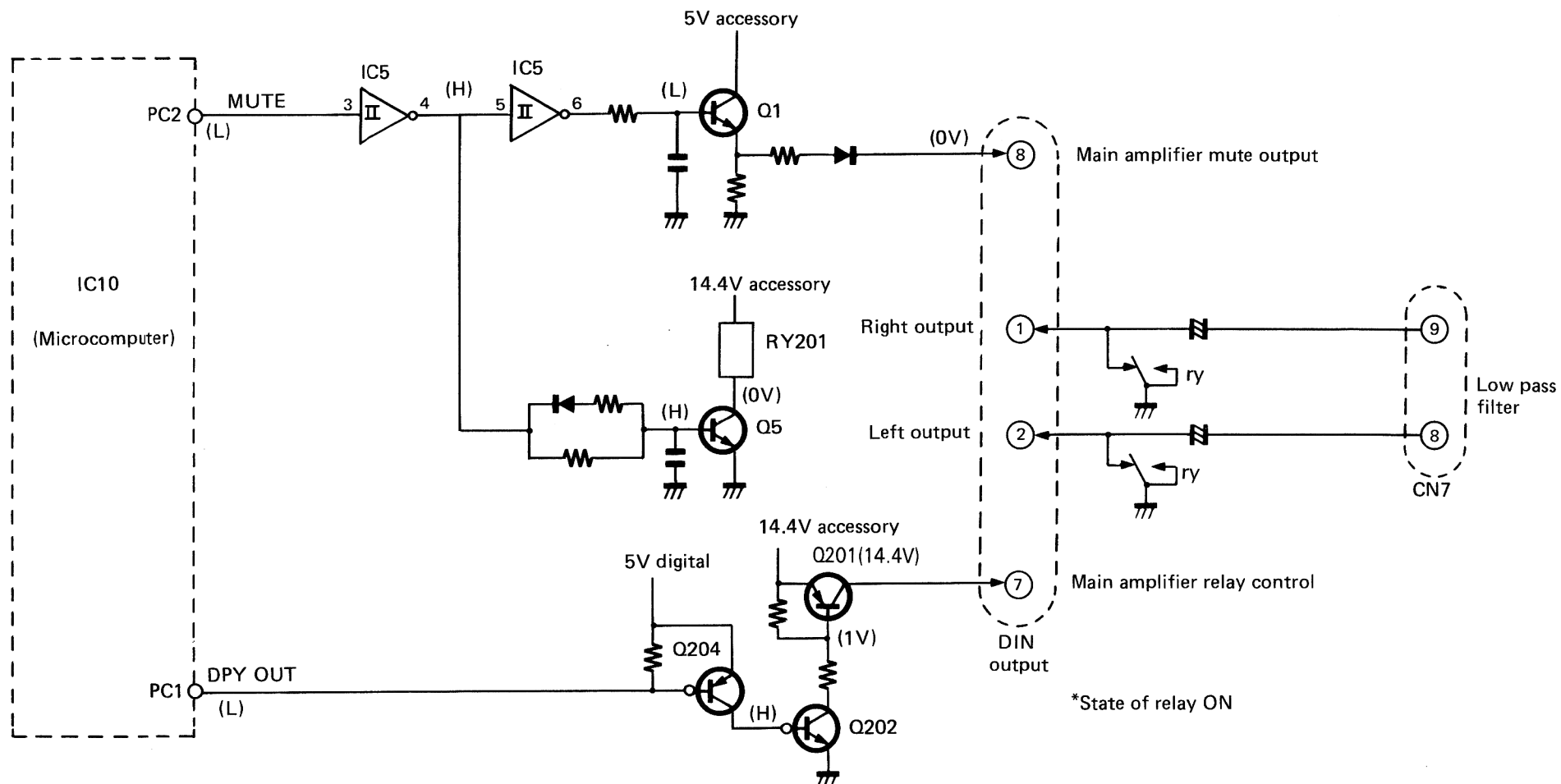
Then, check the mute circuit that it operates properly.





- (1) Check LSI pin 26 DA15 to pin 42 DAO output signals.
- (2) Check CN7 pin 1 PAM output signal. (IC503 pin 17 OUTPUT)
- (3) Check CN7 pin 4 RDG and pin 3 LDG output signals.
- (4) Check CN7 pin 9 ROUT and pin 8 LOOUT audio output signals.
(IC253 → IC252 → LPF)

Fig. 2-7-1 DAC and Lowpass Filter Circuits (Circuit voltages apply where the play mode is in effect.)



1. Check microcomputer PC1 that it is low in the play mode and high when the set is not operating.
2. Check microcomputer PC2 that it is low in the play mode and high during pause.
3. Check DIN pin ⑦ voltage.
(Q204, Q202, Q201)
4. Check DIN pin ⑨ voltage.
(IC5, Q1)
5. Check DIN pin ① and ② for short-circuit.
(RY201, Q5)

Fig. 2-7-2 Mute and System Control Circuits (Circuit voltages apply where the play mode is in effect.)

2-8 Checking Various Operations (Key decoder circuit)

Keyboard commands are received by the microcomputer, which sends control signals suitable to the state to initiate operate. The control signals from the microcomputer are considered nearly normal in view of the checks made so far, check the circuitry from the keyboard to the microcomputer.

See the keyboard interface circuit diagram.

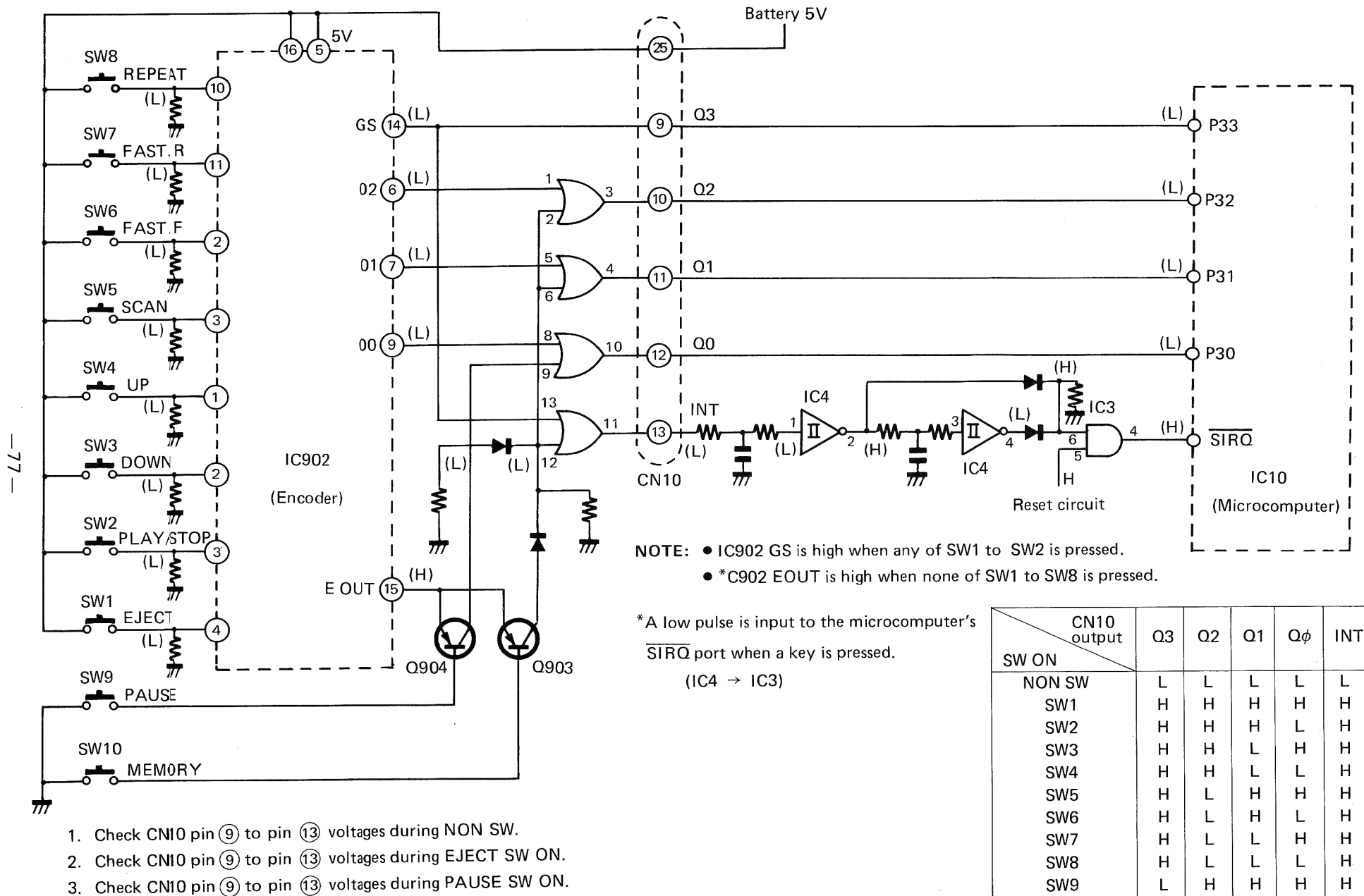


Fig.2-8-1 Keyboard Interface Circuit (Circuit voltages apply where the play mode is in effect with NON SW.)

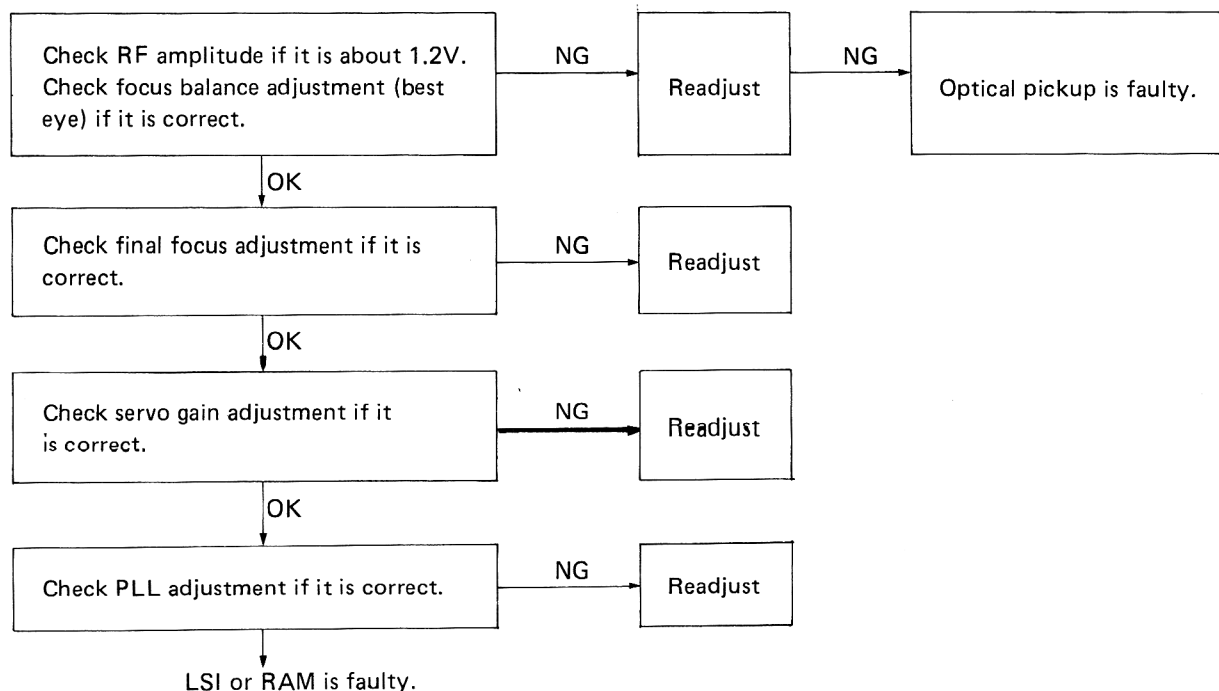
2-9 Readjustments

The set may be considered operating because the operation from reading the TOC to initiating play has been checked and found normal.

- Check the RF amplitude and eye pattern opening if they are sufficient.
- Check if offset adjustment is correct.
- Check if servo gain adjustment is correct.
- Check if the corrective capacity of the LSI is satisfactory.
- * For the adjustment procedures, refer to the adjustment part of the manual.

Check procedures and possible faulty points

Connect an oscilloscope as described in the adjustment procedures, and monitor the following signals.



3. APPENDIX

3-1 Connector Voltage (Stop, Play)

Deck Connector CN1 (12-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	Digital 5V	5V	→	0 when disk is off
2	Accessory 5V	5V	→	
3	GND	0	→	
4	Solenoid ⊕ output	14.4V	→	Battery 14.4V power
5	Solenoid ⊖ output	14.4V	→	0 when door is open
6	Loading motor ⊖ output	0	→	10V during eject
7	Loading motor ⊕ output	0	→	10V during loading
8	Traverse ⊕ output	5V	Waveform	
9	Traverse ⊖ output	5V	Waveform	
10	Inner SW input	5V	→	0 when disk is off.
11	NC	/	/	
12	Anti-drop SW input	0	→	5V before magnet is down

Deck Switch Input CN2 (7-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	NC	/	/	
2	SW3 input	2V or less	→	0 when disk is off
3	Digital 5V	5V	→	
4	GND	0	→	
5	SW2 input	2V or less	→	5V when disk is off
6	Accessory 5V	5V	→	
7	Door SW input	5V	→	2V or less when door is open

Spindle Motor Input/Output CN3 (6-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	FAI (EC) output	1.5V	Waveform	
2	GND	0	→	
3	VCC (10V)	10V	→	
4	PC output	0		
5	FGP input	0	Waveform	
6	DCR	2.5V	→	

Focus Tracking Output CN4 (4-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	Tr ⊕	0	Waveform	
2	Tr ⊖	0	→	
3	Fo ⊕	0	Waveform	
4	Fo ⊖	0	→	

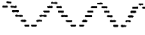
Optical Pickup Input/Output CN5 (11-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	G GND	0	→	Analog + 5V Signal is so low in level that it is invisible.
2	A2	0	→	
3	A3	0	→	
4	A1	0	→	
5	A4	0	→	
6	⊕ 5V	5V	→	
7	RF	4V	→	
8	GND	0	→	
9	PD	⊖ 5V	⊖ 3V	
10	LD	0	⊖ 3V	
11	⊖ 5V	⊖ 5V	→	

LPF Digital Power CN6 (3-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	GND	0	→	Digital 5V
2	⊖ 5V	⊖ 5V	→	
3	⊕ 5V	5V	→	

LPF Control Input/Output CN7 (9-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	PAM	0		De-emphasis disk 5V during play
2	GND	0	→	
3	LDG	Waveform	→	
4	RDG	Waveform	→	
5	D. EMPH	0	0	
6	⊕ 10V	10V	→	
7	⊖ 10V	⊖ 10V	→	
8	ROUT	0	Audio Waveform	
9	LOUT	0	Audio Waveform	

DC/DC Converter CN8 (6-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	V BAT	14.4V	→	Battery power
2	GND	0	→	
3	V ACC	14.4V	→	Accessory power 0 when disk is off
4	V CONT	5V	→	
5	⊖ 5V	– 5V	→	
6	⊖ 10V	– 10V	→	

TRE, BASS, CONT CN9 (6-pin) P3D only

Pin No.	Name of Pin	Stop	Play	Remarks
1	RNF	0	4.5V	When connected to DIN input
2	R ⊖	0	4.5V	
3	ROUT	0	4.5V	
4	LOUT	0	4.5V	
5	L ⊖	0	4.5V	
6	LNF	0	4.5V	

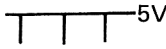
LCD Control Key Input CN10 (26-pin)

Pin No.	Name of Pin	Stop	Play	Remarks
1	LCD LOAD			
2	LCD CLOCK			
3	DATA	Waveform	→	
4	5V	5V	→	Digital 5V
5	NC	/	/	
6	BLANK (LCD)	0	→	High when reset
7	GND	0	→	
8	Illumination (LCD)	14.4V	→	From system main amplifier illumination power
9	Q3	0	→	} High pulse when key is ON
10	Q2	0	→	
11	Q1	0	→	
12	Q ϕ	0	→	
13	INT	0	→	
14	RM (Random memory)	8V	→	} Voltage drops when LED lights.
15	RAUSE	8V	→	
16	STOP	7V	8V	
17	PLAY	8V	7V	
18	REPEAT	8V	→	
19	SCAN	8V	→	
20	DISC	0		Repeats high and low during loading and eject.
21	ALARM (NC)	/	/	
22	GND	0	→	
23	⊕ 10V	10V	→	
24	Illumination	14.4V	→	From system main amplifier illumination power
25	BUP 5V	5V	→	Battery 5V
26	RESET	0	→	10V when reset button is ON

3-2 Microcomputer Pin Voltages (Stop, Play)

Microcomputer MN1554 CAC or MN1554

Sheet 1

Port No.		Port symbol	Signal	Input/Output	Voltage, etc.		Description
Flat	Dual				Stop	Play	
1	9	P12		O	5V	→	During loading { H L During eject { L H Loading motor FWD/REV control
2	10	P13		O	5V	→	
3	11	$\overline{\text{SYNC}}$		NC			
4	12	$\overline{\text{SIRO}}$		I	5V	→	Low key interrupt input when key input is entered
5	13	$\overline{\text{IRO}}$	BLK CK	I	0		IC501 block clock (75 Hz)
6	14	$\overline{\text{SBT}}$	CLD CK	I	5V		IC501 frame clock (7.53 kHz)
7	15	$\overline{\text{SBO}}$		NC			Subcode Q take-in
8	16	$\overline{\text{SBI}}$	SUB Q	I	Waveform	Waveform	
9	17	$\overline{\text{RST}}$		I	5V	→	
10	18	P2 ϕ	CDPSIN	I	0	→	Toyota specifications H . . . Stop
11	19	P21	HAR	I	0	→	H . . . Stop (Traffic information)
12	20	P22	VAC+5V	I	5V	→	ACC power detection
13	21	P23		NC			
14	22	P3 ϕ	Q ϕ	I	0	→	Key input of 4-bit code
15	23	P31	Q1	I	0	→	
16	24	P32	Q2	I	0	→	
17	25	P33	Q3	I	0	→	
18	26	P4 ϕ	LCD LOAD	O			IC901 LCD driver load LCD data sending

Port No.		Port symbol	Signal	Input/Output	Voltage, etc.		Description	
Flat	Dual				Stop	Play		
19	27	P41	LCD CK	O	0		IC901 LCD driver clock	LCD data sending
20	18	P42	DATA	O			LCD, LED data	} LED data sending
21	29	P43		NC				
22	30	P5 ϕ	LED LOAD	O			IC7 LED decoder load	
23	31	P51	LED CK	O			IC7 LED decoder clock	
24	32	P52		NC				
25	33	VDD		I	5V	→	Battery 5V	
26	34	PC ϕ		O	5V	→		
27	35	PC1	CDPY OUT	O	5V	0	CD play signal, system control	
28	36	P53		O	5V	→	Battery 5V detection	
29	37	P6 ϕ		NC				
30	38	P61		NC				
31	39	P62		NC				
32	40	P63		NC				
33	41	P7 ϕ	$\overline{\text{SENSE}}$	I			IC101 Optical servo state signal input	
34	42	P71	$\overline{\text{FLOCK}}$	I	5V	0	IC101 RF signal/no RF signal input; Low for RF signal input	
35	43	P72	TEMP	I	5V	→	IC101 temperature sensor	L: Stop command
36	44	P73	DTC (STAT)	I	0	0 (Pulse sometimes)	IC501 LCV state confirmation input	
37	45	P8 ϕ	Low voltage detection	I	5V	→	IC2 (Reset circuit)	
38	46	P81		NC				

Port No.		Port symbol	Signal	Input/ Output	Voltage, etc.		Description
Flat	Dual				Stop	Play	
39	47	P82	Inner SW	I			Traverse position confirmation
40	48	P83	Door SW	I	0	→	
41	49	P9φ		I			
42	50	P91	Anti-drop	I	5V	→	
43	51	P92	SW2	I	5V	→	
44	52	P93	SW3	I	5V	→	0 when door starts opening 0 when door is open } Loading solenoid control
45	53	PAφ		O	5V	→	
46	54	PA1		O	5V	→	
47	55	PA2		NC			Power circuit control
48	56	PA3		O	5V	→	
49	57	PBφ	CNT4	O	5V	0	
50	58	PB1	CNT3	O	5V	0	} Optical servo control signal
51	59	PB2	CNT2	O	5V	0	
52	60	PB3	CNT1	O	5	0	
53	61	OSC2				→	} Crystal 4 MHz
54	62	OSC1				→	
55	63	X1		I	→	→	Same as $\overline{\text{SENSE}}$ input as to P7φ
56	64	Xφ		NC			
57	1	VSS		O	0	→	
58	2	Pφφ		NC			

Port No.		Port symbol	Signal	Input/ Output	Voltage, etc.		Description
Flat	Dual				Stop	Play	
59	3	P ϕ 1	M DATA	O	0	→	IC501 data IC501 clock IC501 load LCV control and interpolation circuit mode setting
60	4	P ϕ 2	M CLK	O	5	→	
61	5	P ϕ 3	M LD	O	5	→	
62	6	PC2	MUTE	O	0	→	High during skip, etc.
63	7	P1 ϕ	TR FWD	O	0	→	Traverse forced forward signal high
64	8	P11	TR REV	O	0	→	Traverse forced reverse signal high